

The Grails Framework

Authors: The Grails Team

Version: 6.2.2

Table of Contents

[1 Introduction](#)

[1.1 What's new in Grails 6?](#)

[1.1.1 Updated Dependencies](#)

[2 Getting Started](#)

[2.1 Installation Requirements](#)

[2.2 Downloading and Installing](#)

[2.3 Creating an Application](#)

[2.4 Creating a Simple Web Application with Grails](#)

[2.5 Using Interactive Mode](#)

[2.6 Getting Set Up in an IDE](#)

[2.7 Grails Directory Structure and Convention over Configuration](#)

[2.8 Running and Debugging an Application](#)

[2.9 Testing an Application](#)

[2.10 Deploying an Application](#)

[2.11 Supported Java EE Containers](#)

[2.12 Creating Artefacts](#)

[2.13 Generating an Application](#)

[3 Upgrading from the previous versions](#)

[3.1 Upgrading from Grails 5 to Grails 6](#)

[3.2 Upgrading from Grails 4 to Grails 5](#)

[3.3 Upgrading from Grails 3.3.x to Grails 4](#)

[4 Configuration](#)

[4.1 Basic Configuration](#)

[4.1.1 Options for the YAML format Config](#)

[4.1.2 Built in options](#)

[4.1.3 Logging](#)

[4.1.3.1 Logger Names](#)

[4.1.3.2 Masking Request Parameters From Stacktrace Logs](#)

[4.1.3.3 External Configuration File](#)

[4.1.4 GORM](#)

[4.1.5 Configuring an HTTP proxy](#)

[4.2 The Application Class](#)

[4.2.1 Executing the Application Class](#)

[4.2.2 Customizing the Application Class](#)

[4.2.3 The Application LifeCycle](#)

[4.3 Environments](#)

[4.4 The DataSource](#)

[4.4.1 DataSources and Environments](#)

[4.4.2 Automatic Database Migration](#)

[4.4.3 Transaction-aware DataSource Proxy](#)

- [4.4.4 Database Console](#)
- [4.4.5 Multiple Datasources](#)
- [4.5 Versioning](#)
- [4.6 Dependency Resolution](#)
- [5 The Command Line](#)
 - [5.1 Interactive Mode](#)
 - [5.2 Creating Custom Commands](#)
 - [5.3 Creating a Grails Project](#)
- [6 Object Relational Mapping \(GORM\)](#)
 - [6.1 Quick Start Guide](#)
 - [6.1.1 Basic CRUD](#)
 - [6.2 Further Reading on GORM](#)
- [7 The Web Layer](#)
 - [7.1 Controllers](#)
 - [7.1.1 Understanding Controllers and Actions](#)
 - [7.1.2 Controllers and Scopes](#)
 - [7.1.3 Models and Views](#)
 - [7.1.4 Redirects and Chaining](#)
 - [7.1.5 Data Binding](#)
 - [7.1.6 Responding with JSON](#)
 - [7.1.7 More on JSONBuilder](#)
 - [7.1.8 Responding with XML](#)
 - [7.1.9 Uploading Files](#)
 - [7.1.10 Command Objects](#)
 - [7.1.11 Handling Duplicate Form Submissions](#)
 - [7.1.12 Simple Type Converters](#)
 - [7.1.13 Declarative Controller Exception Handling](#)
 - [7.2 Groovy Server Pages](#)
 - [7.3 URL Mappings](#)
 - [7.3.1 Mapping to Controllers and Actions](#)
 - [7.3.2 Mapping to REST resources](#)
 - [7.3.3 Redirects In URL Mappings](#)
 - [7.3.4 Embedded Variables](#)
 - [7.3.5 Mapping to Views](#)
 - [7.3.6 Mapping to Response Codes](#)
 - [7.3.7 Mapping to HTTP methods](#)
 - [7.3.8 Mapping Wildcards](#)
 - [7.3.9 Automatic Link Re-Writing](#)
 - [7.3.10 Applying Constraints](#)
 - [7.3.11 Named URL Mappings](#)
 - [7.3.12 Customizing URL Formats](#)
 - [7.3.13 Namespaced Controllers](#)
 - [7.4 CORS](#)
 - [7.5 Interceptors](#)
 - [7.5.1 Defining Interceptors](#)
 - [7.5.2 Matching Requests with Interceptors](#)
 - [7.5.3 Ordering Interceptor Execution](#)
 - [7.6 Content Negotiation](#)
- [8 Traits](#)
 - [8.1 Traits Provided by Grails](#)

[8.1.1 WebAttributes Trait Example](#)

[9 REST](#)

[9.1 Domain classes as REST resources](#)

[9.2 Mapping to REST resources](#)

[9.3 Linking to REST resources from GSP pages](#)

[9.4 Versioning REST resources](#)

[9.5 Implementing REST controllers](#)

[9.5.1 Extending the RestfulController super class](#)

[9.5.2 Implementing REST Controllers Step by Step](#)

[9.5.3 Generating a REST controller using scaffolding](#)

[9.6 Calling REST Services with HttpClient](#)

[9.7 The REST Profile](#)

[9.8 JSON Views](#)

[9.8.1 Getting Started](#)

[9.8.2 Creating JSON Views](#)

[9.8.3 JSON View Templates](#)

[9.8.4 Rendering Domain Classes with JSON Views](#)

[9.8.5 JSON Views by Convention](#)

[9.9 Customizing Response Rendering](#)

[9.9.1 Customizing the Default Renderers](#)

[9.9.2 Implementing a Custom Renderer](#)

[9.9.3 Using GSP to Customize Rendering](#)

[9.10 Hypermedia as the Engine of Application State](#)

[9.10.1 HAL Support](#)

[9.10.2 Atom Support](#)

[9.10.3 Vnd.Error Support](#)

[9.11 Customizing Binding of Resources](#)

[9.12 RSS and Atom](#)

[10 Asynchronous Programming](#)

[11 Validation](#)

[11.1 Declaring Constraints](#)

[11.2 Validating Constraints](#)

[11.3 Sharing Constraints Between Classes](#)

[11.4 Validation on the Client](#)

[11.5 Validation and Internationalization](#)

[11.6 Applying Validation to Other Classes](#)

[12 The Service Layer](#)

[12.1 Declarative Transactions](#)

[12.1.1 Transactions and Multi-DataSources](#)

[12.1.2 Transactions Rollback and the Session](#)

[12.2 Scoped Services](#)

[12.3 Dependency Injection and Services](#)

[13 Static Type Checking And Compilation](#)

[13.1 The GrailsCompileStatic Annotation](#)

[13.2 The GrailsTypeChecked Annotation](#)

[14 Testing](#)

[14.1 Unit Testing](#)

[14.2 Integration Testing](#)

[14.3 Functional Testing](#)

[15 Internationalization](#)

- [15.1 Understanding Message Bundles](#)
- [15.2 Changing Locales](#)
- [15.3 Reading Messages](#)
- [15.4 Scaffolding and i18n](#)
- [16 Security](#)
 - [16.1 Securing Against Attacks](#)
 - [16.2 Cross Site Scripting \(XSS\) Prevention](#)
 - [16.3 Encoding and Decoding Objects](#)
 - [16.4 Authentication](#)
 - [16.5 Security Plugins](#)
 - [16.5.1 Spring Security](#)
- [17 Plugins](#)
 - [17.1 Creating and Installing Plugins](#)
 - [17.2 Plugin Repositories](#)
 - [17.3 Providing Basic Artefacts](#)
 - [17.4 Evaluating Conventions](#)
 - [17.5 Hooking into Runtime Configuration](#)
 - [17.6 Adding Methods at Compile Time](#)
 - [17.7 Adding Dynamic Methods at Runtime](#)
 - [17.8 Participating in Auto Reload Events](#)
 - [17.9 Understanding Plugin Load Order](#)
 - [17.10 The Artefact API](#)
 - [17.10.1 Asking About Available Artefacts](#)
 - [17.10.2 Adding Your Own Artefact Types](#)
- [18 Grails and Spring](#)
 - [18.1 Configuring Additional Beans](#)
 - [18.2 Runtime Spring with the Beans DSL](#)
 - [18.3 The BeanBuilder DSL Explained](#)
 - [18.4 Property Placeholder Configuration](#)
 - [18.5 Property Override Configuration](#)
 - [18.6 Spring Boot Actuators](#)
- [19 Scaffolding](#)
- [20 Deployment](#)
 - [20.1 Standalone](#)
 - [20.2 Container Deployment \(e.g. Tomcat\)](#)
 - [20.3 Deployment Configuration Tasks](#)
- [21 Contributing to Grails](#)
 - [21.1 Report Issues in Github's issue tracker](#)
 - [21.2 Build From Source and Run Tests](#)
 - [21.3 Submit Patches to Grails Core](#)
 - [21.4 Submit Patches to Grails Documentation](#)

1 Introduction

Many modern web frameworks in the Java space are more complicated than needed and don't embrace the Don't Repeat Yourself (DRY) principles.

Dynamic frameworks like Rails and Django helped pave the way to a more modern way of thinking about web applications. Grails builds on these concepts and dramatically reduces the complexity of building web applications on the Java platform. What makes it different,

however, is that it does so by building on already established Java technologies like Spring and Hibernate.

Grails is a full stack framework and attempts to solve as many pieces of the web development puzzle through the core technology and its associated plugins. Included out the box are things like:

- [GORM](#) - An easy to use Object Mapping library with support for [SQL](#), [MongoDB](#), [Neo4j](#) and [more](#).
- View technologies for [rendering HTML](#) as well as [JSON](#)
- A controller layer built on [Spring Boot](#)
- A plugin system featuring [hundreds of plugins](#).
- Flexible profiles to [create applications with AngularJS, React and more](#).
- An interactive command line environment and build system based on [Gradle](#)
- An embedded [Tomcat](#) container which is configured for on the fly reloading

All of these are made easy to use through the power of the [Groovy](#) language and the extensive use of Domain Specific Languages (DSLs)

This documentation will take you through getting started with Grails and building web applications with the Grails framework.

In addition to this documentation there are [comprehensive guides](#) that walk you through various aspects of the technology.

Finally, Grails is far more than just a web framework and is made up of various sub-projects. The following table summarizes some other key projects in the eco-system with links to documentation.

Table 1. Gra

Project	Description
GORM for Hibernate	An Object Mapping implementation for SQL databases
GORM for MongoDB	An Object Mapping implementation for the MongoDB Document Database
GORM for Neo4j	An Object Mapping implementation for Neo4j Graph Database
JSON Views	A View technology for rendering JSON on the server side
Groovy Server Pages	A View technology for rendering HTML and other markup on the server

1.1 What's new in Grails 6?

This section covers all the new features introduced in Grails 6

Overview

Grails framework 6 updates Spring Boot to version 2.7. We recommend checking the following Spring technologies release notes for more information.

- [Spring Boot 2.7](#)

The minimum Java version required to run Grails 6 has been updated to Java 11.

Support for Micronaut Environments in `application.yml`

The support has been introduced for Micronaut's Environments concept within the `application.yml` file. This feature allows developers to define environment-specific configurations seamlessly, aiding in smoother transitions between different operational environments like development, testing, and production.

1.1.1 Updated Dependencies

Grails 6.2.2 ships with the following dependency upgrades:

- Groovy 3.0.23
- Micronaut 3.10.4
- Micronaut for Spring 4.5.1
- GORM 8.1.2
- Spring Framework 5.3.39
- Spring Boot 2.7.18
- Gradle 7.6.4
- Spock 2.3-groovy-3.0
- Grails Testing Support 3.2.2

2 Getting Started

2.1 Installation Requirements

Before installing Grails you will need a Java Development Kit (JDK) installed with the minimum version denoted in the table below. Download the appropriate JDK for your operating system, run the installer, and then set up an environment variable called `JAVA_HOME` pointing to the location of this installation.

Grails version	JDK version (minimum)
6	11
5	8

To automate the installation of Grails we recommend [SDKMAN](#) which greatly simplifies installing and managing multiple Grails versions.

On some platforms (for example macOS) the Java installation is automatically detected. However in many cases you will want to manually configure the location of Java. For example, if you're using bash or another variant of the Bourne Shell:

```
export JAVA_HOME=/Library/Java/Home
export PATH="$PATH:$JAVA_HOME/bin"
```

On Windows you would have to configure these environment variables in `My Computer/Advanced/Environment Variables`

2.2 Downloading and Installing

The first step to getting up and running with Grails is to install the distribution.

The best way to install Grails on *nix systems is with [SDKMAN](#) which greatly simplifies installing and managing multiple Grails versions.

Install with SDKMAN

To install the latest version of Grails using SDKMAN, run this on your terminal:

```
$ sdk install grails
```

You can also specify a version

```
$ sdk install grails 6.2.2
```

You can find more information about SDKMAN usage on the [SDKMAN Docs](#)

Manual installation

For manual installation follow these steps:

- [Download](#) a binary distribution of Grails and extract the resulting zip file to a location of your choice
- Set the `GRAILS_HOME` environment variable to the location where you extracted the zip

Unix/Linux

- This is typically a matter of adding something like the following `export`

GRAILS_HOME=/path/to/grails to your profile

- This can be done by adding `export PATH="$PATH:$GRAILS_HOME/bin"` to your profile

Windows

- Copy the path to the bin directory inside the grails folder you have downloaded, for example,

```
C:\path_to_grails\bin
```

- Go to Environment Variables, you can typically search or run the command below, the type `env` and then Enter

Start + R

- Edit the Path variable on User Variables / System Variables depending on your choice.
- Paste the copied path in the Path Variable.

If Grails is working correctly you should now be able to type `grails --version` in the terminal window and see output similar to this:

```
Grails Version: 6.2.2
```

2.3 Creating an Application

To create a Grails application you first need to familiarize yourself with the usage of the `grails` command which is used in the following manner:

```
$ grails <<command name>>
```

Run [create-app](#) to create an application:

```
$ grails create-app myapp
```

This will create a new directory inside the current one that contains the project. Navigate to this directory in your console:

```
$ cd myapp
```

2.4 Creating a Simple Web Application with Grails

Step 1: Create a New Project

Open your command prompt or terminal.

Navigate to the directory where you want to create your Grails project:

```
$ cd your_project_directory
```

Create a new Grails project with the following command:

```
$ grails create-app myapp
```

Step 2: Access the Project Directory

Change into the "myapp" directory, which you just created:

```
$ cd myapp
```

Step 3: Start Grails Interactive Console

Start the Grails interactive console by running the "grails" command:

```
$ grails
```

Step 4: Create a Controller

In the Grails interactive console, you can use auto-completion to create a controller. Type the following command to create a controller named "greeting":

```
grails> create-controller greeting
```

This command will generate a new controller named "GreetingController.groovy" within the grails-app/controllers/myapp directory. You might wonder why there is an additional "myapp" directory. This structure aligns with conventions commonly used in Java development, where classes are organized into packages. Grails automatically includes the application name as part of the package structure. If you do not specify a package, Grails defaults to using the application name.

For more detailed information on creating controllers, you can refer to the documentation on the [create-controller](#) page.

Step 5: Edit the Controller

Open the "GreetingController.groovy" file located in the "grails-app/controllers/myapp" directory in a text editor.

Add the following code to the "GreetingController.groovy" file:

```
package myapp

class GreetingController {

    def index() {
        render "Hello, Congratulations for your first Grails application!"
    }
}
```

The action is simply a method. In this particular case, it calls a special method provided by Grails to [render](#) the page.

Step 6: Run the Application

Grails framework now relies on Gradle tasks for running the application. To start the application, use the following Gradle `bootRun` command:

```
$ ./gradlew bootRun
```

Your application will be hosted on port 8080 by default. You can access it in your web browser at:

<http://localhost:8080/>

Now, it's important to know that the welcome page is determined by the following URL mapping:

```

class UrlMappings {
  static mappings = {
    "/*controller/*action?/*id?(.$format)?"{
      constraints {
        // apply constraints here
      }
    }

    "/*"(view: "/index")
    "500"(view: '/error')
    "404"(view: '/notFound')
  }
}

```

This mapping specifies that the root URL ("/") should display the "index.gsp" view, which is located at "grails-app/views/index.gsp." This "index.gsp" file serves as your welcome or landing page. The other entries in the mapping handle error pages for HTTP status codes 500 and 404.

Grails URL Convention Based on Controller and Action Name

Grails follows a URL convention that relies on the names of controllers and their actions. This convention simplifies the creation and access of various pages or functionalities within your web application.

In the provided code example:

```

package myapp

class GreetingController {
  def index() {
    render "Hello, Congratulations for your first Grails application!"
  }
}

```

- The `GreetingController` class represents a controller in Grails.
- Inside the controller, there's an `index` action defined as a method. In Grails, actions are essentially methods within a controller that handle specific tasks or respond to user requests.

Now, let's understand how the Grails URL convention works based on this controller and action:

1. Controller Name in URL:

1. The controller name, in this case, "GreetingController," is used in the URL. However, the convention capitalizes the first letter of the controller name and removes the "Controller" suffix. So, "GreetingController" becomes "greeting" in the URL.

2. Action Name in URL:

1. By default, if you don't specify an action in the URL, Grails assumes the "index" action. So, in this example, accessing the URL `/greeting`

See the end of the [controllers and actions](#) section of the user guide to find out more on default actions.

Optional: Set a Context Path

If you want to set a context path for your application, create a configuration property in the "grails-app/conf/application.yml" file:

```

server:
  servlet:
    context-path: /myapp

```

With this configuration, the application will be available at:

<http://localhost:8080/myapp/>

Alternatively, you can set the context path from the command line when using Gradle to run a Grails application. Here's how you can do it:

```
$ ./gradlew bootRun -Dgrails.server.servlet.context-path=/your-context-path
```

Replace `/your-context-path` with the desired context path for your Grails application. This command sets the context path directly via the `-Dgrails.server.servlet.context-path` system property.

For example, if you want your application to be available at "http://localhost:8080/myapp," you can use the following command:

```
$ ./gradlew bootRun -Dgrails.server.servlet.context-path=/myapp
```

This allows you to configure the context path without modifying the application's configuration files, making it a flexible and convenient option when running your Grails application with Gradle.

Optional: Change Server Port

If port 8080 is already in use, you can start the server on a different port using the `grails.server.port` system-property:

```
$ ./gradlew bootRun -Dgrails.server.port=9090
```

Replace "9090" with your preferred port.

Note for Windows Users

If you encounter an error related to the Java process or filename length, you can use the `--stacktrace` flag or add `grails { pathingJar = true }` to your "build.gradle" file.

It may also be necessary to enclose the system properties in quotes on Windows:

```
./gradlew bootRun "-Dgrails.server.port=9090"
```

Conclusion

Your Grails application will now display a "Hello, Congratulations on your first Grails application!" message when you access it in your web browser.

Remember, you can create multiple controllers and actions to build more complex web applications with Grails. Each action corresponds to a different page accessible through unique URLs based on the controller and action names.

2.5 Using Interactive Mode

The Grails Command-line Interface (CLI) offers an interactive mode, which you can access by entering "grails" in your Terminal application or Linux Command Line.

Once you're in the command-line interface, you can enhance your efficiency by utilizing the TAB key for auto-completion. For instance:

```
grails> create
create-app          create-plugin      create-webapp
create-controller  create-restapi
create-domain-class create-web-plugin
```

This interactive mode provides a convenient way to explore available Grails commands and options, making your Grails development workflow more efficient and user-friendly.

For more information on the capabilities of interactive mode, refer to the section on [Interactive Mode](#) in the user guide.

2.6 Getting Set Up in an IDE

Because Grails is built upon the [Spring Framework](#) (Spring Boot), the [Gradle](#) build tool, and the [Groovy](#) programming language, it is possible to develop Grails application using most popular JVM Integrated Development Environments (IDEs). Some IDEs offer more specialized support for Grails, while others may offer basic support for managing dependencies/plugins, running Gradle tasks, code-completion and syntax highlighting.

1. IntelliJ IDEA

IntelliJ IDEA is a widely used IDE for Grails development. It offers comprehensive support for Groovy and Grails, including features like code-completion, intelligent code analysis, and seamless integration with Grails artefacts.

IntelliJ IDEA also provides powerful database tools that work with Grails' GORM (Grails Object Relational Mapping) seamlessly. It offers both a Community (free) and Ultimate (paid) edition, with the latter offering more advanced Grails support, including an embedded version of the [Grails Forge](#), and view resolution for both GSPs and JSON views.

[IntelliJ IDEA Website](#)

2. Visual Studio Code (VS Code)

Visual Studio Code is a lightweight, open-source code editor developed by Microsoft. While it's not a full-fledged IDE, it offers powerful extensions for Grails and Groovy development. You can install extensions like [code-groovy](#) and [Grails for VS Code](#) to enhance your Grails developer experience.

VS Code provides features such as syntax highlighting, code navigation, and integrated terminal support. It's a great choice for developers who prefer a lightweight and customizable development environment.

[Visual Studio Code \(VS Code\)](#)

3. STS (Spring Tool Suite)

The [Spring Tool Suite](#) (STS) is set of IDE tools designed for Spring Framework development, with versions based on both VS Code and Eclipse. This section focuses on the Eclipse version.

STS can work as an effective Grails developer platform when used with the [Groovy Development Tools](#) plugin (which can be installed using the Eclipse Marketplace). STS does not offer specific support for Grails artefacts or GSP views.

[Spring Tool Suite \(STS\)](#)

4. Netbeans

Apache [Netbeans](#) does not offer specific support for Grails, but it will import Grails applications as Gradle projects and provides reasonable editing support for Groovy and GSP views.

5. TextMate, VIM, and More

There are several excellent text editors that work nicely with Groovy and Grails. Here are some references:

- A [bundle](#) is available for Groovy / Grails support in [Textmate](#).
- A [plugin](#) can be installed via Sublime Package Control for the [Sublime Text Editor](#).
- The [emacs-grails](#) extension offers basic support for Grails development in Emacs.
- See [this post](#) for some helpful tips on how to set up VIM as your Grails editor of choice.

These text editors, along with the provided extensions and configurations, can enhance your Groovy and Grails development experience, offering flexibility and customization to meet your coding preferences.

2.7 Grails Directory Structure and Convention over Configuration

Grails adopts the "convention over configuration" approach to configure itself. In this approach, the name and location of files are used instead of explicit configuration. Therefore, it's essential to become familiar with the directory structure provided by Grails. Here's a breakdown of the key directories and links to relevant sections:

1. grails-app - Top-Level Directory for Groovy Sources

1. `conf` - [Configuration Sources](#)
2. `controllers` - [Web Controllers](#) - Responsible for the "C" in MVC (Model-View-Controller).
3. `domain` - [Application Domain](#) - Represents the "M" in MVC.
4. `i18n` - Supports [Internationalization \(i18n\)](#).
5. `services` - The [Service Layer](#).
6. `taglib` - [Tag Libraries](#).
7. `utils` - Houses Grails-specific utilities.
8. `views` - [Groovy Server Pages \(GSP\)](#) or [JSON Views](#) - Responsible for the "V" in MVC.
9. `commands` - [Custom Grails Commands](#) - Create your own Grails CLI commands.

2. `src/main/groovy` - Supporting Sources

3. `src/test/groovy` - [Unit Tests](#)

4. `src/integration-test/groovy` - [Integration Tests](#) - For testing Grails applications at the integration level.

Understanding this directory structure and its conventions is fundamental to efficient Grails development.

2.8 Running and Debugging an Application

Grails applications can be executed using the built-in application server using the `bootRun` command. By default, it launches a server on port 8080:

```
$ ./gradlew bootRun
```

To specify a different port, you can set the system property `-Dgrails.server.port` as follows:

```
$ ./gradlew bootRun -Dgrails.server.port=8081
```

For debugging a Grails app, you have two options. You can either right-click on the `Application.groovy` class in your IDE and select the appropriate debugging action, or you can run the app with the following command and then connect a remote debugger to it:

```
$ ./gradlew bootRun --debug-jvm
```

For more information on the `bootRun` command, please refer to the [bootRun section of the Grails reference guide](#).

2.9 Testing an Application

Grails offers a convenient feature where you can automatically generate unit and integration tests for your application using the `create-*` commands. These generated tests are stored in the `src/test/groovy` and `src/integration-test/groovy` directories. However, it is your responsibility to populate these tests with the appropriate test logic. You can find comprehensive guidance on crafting valid test logic in the section dedicated to [Unit and Integration Tests](#).

To initiate the execution of your tests, including both unit and integration tests, you can utilize the Gradle `check` task. Follow these steps:

1. Open your terminal or command prompt and navigate to your Grails project's root directory.
2. Execute the following Gradle command:

```
$ ./gradlew check
```

By running the `check` task, you ensure that all tests in your Grails project, including the ones you've created and populated with test logic, are executed. This comprehensive testing approach contributes significantly to the robustness and overall quality of your application.

3. **Viewing Test Reports:** After running your tests, Grails generates test reports that provide valuable insights into the test results. You can typically find these reports in the `build/reports/tests` directory of your Grails project. Open these reports in a web browser to view detailed information about test outcomes, including passed, failed, and skipped tests.

Remember, testing is not just a process; it's a fundamental practice that enhances your Grails application's reliability. Viewing test reports helps you analyze and understand the

test results, making it easier to identify and address any issues.

By following these testing practices and reviewing test reports, you can deliver a high-quality Grails application to your users with confidence.

2.10 Deploying an Application

Grails applications offer multiple deployment options.

For traditional container deployments, such as Tomcat or Jetty, you can generate a Web Application Archive (WAR) file using the Gradle `war` task as follows:

```
$ ./gradlew war
```

This task generates a WAR file with a `-plain` suffix within the `build/libs` directory, ready for deployment according to your container's guidelines.

By default, the `war` task runs in the `production` environment. You can specify a different environment, such as `development`, by overriding it in the Gradle command:

```
$ ./gradlew -Dgrails.env=dev war
```

If you prefer not to use a separate Servlet container, you can create and run the Grails WAR file as a regular Java application:

```
$ ./gradlew bootWar
$ java -jar build/libs/mywar-0.1.war
```

When deploying Grails, ensure that your container's JVM runs with the `-server` option and sufficient memory allocation. Here are recommended VM flags:

```
-server -Xmx1024M
```

2.11 Supported Java EE Containers

The Grails framework requires that runtime containers support Servlet 3.0 and above. By default, Grails framework applications are bundled with an embedded Tomcat server. For more information, please see the [Deployment](#) section of this documentation.

In addition, read the [Grails Guides](#) for tips on how to deploy Grails to various popular Cloud services.

2.12 Creating Artefacts

Grails provides a set of useful CLI commands for various tasks, including the creation of essential artifacts such as controllers and domain classes. These commands simplify the development process, although you can achieve similar results using your preferred Integrated Development Environment (IDE) or text editor.

For instance, to create the foundation of an application, you typically need to generate a domain model using Grails Commands:

```
$ grails create-app myapp
$ cd myapp
$ grails create-domain-class book
```

Executing these commands will result in the creation of a domain class located at `grails-app/domain/myapp/Book.groovy`, as shown in the following code:

```
package myapp

class Book {
}
```

The Grails CLI offers numerous other commands that you can explore in the Grails command line reference guide.

Using interactive mode enhances the development experience by providing auto-complete and making the process smoother.

2.13 Generating an Application

Quick Start with Grails Scaffolding

To quickly initiate your Grails project, you can employ the `runCommand` Gradle task. This task allows you to generate the essential structure of an application swiftly. Specifically, when running the following command, you can create a [controller](#) (including its unit tests) and the associated [views](#) for your application:

```
$ ./gradlew runCommand -Pargs="generate-all myapp.Book"
```

3 Upgrading from the previous versions

3.1 Upgrading from Grails 5 to Grails 6

Upgrade Instructions for Grails and Related Dependencies

To ensure compatibility with Grails 6, you must update the following versions in your project:

1. Java 11 as Baseline:

Starting from Grails 6, Java 11 serves as the baseline requirement for the framework. When upgrading to Grails 6, ensure that your project is configured to use Java 11. This compatibility with Java 11 allows you to take advantage of the latest features, security enhancements, and performance improvements provided by Java 11.

Please make sure to update your project's Java version to 11 before proceeding with the Grails 6 upgrade. Doing so will ensure a seamless transition to the latest version of Grails and enable you to enjoy all the benefits that Java 11 has to offer.

2. The New Grails CLI:

Grails 6 comes with a completely revamped and highly efficient Command Line Interface (CLI) that enables you to generate applications and plugins at a remarkable speed. For instance, you can now use the new CLI to create a new Grails 6 application with the following command:

```
grails create-app my-app
```

The new CLI also allows you to generate plugins easily. For example, to create a new plugin named "my-plugin," you can use the following command:


```
grails create-plugin my-plugin
```

One notable improvement in Grails 6 is that it no longer supports certain commands that performed redundant tasks, such as the outdated `grails run-app` command. Instead, it recommends using the Gradle `bootRun` task for running your application, which offers better performance and functionality.

For example, to run your Grails 6 application, you can use the following command:

```
./gradlew bootRun
```

As a result of these improvements, the new CLI provides a more streamlined and efficient way to work with Grails applications and plugins.

Overall, Grails 6 offers a significantly improved development experience with its new CLI, optimized commands, and advanced features for generating applications and plugins.

3. Setting Grails Version and Grails Gradle Plugin:

To upgrade to Grails 6, it's important to configure the appropriate versions in the `gradle.properties` file as shown below:

```
gradle.properties
grailsVersion=6.0.0
grailsGradlePluginVersion=6.0.0
```

By specifying the above versions, you'll gain access to the latest features, improvements, and bug fixes introduced in Grails 6. Upgrading to this version empowers your application with enhanced performance and improved security. Additionally, it allows you to leverage the latest advancements in the Grails framework for a more efficient and secure development experience.

4. GORM Version:

If your project utilizes GORM, ensure to update the version in the `gradle.properties` file as demonstrated below:

```
gradle.properties
gormVersion=8.0.0
```

By upgrading to GORM 8.0.0, you will benefit from essential updates and optimizations. This upgrade guarantees seamless interactions with your database and enhances your data management experience. Staying current with GORM allows you to take advantage of the latest database features and improvements, thereby optimizing the performance and functionality of your application.

5. Gradle Version:

Grails 6 uses Gradle 7.6.2 which offers performance improvements, bug fixes, and new features over previous versions. Upgrading to the latest Gradle version helps accelerate your build processes and ensures compatibility with other dependencies.

5.1. Upgrade to Gradle 7.6.2

Run the following command to update the Gradle wrapper to the desired version (e.g., Gradle 7.6.2):

```
./gradlew wrapper --gradle-version 7.6.2
```

This command will download the specified Gradle version and update the Gradle wrapper settings in your project.

5.2. Check Gradle Version:

After the command finishes, you can verify that the Gradle version has been updated by checking the `gradle-wrapper.properties` file located in the `gradle/wrapper` directory. The `distributionUrl` in the file should now point to the Gradle 7.6.2 distribution:

```
distributionUrl=https\://services.gradle.org/distributions/gradle-7.6.2-bin.zip
```

5.3. Build the Project:

After updating the Gradle wrapper, you can now build your Grails project using the updated Gradle version:

```
./gradlew build
```

This will initiate the build process with the new Gradle version.

6. Embracing Modern Plugin Management with Grails 6

In Gradle, there are two main ways to add plugins to your project: the `plugins` block and the `apply plugin` statement.

Grails 6 introduces a significant change in how plugins are managed by adopting the Gradle `plugins` block instead of the traditional `apply plugin` statements. This shift streamlines the project's build configuration and brings it more in line with modern Gradle conventions. New Grails projects will now utilize the `plugins` block to manage plugin dependencies and configurations.

Using the `plugins` Block in Grails 6:

With the new approach, adding plugins to a Grails 6 project is more explicit and organized. In your `build.gradle` file, you can declare plugins within the `plugins` block, specifying the plugin's ID and version.

Here's an example of adding the `views-json` plugin using the `plugins` block:

```
build.gradle

plugins {
    id 'org.grails.plugins.views-json' version '3.0.0'
}
```

Managing Multiple Plugins:

The `plugins` block allows you to add multiple plugins, each on its own line. This enhances clarity and makes it easier to manage plugin dependencies.

```
build.gradle

plugins {
    id 'org.grails.plugins.views-json' version '3.0.0'
    // Add other plugins as needed
}
```

Moving Older Applications to the New Approach:

If you are migrating an older Grails application to Grails 6, you can update the plugin declarations from `apply plugin` to the `plugins` block. For example, if your previous application used the `views-json` plugin, you can modify the `build.gradle` as follows:

Before (Using `apply plugin`):

```
build.gradle

apply plugin: 'org.grails.plugins.views-json'
```

After (Using `plugins` Block in Grails 6):

```
build.gradle

plugins {
    id 'org.grails.plugins.views-json' version '3.0.0'
}
```

By migrating to the `plugins` block, your Grails 6 project will adhere to modern Gradle conventions, making it easier to manage plugin dependencies and configurations. This new approach maintains consistency and enhances the overall structure of the project, ensuring a smoother and more efficient development process.

7. GORM for MongoDB Sync Driver:

The GORM for MongoDB is updated to support the latest `mongodb-driver-sync`. If you are using GORM for MongoDB and making use of specific MongoDB Driver or low-level Mongo API features, consider checking the [Upgrading to the 4.0 Driver guide](#).

This update ensures seamless integration with MongoDB, access to new features, and improved performance while interacting with your MongoDB database.

8. Asset Pipeline Plugin:

In Grails 6, there is an update to the Asset Pipeline Plugin, which is now version 4.3.0. The Asset Pipeline Plugin is a crucial component in Grails applications, responsible for managing frontend assets like stylesheets, JavaScript files, and images. The update to version 4.3.0 brings several improvements and new features to enhance the management and processing of frontend assets in your Grails projects.

The asset-pipeline plugin 4.3.0 offers new features for managing and processing your frontend assets, ensuring they are efficiently bundled and served to your users.

9. Spring 5.3:

Grails 6 is built on Spring 5.3.27. If your project uses Spring-specific features, refer to the [Upgrading to Spring 5.3 guide](#).

Spring 5.3 introduces enhancements and fixes to the Spring framework, providing you with the latest improvements in dependency injection, web frameworks, and other Spring-related functionalities.

10. Spring Boot 2.7:

Grails 6 updates to Spring Boot 2.7. For more information, consult the [Spring Boot 2.7 Release Notes](#)

Spring Boot 2.7 comes with new features, performance enhancements, and compatibility improvements, making it a solid foundation for your Grails application.

11. Micronaut 3.9.3:

Grails 6 is shipped with Micronaut 3.9.3. If you are using specific Micronaut features, refer to the [Upgrading to Micronaut 3.x guide](#).

Micronaut 3.9.3 brings new capabilities, improvements, and bug fixes, empowering your application with a powerful and lightweight microservices framework.

12. Micronaut for Spring 4.5.1:

Grails 6 is updated to use Micronaut for Spring 4.5.1. For more information, check out the [release notes](#).

Micronaut for Spring 4.5.1 provides seamless integration between Micronaut and Spring, allowing you to leverage the strengths of both frameworks in your Grails project.

3.2 Upgrading from Grails 4 to Grails 5

Bump up Grails Version

You will need to upgrade your Grails version defined in `gradle.properties` as:

```
gradle.properties
...
grailsVersion=5.2.0
...
```

Apache Groovy 3.0.7

Grails 5.1.1 provide support for Groovy 3. We would recommend you to please check the [Release notes for Groovy 3](#) to update your application in case you are using a specific feature which might not work in Groovy 3.

Define `groovyVersion` in `gradle.properties` to force the application to use Groovy 3.

Grails 5.1 app's `gradle.properties`

```
gradle.properties
...
groovyVersion=3.0.7
...
```

Bump up GORM Version

If you were using [GORM](#), you will need to update the version defined in `gradle.properties` as:

```
gradle.properties
...
gormVersion=7.2.0
...
```

Bump up gradle version

Grails 5.2.x uses gradle 7.2

```
gradle-wrapper.properties
...
distributionUrl=https\://services.gradle.org/distributions/gradle-7.2-bin.zip
```

...

Also you can run this command

```
./gradlew wrapper --gradle-version 7.2
```

GORM for MonogDB Sync Driver

The GORM for MongoDB is updated to support latest mongodb-driver-sync. If you are using GORM for MongoDB and doing something specific to MongoDB Driver or low level Mongo API then you might want to take a look at [Upgrading to the 4.0 Driver](#)

Bump up Asset Pipeline plugin version

The previous version of asset-pipeline is not supported with Grails 5.0 as it is compiled with a version of Groovy which is binary incompatible with Groovy 3. So, please update the plugin version to 3.2.4.

Disabled StringCharArrayAccessor by default

The previous version of Grails use the `StringCharArrayAccessor` which is enabled by default and provides optimized access to `java.lang.String` internals. In Grails 5.0 it is disabled by default but you can enable it by setting a system property with name `stringchararrayaccessor.disabled` and value `false`.

Enabling `StringCharArrayAccessor` would show `IllegalReflectiveAccess` warnings as it uses reflection to do the optimizations.

Changes in profile.yml and feature.yml files in Grails Profiles

The format of how dependencies are defined in features and profiles has been changed. See the section on [Application Profiles](#) for more information.

Deprecation of dot navigation of Grails configuration

In order to reduce complexity, improve performance, and increase maintainability, accessing configuration through dot notation (`config.a.b.c`) has been deprecated. This functionality will be removed in a future release.

Also, you would see a warning message if you are accessing configuration through the dot notation.

The recommended way to access configuration is:

```
grailsApplication.config.getProperty("hola", String.class)
```

Deprecated Classes

- [org.grails.config.NavigableMap](#)
- [org.grails.config.NavigableMapConfig](#)
- [org.grails.config.NavigableMapPropertySource](#)

Spring 5.3

Grails 5.0.0.RC1 is built on Spring 5.3.2 See the [Upgrading to Spring 5.3](#) if you are using

Spring specific features.

Spring Boot 2.4

Grails 5.1.1 updates to Spring Boot 2.6. Please check [Spring Boot 2.6 Release Notes](#) for more information.

Micronaut 3.2.0

Grails 5.1.1 is shipped with Micronaut 3.2.0. Please check the [Upgrading to Micronaut 3.x](#) if you are using a specific feature.

Micronaut for Spring 4.0.1

Grails 5.1.1 is updated to Micronaut for Spring 4.0.1, please check out [release notes](#) for more information.

Gradle 7.x

Compile dependency configuration as well as others have been removed from Gradle 7.x. In previous version they were deprecated.

Replace configurations:

build.gradle

```
...
compile -> implementation
testCompile -> testImplementation
runtime -> runtimeOnly
...
```

More information in Gradle upgrade docs [Gradle upgrade docs](#)

Plugins in multi-project setup

If you have grails plugins as part of multi-project builds you should also replace the `compile` with `implementation` configuration.

Additionally if your main application relied on the dependencies declared by the plugin you need to apply further changes.

To make the dependencies available again you have to declare them with `api` configuration. You also have to apply the `java-library` gradle plugin in your plugin project.

More information [gradle java-library-plugin](#)

3.3 Upgrading from Grails 3.3.x to Grails 4

Bump up Grails Version

You will need to upgrade your Grails version defined in `gradle.properties`.

Grails 3 app's `gradle.properties`

`gradle.properties`

```
...
grailsVersion=3.3.8
```

...

Grails 4 app's `gradle.properties`

`gradle.properties`

```
...
grailsVersion=4.0.4
...
```

Bump up GORM Version

If you were using [GORM](#), you will need to update the version defined in `gradle.properties`.

Grails 3 app's `gradle.properties`

`gradle.properties`

```
...
gormVersion=6.1.10.RELEASE
...
```

Grails 4 app's `gradle.properties`

`gradle.properties`

```
...
gormVersion=7.0.4
...
```

Move GORM DSL Entries to `runtime.groovy`

GORM DSL entries should be move to `runtime.groovy`. For instance, using following GORM configuration in the `application.groovy` is not supported and will break the application:

```
grails.gorm.default.mapping = {
    id generator: 'identity'
}
```

Spring 5 and Spring Boot 2.1

Grails 4.0 is built on Spring 5 and Spring Boot 2.1. See the [migration guide](#) and [release notes](#) if you are using Spring specific features.

Hibernate 5.4 and GORM 7.x

Grails 4.x supports a minimum version of Hibernate 5.4 and GORM 7.x. Several changes have been made to GORM to support the newer version of Hibernate and simplify GORM itself.

The details of these changes are covered in the [GORM upgrade documentation](#).

Spring Boot 2.1 Actuator

Please check the [Spring Boot Actuator](#) documentation since it has changed substantially from Spring Boot 1.5 the version Grails 3.x used.

If you had configuration such as:

`grails-app/conf/application.yml` - Grails 3.3.x

```
endpoints:
  enabled: false
```

```
jmx:
  enabled: true
  unique-names: true
```

replace it with:

grails-app/conf/application.yml - Grails 4.x

```
spring:
  jmx:
    unique-names: true
management:
  endpoints:
    enabled-by-default: false
```

Spring Boot Developer Tools and Spring Loaded

Previous versions of Grails used a reloading agent called [Spring Loaded](#). Since this library is no longer maintained and does not support Java 11 support for Spring Loaded has been removed.

As a replacement, Grails 4 applications include [Spring Boot Developer Tools](#) dependencies in the `build.gradle` build script. If you are migrating a Grails 3.x app, please include the following set of dependencies:

`build.gradle`

```
.
..
...
configurations {
  developmentOnly
  runtimeClasspath {
    extendsFrom developmentOnly
  }
}

dependencies {
  developmentOnly("org.springframework.boot:spring-boot-devtools")
  ...
  ..
}
...
..
.
```

Also you should configure the necessary excludes for Spring Developer Tools in `application.yml`:

```
spring:
  devtools:
    restart:
      exclude:
        - grails-app/views/**
        - grails-app/i18n/**
        - grails-app/conf/**
```

The above configuration prevents the server from restarting when views or message bundles are changed.

You can use Spring Developer Tools in combination with a browser extension such as the [Chrome LiveReload extension](#) to get automatic browser refresh when you change anything in your Grails application.

Spring Boot Gradle Plugin Changes

Grails 4 is built on top of Spring Boot 2.1. Grails 3 apps were built on top of Spring Boot 1.x.

Your Grails 3 app's `build.gradle` may have such configuration:

build.gradle

```
bootRun {
    addResources = true
    ...
}
```

Grails 4 apps are built on top of Spring Boot 2.1. Starting from Spring Boot 2.0, the `addResources` property no longer exists. Instead, you need to set the `sourceResources` property to the source set that you want to use. Typically that's `sourceSets.main`. This is described in the [Spring Boot Gradle plugin's documentation](#).

Your Grails 4 app's `build.gradle` can be configured:

build.gradle

```
bootRun {
    sourceResources sourceSets.main
    ...
}
```

Building executable jars for Grails Plugins

[Spring Boot's new Gradle Plugin](#):

The `bootRepackage` task has been replaced with `bootJar` and `bootWar` tasks for building executable jars and wars respectively. Both tasks extend their equivalent standard Gradle `jar` or `war` task, giving you access to all of the usual configuration options and behaviour.

If you had configuration such as:

build.gradle | Grails 3

```
// enable if you wish to package this plugin as a standalone application
bootRepackage.enabled = false
```

replace it with:

build.gradle | Grails 4

```
// enable if you wish to package this plugin as a standalone application
bootJar.enabled = false
```

Upgrading to Gradle 5

Grails 3 apps by default used [Gradle](#) 3.5. Grails 4 apps use Gradle 5.

To upgrade to Gradle 5 execute:

```
./gradlew wrapper --gradle-version 5.0
```

Due to changes in Gradle 5, [transitive dependencies are no longer resolved](#) for plugins. If your project makes use of a plugin that has transitive dependencies, you will need to add those explicitly to your `build.gradle` file.

If you customized your app's build, other migrations may be necessary. Please check [Gradle Upgrading your build](#) documentation. Especially notice, that default Gradle daemon now starts with 512MB of heap instead of 1GB. Please check [Default memory settings changed](#) documentation.

Groovy language update to 2.5.6

Keep in mind, that with grails 4.0.x there is a minor groovy language upgrade (e.g. 3.3.9).

used groovy 2.4.x), which requires a couple of changes, that are immediately obvious when trying to compile your source code. However there are also issues with changed implementations of core linkedlist functions! Check an overview of the breaking changes here: [Breaking changes of Groovy 2.5](#)

Removed date helper functions

Most common issue is that date util functions have been moved to individual project, e.g new Date().format("ddMMyyyy") no longer works without adding:

build.gradle

```
dependencies {  
    implementation "org.codehaus.groovy:groovy-dateutil:3.0.4"  
}
```

Changed linked list method implementations

Check whether you are using the groovy version of linkedlist implementations:

- [].pop() - will no longer remove the last, but the first element of the list. Replace it with [].removeLast() is recommended.
- [].push(..) - will no longer add to the end, but to the beginning of the list. Replace it with [].add(..) is recommended.

H2 Web Console

Spring Boot 2.1 includes native support for the H2 database web console. Since this is already included in Spring Boot the equivalent feature has been removed from Grails. The H2 console is therefore now available at /h2-console instead of the previous URI of /dbconsole. See [Using H2's Web Console](#) in the Spring Boot documentation for more information.

Upgrade Hibernate

If you were using GORM for Hibernate implementation in your Grails 3 app, you will need to upgrade to Hibernate 5.4.

A Grails 3 build.gradle such as:

build.gradle

```
dependencies {  
    ...  
    implementation "org.grails.plugins:hibernate5"  
    implementation "org.hibernate:hibernate-core:5.1.5.Final"  
}
```

will be in Grails 4:

build.gradle

```
dependencies {  
    ...  
    implementation "org.grails.plugins:hibernate5"  
    implementation "org.hibernate:hibernate-core:5.4.0.Final"  
}
```

Migrating to Geb 2.3

Geb 1.1.x (a JDK 1.7 compatible version) was the version shipped by default with Grails 3. Grails 4 is no longer compatible with Java 1.7. You should migrate to Geb 2.3.

In Grails 3, if your build.gradle looks like:

build.gradle

```
dependencies {
    testCompile "org.grails.plugins:geb:1.1.2"
    testRuntime "org.seleniumhq.selenium:selenium-htmlunit-driver:2.47.1"
    testRuntime "net.sourceforge.htmlunit:htmlunit:2.18"
}
```

In Grails 4, you should replace it with:

build.gradle

```
buildscript {
    repositories {
        ...
    }
    dependencies {
        ...
        classpath "gradle.plugin.com.energizedwork.webdriver-binaries:webdriver-binaries-gradle-plugin:$webdriverBi
    }
}
...
..

repositories {
    ...
}

apply plugin:"idea"
...
...
apply plugin:"com.energizedwork.webdriver-binaries" (1)

dependencies {
    ...
    testCompile "org.grails.plugins:geb" (4)
    testRuntime "org.seleniumhq.selenium:selenium-chrome-driver:$seleniumVersion" (5)
    testRuntime "org.seleniumhq.selenium:selenium-firefox-driver:$seleniumVersion" (5)
    testRuntime "org.seleniumhq.selenium:selenium-safari-driver:$seleniumSafariDriverVersion" (5)

    testCompile "org.seleniumhq.selenium:selenium-remote-driver:$seleniumVersion" (5)
    testCompile "org.seleniumhq.selenium:selenium-api:$seleniumVersion" (5)
    testCompile "org.seleniumhq.selenium:selenium-support:$seleniumVersion" (5)
}

webdriverBinaries {
    chromedriver "$chromedriverVersion" (2)
    geckodriver "$geckodriverVersion" (3)
}

tasks.withType(Test) {
    systemProperty "geb.env", System.getProperty('geb.env')
    systemProperty "geb.build.reportsDir", reporting.file("geb/integrationTest")
    systemProperty "webdriver.chrome.driver", System.getProperty('webdriver.chrome.driver')
    systemProperty "webdriver.gecko.driver", System.getProperty('webdriver.gecko.driver')
}
```

gradle.properties

```
gebVersion=2.3
seleniumVersion=3.12.0
webdriverBinariesVersion=1.4
hibernateCoreVersion=5.1.5.Final
chromedriverVersion=2.44 (2)
geckodriverVersion=0.23.0 (3)
seleniumSafariDriverVersion=3.14.0
```

- 1 Includes [Webdriver binaries Gradle plugin](#).
- 2 Set the appropriate [Webdriver for Chrome version](#).
- 3 Set the appropriate [Webdriver for Firefox version](#).
- 4 Includes the [Grails Geb Plugin dependency](#) which has a transitive dependency to `geb-spock`. This is the dependency necessary to work with [Geb](#) and Spock.
- 5 Selenium and different driver dependencies.

Create also a [Geb Configuration](#) file at `src/integration-test/resources/GebConfig.groovy`.

`src/integration-test/resources/GebConfig.groovy`

```

import org.openqa.selenium.chrome.ChromeDriver
import org.openqa.selenium.chrome.ChromeOptions
import org.openqa.selenium.firefox.FirefoxDriver
import org.openqa.selenium.firefox.FirefoxOptions
import org.openqa.selenium.safari.SafariDriver

environments {

    // You need to configure in Safari -> Develop -> Allowed Remote Automation
    safari {
        driver = { new SafariDriver() }
    }

    // run via "./gradlew -Dgeb.env=chrome iT"
    chrome {
        driver = { new ChromeDriver() }
    }

    // run via "./gradlew -Dgeb.env=chromeHeadless iT"
    chromeHeadless {
        driver = {
            ChromeOptions o = new ChromeOptions()
            o.addArguments('headless')
            new ChromeDriver(o)
        }
    }

    // run via "./gradlew -Dgeb.env=firefoxHeadless iT"
    firefoxHeadless {
        driver = {
            FirefoxOptions o = new FirefoxOptions()
            o.addArguments('-headless')
            new FirefoxDriver(o)
        }
    }

    // run via "./gradlew -Dgeb.env=firefox iT"
    firefox {
        driver = { new FirefoxDriver() }
    }
}

```

Deprecated classes

The following classes, which were deprecated in Grails 3.x, have been removed in Grails 4. Please, check the list below to find a suitable replacement:

Removed Class	Alternative
org.grails.datastore.gorm.validation.constraints.UniqueConstraint	org.grails.datastore.gorm.validation.constraints.
grails.util.BuildScope	
grails.transaction.GrailsTransactionTemplate	grails.gorm.transactions.GrailsTransactionTemplat
org.grails.transaction.transform.RollbackTransform	org.grails.datastore.gorm.transactions.transform.
grails.transaction.NotTransactional	grails.gorm.transactions.NotTransactional
grails.transaction.Rollback	grails.gorm.transactions.Rollback
grails.transaction.Transactional	grails.gorm.transactions.Transactional

`org.grails.config.FlatConfig`

`org.grails.core.metaclass.MetaClassEnhancer`

`org.grails.core.util.ClassPropertyFetcher`

`org.grails.transaction.transform.TransactionalTransform`

`grails.core.ComponentCapableDomainClass`

`grails.core.GrailsDomainClassProperty`

`org.grails.core.DefaultGrailsDomainClassProperty`

`org.grails.core.MetaGrailsDomainClassProperty`

`org.grails.core.support.GrailsDomainConfigurationUtil`

`org.grails.plugins.domain.DomainClassPluginSupport`

`org.grails.plugins.domain.support.GormApiSupport`

`org.grails.plugins.domain.support.GrailsDomainClassCleaner`

`grails.validation.AbstractConstraint`

`grails.validation.AbstractVetoingConstraint`

`grails.validation.CascadingValidator`

`grails.validation.ConstrainedProperty`

`grails.validation.Constraint`

Use traits instead.

`org.grails.datastore.mapping.reflect.ClassPropert`

`org.grails.datastore.gorm.transactions.transform.`

Use the `org.grails.datastore.mapping.model.Mappi`

Use the `org.grails.datastore.mapping.model.Mappi`
`org.grails.datastore.mapping.model.MappingFactory`

Handled by `org.grails.datastore.mapping.model.M`

Use `org.grails.datastore.gorm.validation.constra`

`org.grails.datastore.gorm.validation.constraints`

`grails.gorm.validation.CascadingValidator`

`grails.gorm.validation.ConstrainedProperty`

`grails.gorm.validation.Constraint`

<code>grails.validation.ConstraintFactory</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>grails.validation.VetoingConstraint</code>	<code>grails.gorm.validation.VetoingConstraint</code>
<code>grails.validation.ConstraintException</code>	
<code>org.grails.validation.BlankConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.ConstrainedPropertyBuilder</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.ConstraintDelegate</code>	
<code>org.grails.validation.ConstraintsEvaluatorFactoryBean</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.CreditCardConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.DefaultConstraintEvaluator</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.DomainClassPropertyComparator</code>	
<code>org.grails.validation.EmailConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.GrailsDomainClassValidator</code>	<code>grails.gorm.validation.PersistentEntityValidator</code>
<code>org.grails.validation.InListConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.MatchesConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.MaxConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.MaxSizeConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.MinConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.MinSizeConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>

<code>org.grails.validation.NotEqualConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.NullableConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.RangeConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.ScaleConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.SizeConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.UrlConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.ValidatorConstraint</code>	<code>org.grails.datastore.gorm.validation.constraints.</code>
<code>org.grails.validation.routines.DomainValidator</code>	Replaced by newer version of commons-valid
<code>org.grails.validation.routines.InetAddressValidator</code>	Replaced by newer version of commons-valid
<code>org.grails.validation.routines.RegexValidator</code>	Replaced by newer version of commons-valid
<code>org.grails.validation.routines.ResultPair</code>	Replaced by newer version of commons-valid
<code>org.grails.validation.routines.UrlValidator</code>	Replaced by newer version of commons-valid
<code>grails.web.JSONBuilder</code>	<code>groovy.json.StreamingJsonBuilder</code>

Grails-Java8

For those who have added a dependency on the `grails-java8` plugin, all you should need to do is simply remove the dependency. All of the classes in the plugin have been moved out to their respective projects.

Profiles Deprecation

A few of the profiles supported in Grails 3.x will no longer be maintained going forward and as a result it is no longer possible to create applications when them in the shorthand form. When upgrading existing projects, it will be necessary to supply the version for these profiles.

- `org.grails.profiles:angularjs org.grails.profiles:angularjs:1.1.2`

- `org.grails.profiles:webpack org.grails.profiles:webpack:1.1.6`
- `org.grails.profiles:react-webpack org.grails.profiles:react-webpack:1.0.8`

Scheduled Methods

In Grails 3 no configuration or additional changes were necessary to use the Spring `@Scheduled` annotation. In Grails 4 you must apply the `@EnableScheduling` annotation to your application class in order for scheduling to work.

4 Configuration

It may seem odd that in a framework that embraces "convention-over-configuration" that we tackle this topic now. With Grails' default settings you can actually develop an application without doing any configuration whatsoever, as the quick start demonstrates, but it's important to learn where and how to override the conventions when you need to. Later sections of the user guide will mention what configuration settings you can use, but not how to set them. The assumption is that you have at least read the first section of this chapter!

4.1 Basic Configuration

Configuration in Grails is generally split across 2 areas: build configuration and runtime configuration.

Build configuration is generally done via Gradle and the `build.gradle` file. Runtime configuration is by default specified in YAML in the `grails-app/conf/application.yml` file.

If you prefer to use Grails 2.0-style Groovy configuration then it is possible to specify configuration using Groovy's [ConfigSlurper](#) syntax. Two Groovy configuration files are available: `grails-app/conf/application.groovy` and `grails-app/conf/runtime.groovy`:

1. Use `application.groovy` for configuration that doesn't depend on application classes
2. Use `runtime.groovy` for configuration that does depend on application classes

This separation is necessary because configuration values defined in `application.groovy` are available to the Grails CLI, which needs to be able to load `application.groovy` before the application has been compiled. References to application classes in `application.groovy` will cause an exception when these commands are executed by the CLI:

```
Error occurred running Grails CLI:
startup failed:script14738267015581837265078.groovy: 13: unable to resolve class com.foo.Bar
```

For Groovy configuration the following variables are available to the configuration script:

Variable	Description
<code>userHome</code>	Location of the home directory for the account that is running the Grails application.

grailsHome	Location of the directory where you installed Grails. If the <code>GRAILS_HOME</code> environment variable is set, it is used.
appName	The application name as it appears in <code>build.gradle</code> .
appVersion	The application version as it appears in <code>build.gradle</code> .

For example:

```
my.tmp.dir = "${userHome}/.grails/tmp"
```

Accessing Configuration with GrailsApplication

If you want to read runtime configuration settings, i.e. those defined in `application.yml`, use the [grailsApplication](#) object, which is available as a variable in controllers and tag libraries:

```
class MyController {
    def hello() {
        def recipient = grailsApplication.config.getProperty('foo.bar.hello')

        render "Hello ${recipient}"
    }
}
```

The `config` property of the `grailsApplication` object is an instance of the [Config](#) interface and provides a number of useful methods to read the configuration of the application.

In particular, the `getProperty` method (seen above) is useful for efficiently retrieving configuration properties, while specifying the property type (the default type is `String`) and/or providing a default fallback value.

```
class MyController {
    def hello(Recipient recipient) {
        //Retrieve Integer property 'foo.bar.max.hellos', otherwise use value of 5
        def max = grailsApplication.config.getProperty('foo.bar.max.hellos', Integer, 5)

        //Retrieve property 'foo.bar.greeting' without specifying type (default is String), otherwise use value "Hello"
        def greeting = grailsApplication.config.getProperty('foo.bar.greeting', "Hello")

        def message = (recipient.receivedHelloCount >= max) ?
            "Sorry, you've been greeted the max number of times" : "${greeting}, ${recipient}"
    }

    render message
}
```

Notice that the `Config` instance is a merged configuration based on Spring's [PropertySource](#) concept and reads configuration from the environment, system properties and the local application configuration merging them into a single object.

`GrailsApplication` can be easily injected into services and other Grails artifacts:

```
import grails.core.*

class MyService {
    GrailsApplication grailsApplication

    String greeting() {
        def recipient = grailsApplication.config.getProperty('foo.bar.hello')
```

```

    }
    return "Hello ${recipient}"
}

```

GrailsConfigurationAware Interface

Accessing configuration dynamically at runtime can have a small effect on application performance. An alternative approach is to implement the [GrailsConfigurationAware](#) interface, which provides a `setConfiguration` method that accepts the application configuration as a parameter when the class is initialized. You can then assign relevant configuration properties to instance properties on the class for later usage.

The `Config` instance has the same properties and usage as the injected `GrailsApplication` `Config` object. Here is the service class from the previous example, using `GrailsConfigurationAware` instead of injecting `GrailsApplication`:

```

import grails.core.support.GrailsConfigurationAware

class MyService implements GrailsConfigurationAware {
    String recipient

    String greeting() {
        return "Hello ${recipient}"
    }

    void setConfiguration(Config config) {
        recipient = config.getProperty('foo.bar.hello')
    }
}

```

Spring Value Annotation

You can use Spring's [Value](#) annotation to inject configuration values:

```

import org.springframework.beans.factory.annotation.*

class MyController {
    @Value('${foo.bar.hello}')
    String recipient

    def hello() {
        render "Hello ${recipient}"
    }
}

```

In Groovy code you must use single quotes around the string for the value of the `value` annotation otherwise it is interpreted as a `GString` not a Spring expression.

As you can see, when accessing configuration settings you use the same dot notation as when you define them.

4.1.1 Options for the YAML format Config

The `application.yml` file was introduced in Grails 3.0, and YAML is now the preferred format for configuration files.

Using system properties / command line arguments

Suppose you are using the `JDBC_CONNECTION_STRING` command line argument and you want to access the same in the `yml` file then it can be done in the following manner:

```

production:
  dataSource:
    url: '${JDBC_CONNECTION_STRING}'

```

Similarly system arguments can be accessed.

You will need to have this in `build.gradle` to modify the `bootRun` target if `./gradlew bootRun` is used to start the application

```
bootRun {  
    systemProperties = System.properties  
}
```

For testing the following will need to change the `test` task as follows

```
test {  
    systemProperties = System.properties  
}
```

External configuration

Grails will read `application.(properties|yml)` from the `./config` or the current directory by default. As Grails is a SpringBoot configuration options are available as well, for documentation please consult:

<https://docs.spring.io/spring-boot/docs/2.7.18/reference/html/features.html#features.external-config.files>

4.1.2 Built in options

Grails has a set of core settings that are worth knowing about. Their defaults are suitable for most projects, but it's important to understand what they do because you may need one or more of them later.

Runtime settings

On the runtime front, i.e. `grails-app/conf/application.yml`, there are quite a few more core settings:

- `grails.enable.native2ascii` - Set this to false if you do not require native2ascii conversion of Grails i18n properties files (default: true).
- `grails.views.default.codec` - Sets the default encoding regime for GSPs - can be one of 'none', 'html', or 'base64' (default: 'none'). To reduce risk of XSS attacks, set this to 'html'.
- `grails.views.gsp.encoding` - The file encoding used for GSP source files (default: 'utf-8').
- `grails.mime.file.extensions` - Whether to use the file extension to dictate the mime type in [Content Negotiation](#) (default: true).
- `grails.mime.types` - A map of supported mime types used for [Content Negotiation](#).
- `grails.serverURL` - A string specifying the server URL portion of absolute links, including server name e.g. `grails.serverURL="http://my.yourportal.com"`. See [createLink](#). Also used by redirects.
- `grails.views.gsp.sitemesh.preprocess` - Determines whether SiteMesh preprocessing happens. Disabling this slows down page rendering, but if you need SiteMesh to parse the generated HTML from a GSP view then disabling it is the right option. Don't worry if you don't understand this advanced property: leave it set to true.
- `grails.reload.excludes` and `grails.reload.includes` - Configuring these directives determines the reload behavior for project specific source files. Each directive takes a list of strings that are the class names for project source files that should be excluded from reloading behavior or included accordingly when running the application in development with the `bootRun` task. If

the `grails.reload.includes` directive is configured, then only the classes in that list will be reloaded.

4.1.3 Logging

Since Grails 3.0, logging is handled by the [Logback logging framework](#) and can be configured with the `grails-app/conf/logback.xml` file.

Since Grails 5.1.2 support for groovy configuration (`grails-app/conf/logback.groovy`) has been removed (by logback 1.2.9). It is possible to add back groovy configuration by adding the [logback-groovy-config](#) library to your project.

For more information on configuring logging refer to the [Logback documentation](#) on the subject.

4.1.3.1 Logger Names

Grails artifacts (controllers, services ...) get injected a `log` property automatically.

Prior to Grails 3.3.0, the name of the logger for Grails Artifact followed the convention `grails.app.<type>.<className>`, where `type` is the type of the artifact, for example, `controllers` or `services`, and `className` is the fully qualified name of the artifact.

Grails 3.3.x simplifies logger names. The next examples illustrate the changes:

`BookController.groovy` located at `grails-app/controllers/com/company` NOT annotated with [@Slf4j](#)

Logger Name (Grails 3.3.x or higher)

`com.company.BookController`

Logger Name (Grails 3.2.x or lower)

`grails.app.controllers.com.company.BookController`

`BookController.groovy` located at `grails-app/controllers/com/company` annotated with [@Slf4j](#)

Logger Name (Grails 3.3.x or higher)

`com.company.BookController`

Logger Name (Grails 3.2.x or lower)

`com.company.BookController`

`BookService.groovy` located at `grails-app/services/com/company` NOT annotated with [@Slf4j](#)

Logger Name (Grails 3.3.x or higher)

`com.company.BookService`

Logger Name (Grails 3.2.x or lower)

`grails.app.services.com.company.BookService`

`BookService.groovy` located at `grails-app/services/com/company` annotated with [@Slf4j](#)

Logger Name (Grails 3.3.x or higher)

`com.company.BookService`

Logger Name (Grails 3.2.x or lower)

`com.company.BookService`

`BookDetail.groovy` located at `src/main/groovy/com/company` annotated with [@Slf4j](#)

Logger Name (Grails 3.3.x or higher)

`com.company.BookDetail`

Logger Name (Grails 3.2.x or lower)

`com.company.BookDetail`

4.1.3.2 Masking Request Parameters From Stacktrace Logs

When Grails logs a stacktrace, the log message may include the names and values of all of the request parameters for the current request. To mask out the values of secure request parameters, specify the parameter names in the `grails.exceptionresolver.params.exclude` config property:

`grails-app/conf/application.yml`

```
grails:
  exceptionresolver:
    params:
      exclude:
        - password
        - creditCard
```

Request parameter logging may be turned off altogether by setting the `grails.exceptionresolver.logRequestParameters` config property to `false`. The default value is `true` when the application is running in `DEVELOPMENT` mode and `false` for all other environments.

`grails-app/conf/application.yml`

```
grails:
  exceptionresolver:
    logRequestParameters: false
```

4.1.3.3 External Configuration File

If you set the configuration property `logging.config`, you can instruct Logback to use an external configuration file.

`grails-app/conf/application.yml`

```
logging:
  config: /Users/me/config/logback.groovy
```

Alternatively, you can supply the configuration file location with a system property:

```
$ ./gradlew -Dlogging.config=/Users/me/config/logback.groovy bootRun
```

Or, you could use an environment variable:

```
$ export LOGGING_CONFIG=/Users/me/config/logback.groovy
$ ./gradlew bootRun
```

4.1.4 GORM

Grails provides the following GORM configuration options:

- `grails.gorm.failOnError` - If set to `true`, causes the `save()` method on domain classes to throw a `grails.validation.ValidationException` if [validation](#) fails during a save. This option may also be assigned a list of Strings representing package names. If the value is a list of Strings then the `failOnError` behavior will only be applied to domain classes in those packages (including sub-packages). See the [save](#) method docs for more information.

For example, to enable `failOnError` for all domain classes:

```
grails:
  gorm:
    failOnError: true
```

and to enable `failOnError` for domain classes by package:

```
grails:
  gorm:
    failOnError:
      - com.companyname.somepackage
      - com.companyname.someotherpackage
```

4.1.5 Configuring an HTTP proxy

To setup Grails to use an HTTP proxy there are two steps. Firstly you need to configure the `grails` CLI to be aware of the proxy if you wish to use it to create applications and so on. This can be done using the `GRAILS_OPTS` environment variable, for example on Unix systems:

```
export GRAILS_OPTS="-Dhttps.proxyHost=127.0.0.1 -Dhttps.proxyPort=3128 -Dhttp.proxyUser=test -Dhttp.proxyPassword=t"
```

The default profile repository is resolved over HTTPS so `https.proxyPort` and `https.proxyUser` are used, however the username and password are specified with `http.proxyUser` and `http.proxyPassword`

For Windows systems the environment variable can be configured under `My Computer/Advanced/Environment Variables`.

With this configuration in place the `grails` command can connect and authenticate via a proxy.

Secondly, since Grails uses Gradle as the build system, you need to configure Gradle to authenticate via the proxy. For instructions on how to do this see the [Gradle user guide section on the topic](#).

4.2 The Application Class

Every new Grails application features an `Application` class within the `grails-app/init` directory.

The `Application` class subclasses the [GrailsAutoConfiguration](#) class and features a static `void main` method, meaning it can be run as a regular application.

4.2.1 Executing the Application Class

There are several ways to execute the `Application` class, if you are using an IDE then you can simply right click on the class and run it directly from your IDE which will start your Grails application.

This is also useful for debugging since you can debug directly from the IDE without having to connect a remote debugger when using the `./gradlew bootRun --debug-jvm` command from the command line.

You can also package your application into a runnable WAR file, for example:

```
$ ./gradlew bootWar
$ java -jar build/libs/myapp-0.1.war
```

This is useful if you plan to deploy your application using a container-less approach.

4.2.2 Customizing the Application Class

There are several ways in which you can customize the `Application` class.

Customizing Scanning

By default Grails will scan all known source directories for controllers, domain class etc., however if there are packages in other JAR files you wish to scan you can do so by overriding the `packageNames()` method of the `Application` class:

```
class Application extends GrailsAutoConfiguration {
    @Override
    Collection<String> packageNames() {
        super.packageNames() + ['my.additional.package']
    }
    ...
}
```

Registering Additional Beans

The `Application` class can also be used as a source for Spring bean definitions, simply define a method annotated with the [Bean](#) and the returned object will become a Spring bean. The name of the method is used as the bean name:

```
class Application extends GrailsAutoConfiguration {
    @Bean
    MyType myBean() {
        return new MyType()
    }
    ...
}
```

4.2.3 The Application LifeCycle

The `Application` class also implements the [GrailsApplicationLifeCycle](#) interface which all plugins implement.

This means that the `Application` class can be used to perform the same functions as a plugin. You can override the [regular plugins hooks](#) such as `doWithSpring`, `doWithApplicationContext` and so on by overriding the appropriate method:

```
class Application extends GrailsAutoConfiguration {
    @Override
    Closure doWithSpring() {
        { ->
            mySpringBean(MyType)
        }
    }
}
```

```
} ...
```

4.3 Environments

Per Environment Configuration

Grails supports the concept of per environment configuration. The `application.yml` and `application.groovy` files in the `grails-app/conf` directory can use per-environment configuration using either YAML or the syntax provided by [ConfigSlurper](#). As an example consider the following default `application.yml` definition provided by Grails:

```
environments:
  development:
    dataSource:
      dbCreate: create-drop
      url: jdbc:h2:mem:devDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
  test:
    dataSource:
      dbCreate: update
      url: jdbc:h2:mem:testDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
  production:
    dataSource:
      dbCreate: update
      url: jdbc:h2:prodDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
      properties:
        jmxEnabled: true
        initialSize: 5
  ...
```

The above can be expressed in Groovy syntax in `application.groovy` as follows:

```
dataSource {
  pooled = false
  driverClassName = "org.h2.Driver"
  username = "sa"
  password = ""
}
environments {
  development {
    dataSource {
      dbCreate = "create-drop"
      url = "jdbc:h2:mem:devDb"
    }
  }
  test {
    dataSource {
      dbCreate = "update"
      url = "jdbc:h2:mem:testDb"
    }
  }
  production {
    dataSource {
      dbCreate = "update"
      url = "jdbc:h2:prodDb"
      properties {
        jmxEnabled = true
        initialSize = 5
      }
    }
  }
}
```

Notice how the common configuration is provided at the top level and then an `environments` block specifies per environment settings for the `dbCreate` and `url` properties of the `DataSource`.

Packaging and Running for Different Environments

Grails' [command line](#) has built in capabilities to execute any command within the context of a specific environment. The format is:

```
grails <<environment>> <<command name>>
```

In addition, there are 3 preset environments known to Grails: `dev`, `prod`, and `test` for development, production and test. For example to create a WAR for the `test` environment you

would run:

```
grails test war
```

To target other environments you can pass a `grails.env` variable to any command:

```
./gradlew bootRun -Dgrails.env=UAT
```

Programmatic Environment Detection

Within your code, such as in a Gant script or a bootstrap class you can detect the environment using the [Environment](#) class:

```
import grails.util.Environment

...

switch (Environment.current) {
    case Environment.DEVELOPMENT:
        configureForDevelopment()
        break
    case Environment.PRODUCTION:
        configureForProduction()
        break
}
```

Per Environment Bootstrapping

It's often desirable to run code when your application starts up on a per-environment basis. To do so you can use the `grails-app/init/Bootstrap.groovy` file's support for per-environment execution:

```
def init = { ServletContext ctx ->
    environments {
        production {
            ctx.setAttribute("env", "prod")
        }
        development {
            ctx.setAttribute("env", "dev")
        }
    }
    ctx.setAttribute("foo", "bar")
}
```

Generic Per Environment Execution

The previous `Bootstrap` example uses the `grails.util.Environment` class internally to execute. You can also use this class yourself to execute your own environment specific logic:

```
Environment.executeForCurrentEnvironment {
    production {
        // do something in production
    }
    development {
        // do something only in development
    }
}
```

4.4 The DataSource

Since Grails is built on Java technology setting up a data source requires some knowledge of JDBC (the technology that stands for Java Database Connectivity).

If you use a database other than H2 you need a JDBC driver. For example for MySQL you would need [Connector/J](#).

Drivers typically come in the form of a JAR archive. It's best to use the dependency resolution to resolve the jar if it's available in a Maven repository, for example you could

add a dependency for the MySQL driver like this:

```
dependencies {
    runtimeOnly 'mysql:mysql-connector-java:5.1.29'
}
```

Once you have the JAR resolved you need to get familiar with how Grails manages its database configuration. The configuration can be maintained in either `grails-app/conf/application.groovy` OR `grails-app/conf/application.yml`. These files contain the `dataSource` definition which includes the following settings:

- `driverClassName` - The class name of the JDBC driver
- `username` - The username used to establish a JDBC connection
- `password` - The password used to establish a JDBC connection
- `url` - The JDBC URL of the database
- `dbCreate` - Whether to auto-generate the database from the domain model - one of 'create-drop', 'create', 'update', 'validate', or 'none'
- `pooled` - Whether to use a pool of connections (defaults to true)
- `logSql` - Enable SQL logging to stdout
- `formatSql` - Format logged SQL
- `dialect` - A String or Class that represents the Hibernate dialect used to communicate with the database. See the org.hibernate.dialect package for available dialects.
- `readOnly` - If true makes the DataSource read-only, which results in the connection pool calling `setReadOnly(true)` on each connection
- `transactional` - If false leaves the DataSource's transactionManager bean outside the chained BE1PC transaction manager implementation. This only applies to additional datasources.
- `persistenceInterceptor` - The default datasource is automatically wired up to the persistence interceptor, other datasources are not wired up automatically unless this is set to true
- `properties` - Extra properties to set on the DataSource bean. See the Tomcat Pool documentation. There is also a Javadoc format documentation of the properties.
- `jmxExport` - If false, will disable registration of JMX MBeans for all DataSources. By default JMX MBeans are added for DataSources with `jmxEnabled = true` in properties.
- `type` - The connection pool class if you want to force Grails to use it when there are more than one available.

A typical configuration for MySQL in `application.groovy` may be something like:

```
dataSource {
    pooled = true
    dbCreate = "update"
    url = "jdbc:mysql://localhost:3306/my_database"
    driverClassName = "com.mysql.jdbc.Driver"
    dialect = org.hibernate.dialect.MySQL5InnoDBDialect
    username = "username"
    password = "password"
    type = "com.zaxxer.hikari.HikariDataSource"
    properties {
        jmxEnabled = true
        initialSize = 5
    }
}
```

```

maxActive = 50
minIdle = 5
maxIdle = 25
maxWait = 10000
maxAge = 10 * 60000
timeBetweenEvictionRunsMillis = 5000
minEvictableIdleTimeMillis = 60000
validationQuery = "SELECT 1"
validationQueryTimeout = 3
validationInterval = 15000
testOnBorrow = true
testWhileIdle = true
testOnReturn = false
jdbcInterceptors = "ConnectionState;StatementCache(max=200)"
defaultTransactionIsolation = java.sql.Connection.TRANSACTION_READ_COMMITTED
}
}

```

When configuring the DataSource do not include the type or the def keyword before any of the configuration settings as Groovy will treat these as local variable definitions and they will not be processed. For example the following is invalid:

```

dataSource {
    boolean pooled = true // type declaration results in ignored local variable
    ...
}

```

Example of advanced configuration using extra properties:

```

dataSource {
    pooled = true
    dbCreate = "update"
    url = "jdbc:mysql://localhost:3306/my_database"
    driverClassName = "com.mysql.jdbc.Driver"
    dialect = org.hibernate.dialect.MySQL5InnoDBDialect
    username = "username"
    password = "password"
    type = "com.zaxxer.hikari.HikariDataSource"
    properties {
        // Documentation for Tomcat JDBC Pool
        // http://tomcat.apache.org/tomcat-7.0-doc/jdbc-pool.html#Common_Attributes
        // https://tomcat.apache.org/tomcat-7.0-doc/api/org/apache/tomcat/jdbc/pool/PoolConfiguration.html
        jmxEnabled = true
        initialSize = 5
        maxActive = 50
        minIdle = 5
        maxIdle = 25
        maxWait = 10000
        maxAge = 10 * 60000
        timeBetweenEvictionRunsMillis = 5000
        minEvictableIdleTimeMillis = 60000
        validationQuery = "SELECT 1"
        validationQueryTimeout = 3
        validationInterval = 15000
        testOnBorrow = true
        testWhileIdle = true
        testOnReturn = false
        ignoreExceptionOnPreLoad = true
        // http://tomcat.apache.org/tomcat-7.0-doc/jdbc-pool.html#JDBC_interceptors
        jdbcInterceptors = "ConnectionState;StatementCache(max=200)"
        defaultTransactionIsolation = java.sql.Connection.TRANSACTION_READ_COMMITTED // safe default
        // controls for leaked connections
        abandonWhenPercentageFull = 100 // settings are active only when pool is full
        removeAbandonedTimeout = 120
        removeAbandoned = true
        // use JMX console to change this setting at runtime
        logAbandoned = false // causes stacktrace recording overhead, use only for debugging
        // JDBC driver properties
        // Mysql as example
        dbProperties {
            // Mysql specific driver properties
            // http://dev.mysql.com/doc/connector-j/en/connector-j-reference-configuration-properties.html
            // let Tomcat JDBC Pool handle reconnecting
            autoReconnect=false
            // truncation behaviour
            jdbcCompliantTruncation=false
            // mysql 0-date conversion
            zeroDateTimeBehavior='convertToNull'
            // Tomcat JDBC Pool's StatementCache is used instead, so disable mysql driver's cache
            cachePrepStmts=false
            cacheCallableStmts=false
            // Tomcat JDBC Pool's StatementFinalizer keeps track
            dontTrackOpenResources=true
            // performance optimization: reduce number of SQLExceptions thrown in mysql driver code
            holdResultsOpenOverStatementClose=true
            // enable MySQL query cache - using server prep stmts will disable query caching
            useServerPrepStmts=false
            // metadata caching
            cacheServerConfiguration=true
        }
    }
}

```

```

        cacheResultSetMetadata=true
        metadataCacheSize=100
        // timeouts for TCP/IP
        connectTimeout=15000
        socketTimeout=120000
        // timer tuning (disable)
        maintainTimeStats=false
        enableQueryTimeouts=false
        // misc tuning
        noDatetimeStringSync=true
    }
}

```

More on dbCreate

Hibernate can automatically create the database tables required for your domain model. You have some control over when and how it does this through the `dbCreate` property, which can take these values:

- **create** - Drops the existing schema and creates the schema on startup, dropping existing tables, indexes, etc. first.
- **create-drop** - Same as **create**, but also drops the tables when the application shuts down cleanly.
- **update** - Creates missing tables and indexes, and updates the current schema without dropping any tables or data. Note that this can't properly handle many schema changes like column renames (you're left with the old column containing the existing data).
- **validate** - Makes no changes to your database. Compares the configuration with the existing database schema and reports warnings.
- any other value - does nothing

Setting the `dbCreate` setting to "none" is recommended once your schema is relatively stable and definitely when your application and database are deployed in production. Database changes are then managed through proper migrations, either with SQL scripts or a migration tool like [Flyway](#) or [Liquibase](#). The [Database Migration](#) plugin uses Liquibase.

4.4.1 DataSources and Environments

The previous example configuration assumes you want the same config for all environments: production, test, development etc.

Grails' `DataSource` definition is "environment aware", however, so you can do:

```

dataSource {
    pooled = true
    driverClassName = "com.mysql.jdbc.Driver"
    dialect = org.hibernate.dialect.MySQL5InnoDBDialect
    // other common settings here
}

environments {
    production {
        dataSource {
            url = "jdbc:mysql://liveip.com/liveDb"
            // other environment-specific settings here
        }
    }
}

```

4.4.2 Automatic Database Migration

The `dbCreate` property of the `DataSource` definition is important as it dictates what Grails should do at runtime with regards to automatically generating the database tables from [GORM](#) classes. The options are described in the [DataSource](#) section:

- `create`
- `create-drop`
- `update`
- `validate`
- `no value`

In [development](#) mode `dbCreate` is by default set to "create-drop", but at some point in development (and certainly once you go to production) you'll need to stop dropping and re-creating the database every time you start up your server.

It's tempting to switch to `update` so you retain existing data and only update the schema when your code changes, but Hibernate's update support is very conservative. It won't make any changes that could result in data loss, and doesn't detect renamed columns or tables, so you'll be left with the old one and will also have the new one.

Grails supports migrations with Liquibase or Flyway via plugins.

- [Database Migration](#)
- [Flyway](#)

4.4.3 Transaction-aware DataSource Proxy

The actual `DataSource` bean is wrapped in a transaction-aware proxy so you will be given the connection that's being used by the current transaction or Hibernate `Session` if one is active.

If this were not the case, then retrieving a connection from the `DataSource` would be a new connection, and you wouldn't be able to see changes that haven't been committed yet (assuming you have a sensible transaction isolation setting, e.g. `READ_COMMITTED` or better).

4.4.4 Database Console

The [H2 database console](#) is a convenient feature of H2 that provides a web-based interface to any database that you have a JDBC driver for, and it's very useful to view the database you're developing against. It's especially useful when running against an in-memory database.

You can access the console by navigating to <http://localhost:8080/h2-console> in a browser. See the [Spring Boot H2 Console Documentation](#) for more information on the options available.

The H2 console is disabled by default (unless you are using Spring Boot's developer tools) and must be enabled by configuring the `spring.h2.console.enabled` property with a value of `true`.

The H2 console is only intended for use during development so care should be taken to

ensure that `spring.h2.console.enabled` is not set to `true` in production.

4.4.5 Multiple Datasources

By default all domain classes share a single `DataSource` and a single database, but you have the option to partition your domain classes into two or more data sources.

Configuring Additional DataSources

The default `DataSource` configuration in `grails-app/conf/application.yml` looks something like this:

```
dataSource:
  pooled: true
  jmxExport: true
  driverClassName: org.h2.Driver
  username: sa
  password:

environments:
  development:
    dataSource:
      dbCreate: create-drop
      url: jdbc:h2:mem:devDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
  test:
    dataSource:
      dbCreate: update
      url: jdbc:h2:mem:testDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
  production:
    dataSource:
      dbCreate: update
      url: jdbc:h2:prodDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
      properties:
        jmxEnabled: true
        initialSize: 5
```

This configures a single `DataSource` with the Spring bean named `dataSource`. To configure extra data sources, add a `dataSources` block (at the top level, in an environment block, or both, just like the standard `DataSource` definition) with a custom name. For example, this configuration adds a second `DataSource`, using MySQL in the development environment and Oracle in production:

```
dataSource:
  pooled: true
  jmxExport: true
  driverClassName: org.h2.Driver
  username: sa
  password:

dataSources:
  lookup:
    dialect: org.hibernate.dialect.MySQLInnoDBDialect
    driverClassName: com.mysql.jdbc.Driver
    username: lookup
    password: secret
    url: jdbc:mysql://localhost/lookup
    dbCreate: update

environments:
  development:
    dataSource:
      dbCreate: create-drop
      url: jdbc:h2:mem:devDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
  test:
    dataSource:
      dbCreate: update
      url: jdbc:h2:mem:testDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
  production:
    dataSource:
      dbCreate: update
      url: jdbc:h2:prodDb;MVCC=TRUE;LOCK_TIMEOUT=10000;DB_CLOSE_ON_EXIT=FALSE
      properties:
        jmxEnabled: true
        initialSize: 5
    ...
    dataSources:
      lookup:
        dialect: org.hibernate.dialect.Oracle10gDialect
        driverClassName: oracle.jdbc.driver.OracleDriver
        username: lookup
```

```
password: secret
url: jdbc:oracle:thin:@localhost:1521:lookup
dbCreate: update
```

You can use the same or different databases as long as they're supported by Hibernate.

If you need to inject the `lookup` datasource in a Grails artefact, you can do it like this:

```
DataSource dataSource_lookup
```

While defining multiple data sources, one of them **must** be named "dataSource". This is required because Grails determines which data source is the default by determining which one is named "dataSource".

Configuring Domain Classes

If a domain class has no `DataSource` configuration, it defaults to the standard 'dataSource'. Set the `datasource` property in the `mapping` block to configure a non-default `DataSource`. For example, if you want to use the `ZipCode` domain to use the 'lookup' `DataSource`, configure it like this:

```
class ZipCode {
    String code

    static mapping = {
        datasource 'lookup'
    }
}
```

A domain class can also use two or more data sources. Use the `datasources` property with a list of names to configure more than one, for example:

```
class ZipCode {
    String code

    static mapping = {
        datasources(['lookup', 'auditing'])
    }
}
```

If a domain class uses the default `DataSource` and one or more others, use the special name 'DEFAULT' to indicate the default `DataSource`:

```
class ZipCode {
    String code

    static mapping = {
        datasources(['lookup', 'DEFAULT'])
    }
}
```

If a domain class uses all configured data sources, use the special value 'ALL':

```
class ZipCode {
    String code

    static mapping = {
        datasource 'ALL'
    }
}
```

Namespaces and GORM Methods

If a domain class uses more than one `DataSource` then you can use the namespace implied by each `DataSource` name to make GORM calls for a particular `DataSource`. For example, consider this class which uses two data sources:

```
class ZipCode {
```

```
String code

static mapping = {
    datasources(['lookup', 'auditing'])
}
}
```

The first `DataSource` specified is the default when not using an explicit namespace, so in this case we default to 'lookup'. But you can call GORM methods on the 'auditing' `DataSource` with the `DataSource` name, for example:

```
def zipCode = ZipCode.auditing.get(42)
...
zipCode.auditing.save()
```

As you can see, you add the `DataSource` to the method call in both the static case and the instance case.

Hibernate Mapped Domain Classes

You can also partition annotated Java classes into separate `datasources`. Classes using the default `datasource` are registered in `grails-app/conf/hibernate.cfg.xml`. To specify that an annotated class uses a non-default `datasource`, create a `hibernate.cfg.xml` file for that `datasource` with the file name prefixed with the `datasource` name.

For example if the `Book` class is in the default `datasource`, you would register that in `grails-app/conf/hibernate.cfg.xml`:

```
<?xml version='1.0' encoding='UTF-8'?>
<!DOCTYPE hibernate-configuration PUBLIC
    "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
    'http://hibernate.sourceforge.net/hibernate-configuration-3.0.dtd'>
<hibernate-configuration>
    <session-factory>
        <mapping class='org.example.Book' />
    </session-factory>
</hibernate-configuration>
```

and if the `Library` class is in the "ds2" `datasource`, you would register that in `grails-app/conf/ds2_hibernate.cfg.xml`:

```
<?xml version='1.0' encoding='UTF-8'?>
<!DOCTYPE hibernate-configuration PUBLIC
    "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
    'http://hibernate.sourceforge.net/hibernate-configuration-3.0.dtd'>
<hibernate-configuration>
    <session-factory>
        <mapping class='org.example.Library' />
    </session-factory>
</hibernate-configuration>
```

The process is the same for classes mapped with `hbm.xml` files - just list them in the appropriate `hibernate.cfg.xml` file.

Services

Like Domain classes, by default `Services` use the default `DataSource` and `PlatformTransactionManager`. To configure a `Service` to use a different `DataSource`, use the static `datasource` property, for example:

```
class DataService {
    static datasource = 'lookup'

    void someMethod(...) {
        ...
    }
}
```

A transactional service can only use a single `DataSource`, so be sure to only make changes for

domain classes whose `DataSource` is the same as the Service.

Note that the datasource specified in a service has no bearing on which datasources are used for domain classes; that's determined by their declared datasources in the domain classes themselves. It's used to declare which transaction manager to use.

If you have a `Foo` domain class in `dataSource1` and a `Bar` domain class in `dataSource2`, if `WahooService` uses `dataSource1`, a service method that saves a new `Foo` and a new `Bar` will only be transactional for `Foo` since they share the same datasource. The transaction won't affect the `Bar` instance. If you want both to be transactional you'd need to use two services and XA datasources for two-phase commit, e.g. with the Atomikos plugin.

Transactions across multiple data sources

Grails does not by default try to handle transactions that span multiple data sources.

You can enable Grails to use the Best Effort 1PC pattern for handling transactions across multiple datasources. To do so you must set the

`grails.transaction.chainedTransactionManager.enabled` setting to `true` in `application.yml`:

```
grails:
  transaction:
    chainedTransactionManager:
      enabled: true
```

The [Best Efforts 1PC pattern](#) is fairly general but can fail in some circumstances that the developer must be aware of.

This is a non-XA pattern that involves a synchronized single-phase commit of a number of resources. Because the [2PC](#) is not used, it can never be as safe as an [XA](#) transaction, but is often good enough if the participants are aware of the compromises.

The basic idea is to delay the commit of all resources as late as possible in a transaction so that the only thing that can go wrong is an infrastructure failure (not a business-processing error). Systems that rely on Best Efforts 1PC reason that infrastructure failures are rare enough that they can afford to take the risk in return for higher throughput. If business-processing services are also designed to be idempotent, then little can go wrong in practice.

The BE1PC implementation was added in Grails 2.3.6. . Before this change additional datasources didn't take part in transactions initiated in Grails. The transactions in additional datasources were basically in auto commit mode. In some cases this might be the wanted behavior. One reason might be performance: on the start of each new transaction, the BE1PC transaction manager creates a new transaction to each datasource. It's possible to leave an additional datasource out of the BE1PC transaction manager by setting `transactional = false` in the respective configuration block of the additional `DataSource`. Datasources with `readOnly = true` will also be left out of the chained transaction manager (since 2.3.7).

By default, the BE1PC implementation will add all beans implementing the Spring [PlatformTransactionManager](#) interface to the chained BE1PC transaction manager. For example, a possible [JMSTransactionManager](#) bean in the Grails application context would be added to the Grails BE1PC transaction manager's chain of transaction managers.

You can exclude transaction manager beans from the BE1PC implementation with this configuration option:

```
grails:
  transaction:
```

```
chainedTransactionManager:  
  enabled: true  
  blacklistPattern: '.*'
```

The exclude matching is done on the name of the transaction manager bean. The transaction managers of datasources with `transactional = false` OR `readOnly = true` will be skipped and using this configuration option is not required in that case.

XA and Two-phase Commit

When the Best Efforts 1PC pattern isn't suitable for handling transactions across multiple transactional resources (not only datasources), there are several options available for adding XA/2PC support to Grails applications.

The [Spring transactions documentation](#) contains information about integrating the JTA/XA transaction manager of different application servers. In this case, you can configure a bean with the name `transactionManager` manually in `resources.groovy` OR `resources.xml` file.

4.5 Versioning

Detecting Versions at Runtime

You can detect the application version using Grails' support for application metadata using the [GrailsApplication](#) class. For example within [controllers](#) there is an implicit [grailsApplication](#) variable that can be used:

```
def version = grailsApplication.metadata.getApplicationVersion()
```

You can retrieve the version of Grails that is running with:

```
def grailsVersion = grailsApplication.metadata.getGrailsVersion()
```

or the `GrailsUtil` class:

```
import grails.util.GrailsUtil  
...  
def grailsVersion = GrailsUtil.grailsVersion
```

4.6 Dependency Resolution

Dependency resolution is handled by the [Gradle build tool](#), all dependencies are defined in the `build.gradle` file. Refer to the [Gradle user guide](#) for more information.

5 The Command Line

The Grails New Command-Line Interface (CLI) has undergone significant changes compared to its previous versions, primarily focusing on code generation. One notable alteration is the removal of APIs for invoking Gradle for tasks related to building using Gradle Tooling APIs. This shift in responsibility aligns with the framework's evolution and its integration with the Gradle build system.

Accessing the Grails CLI

The Grails CLI (Command Line Interface) can be swiftly and effortlessly accessed by simply typing the following command into your terminal or command prompt:

grails

This command allows developers to quickly initiate the Grails CLI and begin working with the framework, making it an easy entry point for those looking to start their Grails projects.

The New Grails CLI! is the preferred method for initiating new Grails projects. This command-line interface offers various options for creating projects, enabling you to select your preferred build tools, test frameworks, GORM implementation, and more. Additionally, the CLI provides commands for generating essential components like controllers and domain classes.

The Grails Forge Website

You can also begin your Grails application without the need to install the Grails CLI by visiting [the Grails Forge website](#). This web-based platform allows you to initiate Grails projects conveniently, bypassing the installation of the CLI.

Understanding the New Grails Command-line Interface (CLI)

Once the Grails CLI has been successfully installed, you can activate it using the "grails" command. For example:

```
grails create-app myapp
```

A Grails framework CLI project is recognizable by the presence of the "grails-cli.yml" file, which is automatically generated at the project's root if it was created using the CLI. This file contains information about the project's profile, default package, and other variables.

Here is an example of a "grails-cli.yml" configuration for a default Grails web application:

```
applicationType: web
defaultPackage: com.example
testFramework: spock
sourceLanguage: groovy
buildTool: gradle
gormImpl: gorm-hibernate5
servletImpl: spring-boot-starter-tomcat
features:
  - app-name
  - asset-pipeline-grails
  - base
  - geb
  - gorm-hibernate5
  - gradle
  - grails-application
  - grails-console
  - grails-dependencies
  - grails-gorm-testing-support
  - grails-gradle-plugin
  - grails-gsp
  - grails-url-mappings
  - grails-web
  - grails-web-testing-support
  - h2
  - logback
  - micronaut-inject-groovy
  - readme
  - scaffolding
  - spock
  - spring-boot-autoconfigure
  - spring-boot-starter
  - spring-boot-starter-tomcat
  - yaml
```

This "grails-cli.yml" configuration sets the default values for various aspects of the Grails web application, including the application type, default package, test framework, source language, build tool, GORM implementation, servlet implementation, and a list of enabled features.

Grails Default Package Configuration

The project's default package is determined based on the project's name. For instance, running the following command:

```
grails create-app myapp
```

Will set the default package to `myapp`.

If you wish to specify your own default package when creating the application, you can do so by prefixing the application name with the package like this:

```
grails create-app com.example.myapp
```

In this case, "com.example" becomes the default package for your project.

Gradle Build Tool

Grails now utilizes the Gradle Build System for project management. The project's build configuration is specified in the `build.gradle` file, where you define critical aspects of your project such as its version, required dependencies, and the repositories from which these dependencies should be sourced. Here's an example of how this is done:

```
plugins {
    id 'org.grails.grails-web' version 'x.y.z' // Grails plugin
}

repositories {
    mavenCentral()
}

dependencies {
    implementation 'org.springframework.boot:spring-boot-starter'
    implementation 'org.grails:grails-core'
    // Add more dependencies as needed...
}

grails {
    pathingJar = true
}
```

Utilizing Gradle Build Tool

To interact with your Grails project and perform various tasks related to building and running it, you should employ Gradle commands. Here are a few examples:

- Building the Grails application:

```
gradle build
```

- Running the Grails application:

```
gradle bootRun
```

- Listing available Gradle tasks:

```
gradle tasks
```

By invoking these Gradle commands, you can effectively manage your Grails application's lifecycle.

It is important to remember that Grails leverages the power of Gradle for streamlined project management, including build automation and dependency resolution. This approach allows for greater flexibility and control over your Grails projects.

5.1 Interactive Mode

When you execute the `grails` command without any arguments, the Grails Command Line Interface (CLI) enters interactive mode. In this mode, it functions like a shell, allowing you to run multiple CLI commands without the need to re-initialize the CLI runtime. This mode is particularly useful when working with code-generation commands (such as `create-controller`), creating multiple projects, or exploring various CLI features.

One of the advantages of interactive mode is the availability of tab-completion. You can simply press the TAB key to view possible options for a given command or flag. Here's an example of the available options in interactive mode:

```
grails>
--help          --verbose          -V          -v          create-app          creat
--stacktrace    --version          -h          -x          create-controller  creat
```

Help and Information

You can access general usage information for Grails commands using the help flag associated with a specific command.

```

grails> create-app -h
Usage: grails create-app [-hivVx] [--list-features] [-g=GORM Implementation] [--jdk=<javaVersion>]
                        [-s=Servlet Implementation] [-t=TEST] [-f=FEATURE[,FEATURE...]]... [NAME]

Creates an application
  [NAME]                The name of the application to create.
  -f, --features=FEATURE[,FEATURE...]
                        The features to use. Possible values: h2, scaffolding, gorm-hibernate5,
                        spring-boot-starter-jetty, springloaded, spring-boot-starter-tomcat,
                        micronaut-http-client, cache-ehcache, hibernate-validator, postgres,
                        mysql, cache, database-migration, grails-gsp, hamcrest, gorm-mongodb,
                        assertj, mockito, spring-boot-starter-undertow, micronaut-inject-groovy,
                        github-workflow-java-ci, jrebel, testcontainers, sqlserver,
                        grails-console, views-markup, asset-pipeline-grails, views-json,
                        gorm-neo4j, asciidoctor, embedded-mongodb, grails-web-console,
                        logbackGroovy, mongo-sync, shade, geb, properties
  -g, --gorm=GORM Implementation
                        Which GORM Implementation to configure. Possible values: hibernate,
                        mongodb, neo4j.
  -h, --help            Show this help message and exit.
  -i, --inplace         Create a service using the current directory
                        --jdk, --java-version=<javaVersion>
                        The JDK version the project should target
                        --list-features
                        Output the available features and their descriptions
  -s, --servlet=Servlet Implementation
                        Which Servlet Implementation to configure. Possible values: none, tomcat,
                        jetty, undertow.
  -t, --test=TEST       Which test framework to use. Possible values: junit, spock.
  -v, --verbose         Create verbose output.
  -V, --version         Print version information and exit.
  -x, --stacktrace      Show full stack trace when exceptions occur.

```

You can also obtain a list of available features by using the `--list-features` flag with any of the create commands.

```

grails> create-app --list-features
Available Features
(+) denotes the feature is included by default
Name                                     Description
-----
CI/CD
github-workflow-java-ci [PREVIEW]      Adds a Github Actions Workflow to Build and Test Grails Application

Cache
cache                                   The Grails Cache plugin provides powerful and easy to use caching functionality
cache-ehcache                           The Grails Cache Ehcache plugin extends the Cache plugin and uses Ehcache as the cache provider

Client
micronaut-http-client                   Adds support for the Micronaut HTTP client

Configuration
properties                              Creates a properties configuration file

Database
database-migration                      Adds support for Liquibase database migrations. The Database Migration plugin
embedded-mongoddb                       Executes an embedded mongo database for integration or functional testing
gorm-hibernate5 (+)                    Adds support for Hibernate5 using GORM

```

gorm-mongodb	Configures GORM for MongoDB for Groovy applications
gorm-neo4j	Configures GORM for Neo4j for Groovy applications
h2 (+)	Adds the H2 driver and default config
mongo-sync	Adds support for the MongoDB Synchronous Driver
mysql	Adds the MySQL driver and default config
postgres	Adds the PostgreSQL driver and default config
sqlserver	Adds the SQL Server driver and default config
testcontainers	Use Testcontainers to run a database or other software in a Docker container
Development Tools	
assertj	AssertJ fluent assertions framework
hamcrest	Hamcrest matchers for JUnit
jrebel	Adds support for class reloading with JRebel (requires separate JRebel instal
springloaded	Adds support for class reloading with Spring Loaded
Documentation	
asciidoctor	Adds support for creating Asciidoctor documentation
Logging	
logbackGroovy	Gives you the ability to use groovy to configure logback instead of XML.
Management	
grails-web-console	A web-based Groovy console for interactive runtime application management and
Other	
geb (+)	This plugins configure Geb for Grails framework to write automation tests.
grails-console (+)	Starts the Grails console, which is an extended version of the regular Groovy
micronaut-inject-groovy (+)	micronaut-inject-groovy
scaffolding (+)	The Grails® framework Scaffolding plugin replicates much of the functionality
Packaging	
shade	Adds the ability to build a Fat/Shaded JAR
Server	
spring-boot-starter-jetty	spring-boot-starter-jetty
spring-boot-starter-tomcat (+)	spring-boot-starter-tomcat
spring-boot-starter-undertow	spring-boot-starter-undertow
Validation	
hibernate-validator	Adds support for the Hibernate Validator
mockito	Mockito test mocking framework for JUnit
View Rendering	
asset-pipeline-grails (+)	The Asset-Pipeline is a plugin used for managing and processing static assets
grails-gsp (+)	grails-gsp
views-json	JSON views are written in Groovy, end with the file extension gson and reside
views-markup	Markup views are written in Groovy, end with the file extension gml and resid

5.2 Creating Custom Commands

In Grails, a custom command is a piece of functionality that you can add to your Grails application and execute via the command-line interface (CLI). These commands are not part of the core Grails framework but are extensions you can create to perform specific tasks or operations that are unique to your application's requirements. Custom commands are a powerful way to automate various tasks, interact with your application, and perform administrative functions from the command line. When you run custom commands, they cause the Grails environment to start, giving you full access to the application context and the runtime, allowing you to work with the application's resources, services, and configuration as needed within your custom command.

There are several reasons why you might want to write a custom command for your Grails application:

- **Automating Tasks:** Custom commands allow you to automate routine tasks, such as data migration, database updates, or batch processing, by encapsulating the logic in a command that can be executed on-demand.
- **Administrative Operations:** You can use custom commands for administrative tasks like user management, system maintenance, and configuration management, making it easier to manage your application in different environments.
- **Integration:** Custom commands can be used to integrate your Grails application with other systems or services. For example, you can create a command to synchronize data with an external API.

- **Customized Workflows:** If your application has unique workflows or processes, custom commands provide a way to execute these workflows from the command line.

In Grails, you can create custom commands by implementing the [GrailsApplicationCommand](#) trait. By default, this trait requires your command to implement the `handle()` method as following:

```
boolean handle()
```

Commands defined this way still have access to the execution context via a variable called "executionContext."

Here's a step-by-step guide on how to create custom commands using the `GrailsApplicationCommand` trait with examples, and how to run these commands.

In Grails, you can create custom commands by implementing the `GrailsApplicationCommand` trait. Custom commands allow you to add functionality to your Grails application that can be executed via the command-line interface (CLI). Here's a step-by-step guide on how to create custom commands using the `GrailsApplicationCommand` trait with examples, and how to run these commands.

Step 1: Create a Custom Command

To create a custom command, you need to create a Groovy class that implements the `GrailsApplicationCommand` trait. This trait provides methods for command execution. Let's create a simple example command that greets the user:

```
// grails-app/commands/com/example/GreetCommand.groovy
package com.example

import grails.dev.commands.GrailsApplicationCommand

class GreetCommand implements GrailsApplicationCommand {

    String getName() {
        return "greet"
    }

    String getDescription() {
        return "Greet the user"
    }

    boolean handle() {
        println("Hello, user!")
        return true // Return true to indicate successful execution
    }
}
```

In this example, we've created a `GreetCommand` class that implements the `GrailsApplicationCommand` trait. It provides a `getName()` method to define the command name, a `getDescription()` method for a brief description, and the `run()` method to specify the code to execute when the command is run.

Step 2: Build Your Grails Application

Before you can use the `runCommand` task, ensure you have built your Grails application using the following command:

```
./gradlew assemble
```

This command will compile your application and make it ready for running custom commands.

Step 3: Run the Custom Command

To run the custom command, use Gradle's `runCommand` task. Open your terminal and navigate to your Grails application's root directory. Then, run the custom command with the following Gradle command:

```
./gradlew runCommand -Pargs="greet"
```

In the command above, replace "greet" with the name of your custom command. This will execute the `GreetCommand`, and you will see the output.

Here's the expected final output when you run the greet command:

```
Hello, user!
```

Additional Features: Command Arguments and Options

Grails also supports command-line arguments and options for custom commands. You can define these in your command class by implementing the `GrailsApplicationCommand` interface. Here's an example of a command that takes a name as an argument and an optional `--loud` option to make the greeting louder:

```
// grails-app/commands/com/example/GreetCommand.groovy
package com.example

import grails.dev.commands.GrailsApplicationCommand

class GreetCommand implements GrailsApplicationCommand {

    String getName() {
        return "greet"
    }

    String getDescription() {
        return "Greet the user with options"
    }

    boolean handle() {
        def args = commandLine.args
        String name = args.size() > 0 ? args[0] : "user"
        boolean loud = args.contains("--loud")

        if (loud) {
            println("HELLO, $name! (LOUD)")
        } else {
            println("Hello, $name!")
        }

        return true
    }
}
```

Now you can run the `greet` command with arguments and options:

```
# Greet the user with the default message
./gradlew runCommand -Pargs="greet"

# Greet a specific user
./gradlew runCommand -Pargs="greet Alice"

# Greet loudly
./gradlew runCommand -Pargs="greet --loud"

# Greet a specific user loudly
./gradlew runCommand -Pargs="greet Alice --loud"
```

This allows you to create more versatile and interactive custom commands for your Grails application.

In summary, creating custom commands in Grails using the `GrailsApplicationCommand` trait is a powerful way to extend your application's functionality beyond the web interface. You can define the command's name, description, and logic, and then execute it from the command line, optionally passing arguments and options as needed.

Using the `executionContext` in the Grails Custom Commands

In Grails, the `executionContext` is a runtime context object that provides valuable information about the current execution environment of a Grails application. It includes details such as the application's environment (e.g., development, production, test) and allows developers to access this context within custom commands.

Custom commands in Grails can use the `executionContext` to make informed decisions and perform specific tasks based on the current runtime environment. For example, developers can write conditional logic in custom commands that execute differently in production, development, or testing environments. This flexibility enables custom commands to adapt and behave differently depending on the context in which they are run, making them versatile tools for managing and extending Grails applications.

Suppose you have a Grails application that manages customer data, and you want to create a custom command to perform data backup. In this scenario, you may want the backup process to behave differently depending on whether you're running it in a development, staging, or production environment.

Here's an example of how you can create a custom command that uses the `executionContext` to determine the backup behavior:

```
// grails-app/commands/com/example/BackupCommand.groovy
package com.example

import grails.dev.commands.GrailsApplicationCommand

class BackupCommand implements GrailsApplicationCommand {

    String getName() {
        return "backup"
    }

    String getDescription() {
        return "Backup customer data"
    }

    boolean handle() {
        // Access the executionContext to determine the environment
        def environment = executionContext.environment

        if (environment == "production") {
            // Perform a full backup in the production environment
            println("Performing a full backup of customer data (Production)")
            // Add production-specific backup logic here
        } else {
            // Perform a partial backup in other environments
            println("Performing a partial backup of customer data (Non-production)")
            // Add non-production backup logic here
        }

        return true // Return true to indicate successful execution
    }
}
```

In this example:

- The custom command, named `BackupCommand`, is created to back up customer data.
- It checks the `executionContext` to determine the current environment.
- If the environment is "production," it performs a full backup with production-specific logic.
- In all other environments, it performs a partial backup with non-production logic.

When you run this custom command using `./gradlew runCommand -Pargs="backup"`, it will adapt its behavior based on whether you're in a production or non-production environment, demonstrating how the `executionContext` can be used to make environment-specific decisions in a realistic scenario.

How to Create a Custom Command from a Grails Plugin

You can create custom commands not only within your Grails application but also from a Grails plugin. Here's how to do it:

1. **Create a Grails Plugin:** If you don't already have a Grails plugin, you can create one using Grails' plugin generation commands. For example:

```
grails create-plugin my-plugin
```

2. **Define the Command:** Inside your Grails plugin, define the custom command by creating a Groovy class that implements the `GrailsApplicationCommand` trait or interface, providing the necessary methods like `getName()`, `getDescription()`, and `handle()`.
3. **Build and Package the Plugin:** To publish the plugin, you should use the Gradle maven-publish plugin. Update your plugin's `build.gradle` file to include the following configuration:

```
publishing {
    publications {
        mavenJava(MavenPublication) {
            from components.java
        }
    }
    repositories {
        maven {
            url "file://path/to/your/local/repo" // Adjust the path accordingly
        }
    }
}
```

Then, you can publish the plugin to your local repository:

```
./gradlew publishToMavenLocal
```

4. **Add the Plugin as a Dependency:** Instead of using the `grails install-plugin` command, you should add the plugin as a dependency in your Grails application's `build.gradle` file. Include the following dependency:

```
dependencies {
    // ...
    implementation 'com.example:my-plugin:1.0.0' // Replace with your plugin's group and version
    // ...
}
```

Make sure to replace `"com.example:my-plugin:1.0.0"` with the appropriate group and version for your plugin

5. **Run the Custom Command:** Now, you can run the custom command from your Grails application's root directory using the Gradle `runCommand` task, as previously explained:

```
./gradlew runCommand -Pargs="your-command-name"
```

Replace `"your-command-name"` with the name of the custom command you defined in your plugin.

By following these steps, you can create and run custom commands from a Grails plugin, extending the functionality of your Grails application as needed. This approach allows you to modularize your custom functionality and share it across multiple Grails projects if necessary.

5.3 Creating a Grails Project

Creating a project is the primary usage of the CLI. The primary command for creating a new project is `create-app`, which creates a standard Grails web application that communicates over HTTP. For other types of application, see the documentation below.

Command	Description	Options
<code>create-app / create-webapp</code>	Creates a Grails web application	• <code>-jdk, --java-version</code> • <code>-s, --servlet</code> • <code>-g, --gorm</code> • <code>-t, --test</code> • <code>-f, --features</code> • <code>-i, --inplace</code>
<code>create-restapi</code>	Creates a Grails REST API application	• <code>-jdk, --java-version</code> • <code>-s, --servlet</code> • <code>-g, --gorm</code> • <code>-t, --test</code> • <code>-f, --features</code> • <code>-i, --inplace</code>
<code>create-plugin</code>	Creates a Grails Plugin application	• <code>-jdk, --java-version</code> • <code>-s, --servlet</code> • <code>-g, --gorm</code> • <code>-t, --test</code> • <code>-f, --features</code> • <code>-i, --inplace</code>
<code>create-web-plugin</code>	Creates a Grails Web Plugin application	• <code>-jdk, --java-version</code> • <code>-s, --servlet</code> • <code>-g, --gorm</code> • <code>-t, --test</code> • <code>-f, --features</code> • <code>-i, --inplace</code>

| Flag | Description | Example

The `create-` command flags

The "create-*" commands are used to produce a fundamental Grails project, allowing for the inclusion of optional flags to select additional features, to customize GORM settings, an embedded servlet, the test framework, and the Java version.

Flag	Description	Example
<code>-jdk,</code> <code>--java-version</code>	The JDK version the project should target	<code>--java-version 11</code>
<code>-s, --servlet</code>	Which Servlet Implementation to configure. Possible values: none, tomcat, jetty, undertow.	<code>--servlet=tomcat</code>
<code>-g, --gorm</code>	Which GORM Implementation to configure. Possible values: hibernate, mongodb, neo4j.	<code>--gorm hibernate</code>
<code>-t, --test</code>	Which test framework to use. Possible values: junit, spock.	<code>--test spock</code>
<code>-f, --features</code>	The features to use. Possible values: h2, gorm-hibernate5, spring-boot-starter-jetty, springloaded, micronaut-http-client, cache-ehcache, hibernate-validator, postgres, mysql, cache, database-migration, grails-gsp, hamcrest, gorm-mongodb, assertj, mockito, spring-boot-starter-undertow, micronaut-inject-groovy, github-workflow-java-ci, jrebel, testcontainers, sqlserver, grails-console, views-markup, views-json, gorm-neo4j, asciidoctor, embedded-mongodb, grails-web-console,	<code>--features github-workflow-java-ci,micronaut-http-client</code>

logbackGroovy,
mongo-sync, shade,
properties

`-i, --inplace` Create a project using the
current directory `--inplace`

6 Object Relational Mapping (GORM)

Domain classes are core to any business application. They hold state about business processes and hopefully also implement behavior. They are linked together through relationships; one-to-one, one-to-many, or many-to-many.

GORM is Grails' object relational mapping (ORM) implementation. Under the hood it uses Hibernate (a very popular and flexible open source ORM solution) and thanks to the dynamic nature of Groovy with its static and dynamic typing, along with the convention of Grails, there is far less configuration involved in creating Grails domain classes.

You can also write Grails domain classes in Java. See the section on Hibernate Integration for how to write domain classes in Java but still use dynamic persistent methods. Below is a preview of GORM in action:

```
def book = Book.findByTitle("Groovy in Action")

book
  .addToAuthors(name:"Dierk Koenig")
  .addToAuthors(name:"Guillaume LaForge")
  .save()
```

6.1 Quick Start Guide

A domain class can be created with the [create-domain-class](#) command:

```
grails create-domain-class myapp.Person
```

If no package is specified with the create-domain-class script, Grails automatically uses the application name as the package name.

This will create a class at the location `grails-app/domain/myapp/Person.groovy` such as the one below:

```
package myapp

class Person {
}
```

If you have the `dbCreate` property set to "update", "create" or "create-drop" on your [DataSource](#), Grails will automatically generate/modify the database tables for you.

You can customize the class by adding properties:

```
class Person {
    String name
    Integer age
    Date lastVisit
}
```

Once you have a domain class try and manipulate it with the [shell](#) or [console](#) by typing:

grails console

This loads an interactive GUI where you can run Groovy commands with access to the Spring ApplicationContext, GORM, etc.

6.1.1 Basic CRUD

Try performing some basic CRUD (Create/Read/Update/Delete) operations.

Create

To create a domain class use Map constructor to set its properties and call [save](#):

```
def p = new Person(name: "Fred", age: 40, lastVisit: new Date())
p.save()
```

The [save](#) method will persist your class to the database using the underlying Hibernate ORM layer.

Read

Grails transparently adds an implicit `id` property to your domain class which you can use for retrieval:

```
def p = Person.get(1)
assert 1 == p.id
```

This uses the [get](#) method that expects a database identifier to read the `Person` object back from the database. You can also load an object in a read-only state by using the [read](#) method:

```
def p = Person.read(1)
```

In this case the underlying Hibernate engine will not do any dirty checking and the object will not be persisted. Note that if you explicitly call the [save](#) method then the object is placed back into a read-write state.

In addition, you can also load a proxy for an instance by using the [load](#) method:

```
def p = Person.load(1)
```

This incurs no database access until a method other than `getId()` is called. Hibernate then initializes the proxied instance, or throws an exception if no record is found for the specified id.

Update

To update an instance, change some properties and then call [save](#) again:

```
def p = Person.get(1)
p.name = "Bob"
p.save()
```

Delete

To delete an instance use the [delete](#) method:

```
def p = Person.get(1)
p.delete()
```

6.2 Further Reading on GORM

For more information on using GORM see the [dedicated documentation](#) for the GORM project.

7 The Web Layer

7.1 Controllers

A controller handles requests and creates or prepares the response. A controller can generate the response directly or delegate to a view. To create a controller, simply create a class whose name ends with `Controller` in the `grails-app/controllers` directory (in a subdirectory if it's in a package).

The default [URL Mapping](#) configuration ensures that the first part of your controller name is mapped to a URI and each action defined within your controller maps to URIs within the controller name URI.

7.1.1 Understanding Controllers and Actions

Creating a controller

Controllers can be created with the [create-controller](#) or [generate-controller](#) command. For example try running the following command from the root of a Grails project:

```
grails create-controller book
```

The command will create a controller at the location

```
grails-app/controllers/myapp/BookController.groovy:
```

```
package myapp

class BookController {

    def index() { }
}
```

where "myapp" will be the name of your application, the default package name if one isn't specified.

`BookController` by default maps to the `/book` URI (relative to your application root).

The `create-controller` and `generate-controller` commands are just for convenience and you can just as easily create controllers using your favorite text editor or IDE

Creating Actions

A controller can have multiple public action methods; each one maps to a URI:

```
class BookController {

    def list() {

        // do controller logic
        // create model
    }
}
```

```

    return model
}
}

```

This example maps to the `/book/list` URI by default thanks to the property being named `list`.

The Default Action

A controller has the concept of a default URI that maps to the root URI of the controller, for example `/book` for `BookController`. The action that is called when the default URI is requested is dictated by the following rules:

- If there is only one action, it's the default
- If you have an action named `index`, it's the default
- Alternatively you can set it explicitly with the `defaultAction` property:

```
static defaultAction = "list"
```

7.1.2 Controllers and Scopes

Available Scopes

Scopes are hash-like objects where you can store variables. The following scopes are available to controllers:

- [servletContext](#) - Also known as application scope, this scope lets you share state across the entire web application. The `servletContext` is an instance of [ServletContext](#)
- [session](#) - The session allows associating state with a given user and typically uses cookies to associate a session with a client. The session object is an instance of [HttpSession](#)
- [request](#) - The request object allows the storage of objects for the current request only. The request object is an instance of [HttpServletRequest](#)
- [params](#) - Mutable map of incoming request query string or POST parameters
- [flash](#) - See below

Accessing Scopes

Scopes can be accessed using the variable names above in combination with Groovy's array index operator, even on classes provided by the Servlet API such as the [HttpServletRequest](#):

```

class BookController {
    def find() {
        def findBy = params["findBy"]
        def appContext = request["foo"]
        def loggedUser = session["logged_user"]
    }
}

```

You can also access values within scopes using the de-reference operator, making the syntax even more clear:

```

class BookController {
    def find() {
        def findBy = params.findBy
        def appContext = request.foo
    }
}

```



```

    def loggedUser = session.logged_user
  }
}

```

This is one of the ways that Grails unifies access to the different scopes.

Using Flash Scope

Grails supports the concept of [flash](#) scope as a temporary store to make attributes available for this request and the next request only. Afterwards the attributes are cleared. This is useful for setting a message directly before redirecting, for example:

```

def delete() {
  def b = Book.get(params.id)
  if (!b) {
    flash.message = "User not found for id ${params.id}"
    redirect(action: list)
  }
  ... // remaining code
}

```

When the `delete` action is requested, the `message` value will be in scope and can be used to display an information message. It will be removed from the `flash` scope after this second request.

Note that the attribute name can be anything you want, and the values are often strings used to display messages, but can be any object type.

Scoped Controllers

Newly created applications have the `grails.controllers.defaultScope` property set to a value of "singleton" in `application.yml`. You may change this value to any of the supported scopes listed below. If the property is not assigned a value at all, controllers will default to "prototype" scope.

Supported controller scopes are:

- `prototype` (default) - A new controller will be created for each request (recommended for actions as Closure properties)
- `session` - One controller is created for the scope of a user session
- `singleton` - Only one instance of the controller ever exists (recommended for actions as methods)

To enable one of the scopes, add a static `scope` property to your class with one of the valid scope values listed above, for example

```
static scope = "singleton"
```

You can define the default strategy in `application.yml` with the `grails.controllers.defaultScope` key, for example:

```

grails:
  controllers:
    defaultScope: singleton

```

Use scoped controllers wisely. For instance, we don't recommend having any properties in a singleton-scoped controller since they will be shared for *all* requests.

7.1.3 Models and Views

Returning the Model

A model is a Map that the view uses when rendering. The keys within that Map correspond to variable names accessible by the view. There are a couple of ways to return a model. First, you can explicitly return a Map instance:

```
def show() {  
    [book: Book.get(params.id)]  
}
```

The above does *not* reflect what you should use with the scaffolding views - see the [scaffolding section](#) for more details.

A more advanced approach is to return an instance of the Spring [ModelAndView](#) class:

```
import org.springframework.web.servlet.ModelAndView  
  
def index() {  
    // get some books just for the index page, perhaps your favorites  
    def favoriteBooks = ...  
  
    // forward to the list view to show them  
    return new ModelAndView("/book/list", [ bookList : favoriteBooks ])  
}
```

One thing to bear in mind is that certain variable names can not be used in your model:

- attributes
- application

Currently, no error will be reported if you do use them, but this will hopefully change in a future version of Grails.

Selecting the View

In both of the previous two examples there was no code that specified which [view](#) to render. So how does Grails know which one to pick? The answer lies in the conventions. Grails will look for a view at the location `grails-app/views/book/show.gsp` for this `show` action:

```
class BookController {  
    def show() {  
        [book: Book.get(params.id)]  
    }  
}
```

To render a different view, use the [render](#) method:

```
def show() {  
    def map = [book: Book.get(params.id)]  
    render(view: "display", model: map)  
}
```

In this case Grails will attempt to render a view at the location `grails-app/views/book/display.gsp`. Notice that Grails automatically qualifies the view location with the `book` directory of the `grails-app/views` directory. This is convenient, but to access shared views, you use an absolute path instead of a relative one:

```
def show() {  
    def map = [book: Book.get(params.id)]  
    render(view: "/shared/display", model: map)  
}
```

In this case Grails will attempt to render a view at the location `grails-app/views/shared/display.gsp`.

Grails also supports JSPs as views, so if a GSP isn't found in the expected location but a JSP

is, it will be used instead.

Unlike GSPs, JSPs must be located in the directory path

`/src/main/webapp/WEB-INF/grails-app/views.`

Additionally, to ensure JSPs work as intended, don't forget to include the required dependencies for JSP and JSTL implementations in your `build.gradle` file.

Selecting Views For Namespaced Controllers

If a controller defines a namespace for itself with the [namespace](#) property that will affect the root directory in which Grails will look for views which are specified with a relative path.

The default root directory for views rendered by a namespaced controller is `grails-app/views/<namespace name>/<controller name>.` If the view is not found in the namespaced directory then Grails will fallback to looking for the view in the non-namespaced directory.

See the example below.

```
class ReportingController {
    static namespace = 'business'

    def humanResources() {
        // This will render grails-app/views/business/reporting/humanResources.gsp
        // if it exists.

        // If grails-app/views/business/reporting/humanResources.gsp does not
        // exist the fallback will be grails-app/views/reporting/humanResources.gsp.

        // The namespaced GSP will take precedence over the non-namespaced GSP.

        [numberOfEmployees: 9]
    }

    def accountsReceivable() {
        // This will render grails-app/views/business/reporting/numberCrunch.gsp
        // if it exists.

        // If grails-app/views/business/reporting/numberCrunch.gsp does not
        // exist the fallback will be grails-app/views/reporting/numberCrunch.gsp.

        // The namespaced GSP will take precedence over the non-namespaced GSP.

        render view: 'numberCrunch', model: [numberOfEmployees: 13]
    }
}
```

Rendering a Response

Sometimes it's easier (for example with Ajax applications) to render snippets of text or code to the response directly from the controller. For this, the highly flexible `render` method can be used:

```
render "Hello World!"
```

The above code writes the text "Hello World!" to the response. Other examples include:

```
// write some markup
render {
    for (b in books) {
        div(id: b.id, b.title)
    }
}

// render a specific view
render(view: 'show')

// render a template for each item in a collection
render(template: 'book_template', collection: Book.list())

// render some text with encoding and content type
render(text: "<xml>some xml</xml>", contentType: "text/xml", encoding: "UTF-8")
```

If you plan on using Groovy's `MarkupBuilder` to generate HTML for use with the `render` method be careful of naming clashes between HTML elements and Grails tags, for example:

```
import groovy.xml.MarkupBuilder

...
def login() {
    def writer = new StringWriter()
    def builder = new MarkupBuilder(writer)
    builder.html {
        head {
            title 'Log in'
        }
        body {
            h1 'Hello'
            form {
            }
        }
    }

    def html = writer.toString()
    render html
}
```

This will actually [call the form tag](#) (which will return some text that will be ignored by the `MarkupBuilder`). To correctly output a `<form>` element, use the following:

```
def login() {
    // ...
    body {
        h1 'Hello'
        builder.form {
        }
    }
    // ...
}
```

7.1.4 Redirects and Chaining

Redirects

Actions can be redirected using the [redirect](#) controller method:

```
class OverviewController {

    def login() {}

    def find() {
        if (!session.user)
            redirect(action: 'login')
        return
    }
    ...
}
```

Internally the [redirect](#) method uses the [HttpServletResponse](#) object's `sendRedirect` method.

The `redirect` method expects one of:

- The name of an action (and controller name if the redirect isn't to an action in the current controller):

```
// Also redirects to the index action in the home controller
redirect(controller: 'home', action: 'index')
```

- A URI for a resource relative the application context path:

```
// Redirect to an explicit URI
redirect(uri: "/login.html")
```

- Or a full URL:

```
// Redirect to a URL
redirect(url: "http://grails.org")
```

- A [domain class](#) instance:

```
// Redirect to the domain instance
Book book = ... // obtain a domain instance
redirect book
```

In the above example Grails will construct a link using the domain class `id` (if present).

Parameters can optionally be passed from one action to the next using the `params` argument of the method:

```
redirect(action: 'myaction', params: [myparam: "myvalue"])
```

These parameters are made available through the [params](#) dynamic property that accesses request parameters. If a parameter is specified with the same name as a request parameter, the request parameter is overridden and the controller parameter is used.

Since the `params` object is a Map, you can use it to pass the current request parameters from one action to the next:

```
redirect(action: "next", params: params)
```

Finally, you can also include a fragment in the target URI:

```
redirect(controller: "test", action: "show", fragment: "profile")
```

which will (depending on the URL mappings) redirect to something like `/myapp/test/show#profile`.

Chaining

Actions can also be chained. Chaining allows the model to be retained from one action to the next. For example calling the `first` action in this action:

```
class ExampleChainController {
    def first() {
        chain(action: second, model: [one: 1])
    }
    def second () {
        chain(action: third, model: [two: 2])
    }
    def third() {
        [three: 3]
    }
}
```

results in the model:

```
[one: 1, two: 2, three: 3]
```

The model can be accessed in subsequent controller actions in the chain using the `chainModel` map. This dynamic property only exists in actions following the call to the `chain` method:

```
class ChainController {
    def nextInChain() {
        def model = chainModel.myModel
        ...
    }
}
```

Like the `redirect` method you can also pass parameters to the `chain` method:

```
chain(action: "action1", model: [one: 1], params: [myparam: "param1"])
```

The chain method uses the HTTP session and hence should only be used if your application is stateful.

7.1.5 Data Binding

Data binding is the act of "binding" incoming request parameters onto the properties of an object or an entire graph of objects. Data binding should deal with all necessary type conversion since request parameters, which are typically delivered by a form submission, are always strings whilst the properties of a Groovy or Java object may well not be.

Map Based Binding

The data binder is capable of converting and assigning values in a Map to properties of an object. The binder will associate entries in the Map to properties of the object using the keys in the Map that have values which correspond to property names on the object. The following code demonstrates the basics:

grails-app/domain/Person.groovy

```
class Person {
    String firstName
    String lastName
    Integer age
}

def bindingMap = [firstName: 'Peter', lastName: 'Gabriel', age: 63]

def person = new Person(bindingMap)

assert person.firstName == 'Peter'
assert person.lastName == 'Gabriel'
assert person.age == 63
```

To update properties of a domain object you may assign a Map to the `properties` property of the domain class:

```
def bindingMap = [firstName: 'Peter', lastName: 'Gabriel', age: 63]

def person = Person.get(someId)
person.properties = bindingMap

assert person.firstName == 'Peter'
assert person.lastName == 'Gabriel'
assert person.age == 63
```

The binder can populate a full graph of objects using Maps of Maps.

```
class Person {
    String firstName
    String lastName
    Integer age
    Address homeAddress
}

class Address {
    String county
    String country
}

def bindingMap = [firstName: 'Peter', lastName: 'Gabriel', age: 63, homeAddress: [county: 'Surrey', country: 'England']]

def person = new Person(bindingMap)

assert person.firstName == 'Peter'
assert person.lastName == 'Gabriel'
assert person.age == 63
assert person.homeAddress.county == 'Surrey'
assert person.homeAddress.country == 'England'
```

Binding To Collections And Maps

The data binder can populate and update Collections and Maps. The following code shows a simple example of populating a `List` of objects in a domain class:

```
class Band {
    String name
    static hasMany = [albums: Album]
    List albums
}

class Album {
    String title
    Integer numberOfTracks
}

def bindingMap = [name: 'Genesis',
                  'albums[0]': [title: 'Foxtrot', numberOfTracks: 6],
                  'albums[1]': [title: 'Nursery Cryme', numberOfTracks: 7]]

def band = new Band(bindingMap)

assert band.name == 'Genesis'
assert band.albums.size() == 2
assert band.albums[0].title == 'Foxtrot'
assert band.albums[0].numberOfTracks == 6
assert band.albums[1].title == 'Nursery Cryme'
assert band.albums[1].numberOfTracks == 7
```

That code would work in the same way if `albums` were an array instead of a `List`.

Note that when binding to a `Set` the structure of the `Map` being bound to the `Set` is the same as that of a `Map` being bound to a `List` but since a `Set` is unordered, the indexes don't necessarily correspond to the order of elements in the `Set`. In the code example above, if `albums` were a `Set` instead of a `List`, the `bindingMap` could look exactly the same but 'Foxtrot' might be the first album in the `Set` or it might be the second. When updating existing elements in a `Set` the `Map` being assigned to the `Set` must have `id` elements in it which represent the element in the `Set` being updated, as in the following example:

```
/*
 * The value of the indexes 0 and 1 in albums[0] and albums[1] are arbitrary
 * values that can be anything as long as they are unique within the Map.
 * They do not correspond to the order of elements in albums because albums
 * is a Set.
 */
def bindingMap = ['albums[0]': [id: 9, title: 'The Lamb Lies Down On Broadway'],
                  'albums[1]': [id: 4, title: 'Selling England By The Pound']]

def band = Band.get(someBandId)

/*
 * This will find the Album in albums that has an id of 9 and will set its title
 * to 'The Lamb Lies Down On Broadway' and will find the Album in albums that has
 * an id of 4 and set its title to 'Selling England By The Pound'. In both
 * cases if the Album cannot be found in albums then the album will be retrieved
 * from the database by id, the Album will be added to albums and will be updated
 * with the values described above. If a Album with the specified id cannot be
 * found in the database, then a binding error will be created and associated
 * with the band object. More on binding errors later.
 */
band.properties = bindingMap
```

When binding to a `Map` the structure of the binding `Map` is the same as the structure of a `Map` used for binding to a `List` or a `Set` and the index inside of square brackets corresponds to the key in the `Map` being bound to. See the following code:

```
class Album {
    String title
    static hasMany = [players: Player]
    Map players
}

class Player {
    String name
}

def bindingMap = [title: 'The Lamb Lies Down On Broadway',
                  'players[guitar]': [name: 'Steve Hackett'],
                  'players[vocals]': [name: 'Peter Gabriel'],
                  'players[keyboards]': [name: 'Tony Banks']]

def album = new Album(bindingMap)
```

```

assert album.title == 'The Lamb Lies Down On Broadway'
assert album.players.size() == 3
assert album.players.guitar.name == 'Steve Hackett'
assert album.players.vocals.name == 'Peter Gabriel'
assert album.players.keyboards.name == 'Tony Banks'

```

When updating an existing `Map`, if the key specified in the binding `Map` does not exist in the `Map` being bound to then a new value will be created and added to the `Map` with the specified key as in the following example:

```

def bindingMap = [title: 'The Lamb Lies Down On Broadway',
                  'players[guitar]': [name: 'Steve Hackett'],
                  'players[vocals]': [name: 'Peter Gabriel'],
                  'players[keyboards]': [name: 'Tony Banks']]

def album = new Album(bindingMap)

assert album.title == 'The Lamb Lies Down On Broadway'
assert album.players.size() == 3
assert album.players.guitar.name == 'Steve Hackett'
assert album.players.vocals.name == 'Peter Gabriel'
assert album.players.keyboards.name == 'Tony Banks'

def updatedBindingMap = ['players[drums]': [name: 'Phil Collins'],
                        'players[keyboards]': [name: 'Anthony George Banks']]

album.properties = updatedBindingMap

assert album.title == 'The Lamb Lies Down On Broadway'
assert album.players.size() == 4
assert album.players.guitar.name == 'Steve Hackett'
assert album.players.vocals.name == 'Peter Gabriel'
assert album.players.keyboards.name == 'Anthony George Banks'
assert album.players.drums.name == 'Phil Collins'

```

Binding Request Data to the Model

The [params](#) object that is available in a controller has special behavior that helps convert dotted request parameter names into nested `Maps` that the data binder can work with. For example, if a request includes request parameters named `person.homeAddress.country` and `person.homeAddress.city` with values 'USA' and 'St. Louis' respectively, `params` would include entries like these:

```
[person: [homeAddress: [country: 'USA', city: 'St. Louis']]]
```

There are two ways to bind request parameters onto the properties of a domain class. The first involves using a domain classes' `Map` constructor:

```

def save() {
    def b = new Book(params)
    b.save()
}

```

The data binding happens within the code `new Book(params)`. By passing the [params](#) object to the domain class constructor Grails automatically recognizes that you are trying to bind from request parameters. So if we had an incoming request like:

```
/book/save?title=The%20Stand&author=Stephen%20King
```

Then the `title` and `author` request parameters would automatically be set on the domain class. You can use the [properties](#) property to perform data binding onto an existing instance:

```

def save() {
    def b = Book.get(params.id)
    b.properties = params
    b.save()
}

```

This has the same effect as using the implicit constructor.

When binding an empty `String` (a `String` with no characters in it, not even spaces), the data binder will convert the empty `String` to null. This simplifies the most common case where

the intent is to treat an empty form field as having the value null since there isn't a way to actually submit a null as a request parameter. When this behavior is not desirable the application may assign the value directly.

The mass property binding mechanism will by default automatically trim all Strings at binding time. To disable this behavior set the `grails.databinding.trimStrings` property to false in `grails-app/conf/application.groovy`.

```
// the default value is true
grails.databinding.trimStrings = false

// ...
```

The mass property binding mechanism will by default automatically convert all empty Strings to null at binding time. To disable this behavior set the `grails.databinding.convertEmptyStringsToNull` property to false in `grails-app/conf/application.groovy`.

```
// the default value is true
grails.databinding.convertEmptyStringsToNull = false

// ...
```

The order of events is that the String trimming happens and then null conversion happens so if `trimStrings` is true and `convertEmptyStringsToNull` is true, not only will empty Strings be converted to null but also blank Strings. A blank String is any String such that the `trim()` method returns an empty String.

These forms of data binding in Grails are very convenient, but also indiscriminate. In other words, they will bind *all* non-transient, typed instance properties of the target object, including ones that you may not want bound. Just because the form in your UI doesn't submit all the properties, an attacker can still send malign data via a raw HTTP request. Fortunately, Grails also makes it easy to protect against such attacks - see the section titled "Data Binding and Security concerns" for more information.

Data binding and Single-ended Associations

If you have a one-to-one or many-to-one association you can use Grails' data binding capability to update these relationships too. For example if you have an incoming request such as:

```
/book/save?author.id=20
```

Grails will automatically detect the `.id` suffix on the request parameter and look up the `Author` instance for the given id when doing data binding such as:

```
def b = new Book(params)
```

An association property can be set to null by passing the literal string "null". For example:

```
/book/save?author.id=null
```

Data Binding and Many-ended Associations

If you have a one-to-many or many-to-many association there are different techniques for data binding depending of the association type.

If you have a set based association (the default for a `hasMany`) then the simplest way to populate an association is to send a list of identifiers. For example consider the usage of `<g:select>` below:

```
<g:select name="books"
         from="${Book.list()}"
```

```
size="5" multiple="yes" optionKey="id"
value="${author?.books}" />
```

This produces a select box that lets you select multiple values. In this case if you submit the form Grails will automatically use the identifiers from the select box to populate the `books` association.

However, if you have a scenario where you want to update the properties of the associated objects the this technique won't work. Instead you use the subscript operator:

```
<g:textField name="books[0].title" value="the Stand" />
<g:textField name="books[1].title" value="the Shining" />
```

However, with `set` based association it is critical that you render the mark-up in the same order that you plan to do the update in. This is because a `set` has no concept of order, so although we're referring to `books[0]` and `books[1]` it is not guaranteed that the order of the association will be correct on the server side unless you apply some explicit sorting yourself.

This is not a problem if you use `List` based associations, since a `List` has a defined order and an index you can refer to. This is also true of `Map` based associations.

Note also that if the association you are binding to has a size of two and you refer to an element that is outside the size of association:

```
<g:textField name="books[0].title" value="the Stand" />
<g:textField name="books[1].title" value="the Shining" />
<g:textField name="books[2].title" value="Red Madder" />
```

Then Grails will automatically create a new instance for you at the defined position.

You can bind existing instances of the associated type to a `List` using the same `.id` syntax as you would use with a single-ended association. For example:

```
<g:select name="books[0].id" from="${bookList}"
value="${author?.books[0]?.id}" />

<g:select name="books[1].id" from="${bookList}"
value="${author?.books[1]?.id}" />

<g:select name="books[2].id" from="${bookList}"
value="${author?.books[2]?.id}" />
```

Would allow individual entries in the `books List` to be selected separately.

Entries at particular indexes can be removed in the same way too. For example:

```
<g:select name="books[0].id"
from="${Book.list()}"
value="${author?.books[0]?.id}"
noSelection="['null': '']"/>
```

Will render a select box that will remove the association at `books[0]` if the empty option is chosen.

Binding to a `Map` property works the same way except that the list index in the parameter name is replaced by the map key:

```
<g:select name="images[cover].id"
from="${Image.list()}"
value="${book?.images[cover]?.id}"
noSelection="['null': '']"/>
```

This would bind the selected image into the `Map` property `images` under a key of `"cover"`.

When binding to `Maps`, `Arrays` and `Collections` the data binder will automatically grow the size of the collections as necessary.

The default limit to how large the binder will grow a collection is 256. If the data binder encounters an entry that requires the collection be grown beyond that limit, the entry is ignored. The limit may be configured by assigning a value to the

`grails.databinding.autoGrowCollectionLimit` property in `application.groovy`.

`grails-app/conf/application.groovy`

```
// the default value is 256
grails.databinding.autoGrowCollectionLimit = 128

// ...
```

Data binding with Multiple domain classes

It is possible to bind data to multiple domain objects from the [params](#) object.

For example so you have an incoming request to:

```
/book/save?book.title=The%20Stand&author.name=Stephen%20King
```

You'll notice the difference with the above request is that each parameter has a prefix such as `author.` or `book.` which is used to isolate which parameters belong to which type. Grails' `params` object is like a multi-dimensional hash and you can index into it to isolate only a subset of the parameters to bind.

```
def b = new Book(params.book)
```

Notice how we use the prefix before the first dot of the `book.title` parameter to isolate only parameters below this level to bind. We could do the same with an `Author` domain class:

```
def a = new Author(params.author)
```

Data Binding and Action Arguments

Controller action arguments are subject to request parameter data binding. There are 2 categories of controller action arguments. The first category is command objects. Complex types are treated as command objects. See the [Command Objects](#) section of the user guide for details. The other category is basic object types. Supported types are the 8 primitives, their corresponding type wrappers and [java.lang.String](#). The default behavior is to map request parameters to action arguments by name:

```
class AccountingController {

    // accountNumber will be initialized with the value of params.accountNumber
    // accountType will be initialized with params.accountType
    def displayInvoice(String accountNumber, int accountType) {
        // ...
    }
}
```

For primitive arguments and arguments which are instances of any of the primitive type wrapper classes a type conversion has to be carried out before the request parameter value can be bound to the action argument. The type conversion happens automatically. In a case like the example shown above, the `params.accountType` request parameter has to be converted to an `int`. If type conversion fails for any reason, the argument will have its default value per normal Java behavior (null for type wrapper references, false for booleans and zero for numbers) and a corresponding error will be added to the `errors` property of the defining controller.

```
/accounting/displayInvoice?accountNumber=B59786&accountType=bogusValue
```

Since "bogusValue" cannot be converted to type `int`, the value of `accountType` will be zero, the controller's `errors.hasErrors()` will be true, the controller's `errors.errorCount` will be equal

to 1 and the controller's `errors.getFieldError('accountType')` will contain the corresponding error.

If the argument name does not match the name of the request parameter then the `@grails.web.RequestParameter` annotation may be applied to an argument to express the name of the request parameter which should be bound to that argument:

```
import grails.web.RequestParameter

class AccountingController {

    // mainAccountNumber will be initialized with the value of params.accountNumber
    // accountType will be initialized with params.accountType
    def displayInvoice(@RequestParameter('accountNumber') String mainAccountNumber, int accountType) {
        // ...
    }
}
```

Data binding and type conversion errors

Sometimes when performing data binding it is not possible to convert a particular String into a particular target type. This results in a type conversion error. Grails will retain type conversion errors inside the [errors](#) property of a Grails domain class. For example:

```
class Book {
    ...
    URL publisherURL
}
```

Here we have a domain class `Book` that uses the `java.net.URL` class to represent URLs. Given an incoming request such as:

```
/book/save?publisherURL=a-bad-url
```

it is not possible to bind the string `a-bad-url` to the `publisherURL` property as a type mismatch error occurs. You can check for these like this:

```
def b = new Book(params)

if (b.hasErrors()) {
    println "The value ${b.errors.getFieldError('publisherURL').rejectedValue}" +
        " is not a valid URL!"
}
```

Although we have not yet covered error codes (for more information see the section on [validation](#)), for type conversion errors you would want a message from the `grails-app/i18n/messages.properties` file to use for the error. You can use a generic error message handler such as:

```
typeMismatch.java.net.URL=The field {0} is not a valid URL
```

Or a more specific one:

```
typeMismatch.Book.publisherURL=The publisher URL you specified is not a valid URL
```

The BindUsing Annotation

The [BindUsing](#) annotation may be used to define a custom binding mechanism for a particular field in a class. Any time data binding is being applied to the field the closure value of the annotation will be invoked with 2 arguments. The first argument is the object that data binding is being applied to and the second argument is [DataBindingSource](#) which is the data source for the data binding. The value returned from the closure will be bound to the property. The following example would result in the upper case version of the `name` value in the source being applied to the `name` field during data binding.

```
import grails.databinding.BindUsing
```

```

class SomeClass {
    @BindUsing({obj, source ->

        //source is DataSourceBinding which is similar to a Map
        //and defines getAt operation but source.name cannot be used here.
        //In order to get name from source use getAt instead as shown below.

        source['name']?.toUpperCase()
    })
    String name
}

```

Note that data binding is only possible when the name of the request parameter matches with the field name in the class. Here, `name` from request parameters matches with `name` from `SomeClass`.

The [BindUsing](#) annotation may be used to define a custom binding mechanism for all of the fields on a particular class. When the annotation is applied to a class, the value assigned to the annotation should be a class which implements the [BindingHelper](#) interface. An instance of that class will be used any time a value is bound to a property in the class that this annotation has been applied to.

```

@BindUsing(SomeClassWhichImplementsBindingHelper)
class SomeClass {
    String someProperty
    Integer someOtherProperty
}

```

The BindInitializer Annotation

The [BindInitializer](#) annotation may be used to initialize an associated field in a class if it is undefined. Unlike the [BindUsing](#) annotation, databinding will continue binding all nested properties on this association.

```

import grails.databinding.BindInitializer

class Account{}

class User {
    Account account

    // BindInitializer expects you to return a instance of the type
    // where it's declared on. You can use source as a parameter, in this case user.
    @BindInitializer({user-> new Contact(account:user.account) })
    Contact contact
}

class Contact{
    Account account
    String firstName
}

```

`@BindInitializer` only makes sense for associated entities, as per this use case.

Custom Data Converters

The binder will do a lot of type conversion automatically. Some applications may want to define their own mechanism for converting values and a simple way to do this is to write a class which implements [ValueConverter](#) and register an instance of that class as a bean in the Spring application context.

```

package com.myapp.converters

import grails.databinding.converters.ValueConverter

/**
 * A custom converter which will convert String of the
 * form 'city:state' into an Address object.
 */
class AddressValueConverter implements ValueConverter {

    boolean canConvert(value) {
        value instanceof String
    }

    def convert(value) {

```

```

        def pieces = value.split(':')
        new com.myapp.Address(city: pieces[0], state: pieces[1])
    }

    Class<?> getTargetType() {
        com.myapp.Address
    }
}

```

An instance of that class needs to be registered as a bean in the Spring application context. The bean name is not important. All beans that implemented `ValueConverter` will be automatically plugged in to the data binding process.

grails-app/conf/spring/resources.groovy

```

beans = {
    addressValueConverter com.myapp.converters.AddressValueConverter
    // ...
}

class Person {
    String firstName
    Address homeAddress
}

class Address {
    String city
    String state
}

def person = new Person()
person.properties = [firstName: 'Jeff', homeAddress: "O'Fallon:Missouri"]
assert person.firstName == 'Jeff'
assert person.homeAddress.city = "O'Fallon"
assert person.homeAddress.state = 'Missouri'

```

Date Formats For Data Binding

A custom date format may be specified to be used when binding a String to a Date value by applying the [BindingFormat](#) annotation to a Date field.

```

import grails.databinding.BindingFormat

class Person {
    @BindingFormat('MMddyyyy')
    Date birthDate
}

```

A global setting may be configured in `application.groovy` to define date formats which will be used application wide when binding to Date.

grails-app/conf/application.groovy

```

grails.databinding.dateFormats = ['MMddyyyy', 'yyyy-MM-dd HH:mm:ss.S', "yyyy-MM-dd'T'hh:mm:ss'Z'"]

```

The formats specified in `grails.databinding.dateFormats` will be attempted in the order in which they are included in the List. If a property is marked with `@BindingFormat`, the `@BindingFormat` will take precedence over the values specified in `grails.databinding.dateFormats`.

The formats configured by default are:

- yyyy-MM-dd HH:mm:ss.S
- yyyy-MM-dd'T'hh:mm:ss'Z'
- yyyy-MM-dd HH:mm:ss.S z
- yyyy-MM-dd'T'HH:mm:ss.SSSX

Custom Formatted Converters

You may supply your own handler for the [BindingFormat](#) annotation by writing a class which implements the [FormattedValueConverter](#) interface and registering an instance of that class as a bean in the Spring application context. Below is an example of a trivial custom String formatter that might convert the case of a String based on the value assigned to the BindingFormat annotation.

```
package com.myapp.converters

import grails.databinding.converters.FormattedValueConverter

class FormattedStringValueConverter implements FormattedValueConverter {
    def convert(value, String format) {
        if('UPPERCASE' == format) {
            value = value.toUpperCase()
        } else if('LOWERCASE' == format) {
            value = value.toLowerCase()
        }
        value
    }

    Class getTargetType() {
        // specifies the type to which this converter may be applied
        String
    }
}
```

An instance of that class needs to be registered as a bean in the Spring application context. The bean name is not important. All beans that implemented FormattedValueConverter will be automatically plugged in to the data binding process.

grails-app/conf/spring/resources.groovy

```
beans = {
    formattedStringConverter com.myapp.converters.FormattedStringValueConverter
    // ...
}
```

With that in place the BindingFormat annotation may be applied to String fields to inform the data binder to take advantage of the custom converter.

```
import grails.databinding.BindingFormat

class Person {
    @BindingFormat('UPPERCASE')
    String someUpperCaseString

    @BindingFormat('LOWERCASE')
    String someLowerCaseString

    String someOtherString
}
```

Localized Binding Formats

The BindingFormat annotation supports localized format strings by using the optional code attribute. If a value is assigned to the code attribute that value will be used as the message code to retrieve the binding format string from the messageSource bean in the Spring application context and that lookup will be localized.

```
import grails.databinding.BindingFormat

class Person {
    @BindingFormat(code='date.formats.birthdays')
    Date birthDate
}

# grails-app/conf/i18n/messages.properties
date.formats.birthdays=MMddyyyy

# grails-app/conf/i18n/messages_es.properties
date.formats.birthdays=ddMMyyyy
```

Structured Data Binding Editors

A structured data binding editor is a helper class which can bind structured request

parameters to a property. The common use case for structured binding is binding to a `Date` object which might be constructed from several smaller pieces of information contained in several request parameters with names like `birthday_month`, `birthday_date` and `birthday_year`. The structured editor would retrieve all of those individual pieces of information and use them to construct a `Date`.

The framework provides a structured editor for binding to `Date` objects. An application may register its own structured editors for whatever types are appropriate. Consider the following classes:

```
src/main/groovy/databinding/Gadget.groovy
```

```
package databinding

class Gadget {
    Shape expandedShape
    Shape compressedShape
}
```

```
src/main/groovy/databinding/Shape.groovy
```

```
package databinding

class Shape {
    int area
}
```

A `Gadget` has 2 `Shape` fields. A `Shape` has an `area` property. It may be that the application wants to accept request parameters like `width` and `height` and use those to calculate the `area` of a `Shape` at binding time. A structured binding editor is well suited for that.

The way to register a structured editor with the data binding process is to add an instance of the [grails.databinding.TypedStructuredBindingEditor](#) interface to the Spring application context. The easiest way to implement the `TypedStructuredBindingEditor` interface is to extend the [org.grails.databinding.converters.AbstractStructuredBindingEditor](#) abstract class and override the `getPropertyValue` method as shown below:

```
src/main/groovy/databinding/converters/StructuredShapeEditor.groovy
```

```
package databinding.converters

import databinding.Shape

import org.grails.databinding.converters.AbstractStructuredBindingEditor

class StructuredShapeEditor extends AbstractStructuredBindingEditor<Shape> {

    public Shape getPropertyValue(Map values) {
        // retrieve the individual values from the Map
        def width = values.width as int
        def height = values.height as int

        // use the values to calculate the area of the Shape
        def area = width * height

        // create and return a Shape with the appropriate area
        new Shape(area: area)
    }
}
```

An instance of that class needs to be registered with the Spring application context:

```
grails-app/conf/spring/resources.groovy
```

```
beans = {
    shapeEditor databinding.converters.StructuredShapeEditor
    // ...
}
```

When the data binder binds to an instance of the `Gadget` class it will check to see if there are request parameters with names `compressedShape` and `expandedShape` which have a value of "struct" and if they do exist, that will trigger the use of the `StructuredShapeEditor`. The individual components of the structure need to have parameter names of the form

propertyName_structuredElementName. In the case of the `Gadget` class above that would mean that the `compressedShape` request parameter should have a value of "struct" and the `compressedShape_width` and `compressedShape_height` parameters should have values which represent the width and the height of the compressed shape. Similarly, the `expandedShape` request parameter should have a value of "struct" and the `expandedShape_width` and `expandedShape_height` parameters should have values which represent the width and the height of the expanded Shape.

grails-app/controllers/demo/DemoController.groovy

```
class DemoController {

    def createGadget(Gadget gadget) {
        /*
         /demo/createGadget?expandedShape=struct&expandedShape_width=80&expandedShape_height=30
         &compressedShape=struct&compressedShape_width=10&compressedShape_height=30

        */

        // with the request parameters shown above gadget.expandedShape.area would be 2400
        // and gadget.compressedShape.area would be 30
        // ...
    }
}
```

Typically the request parameters with "struct" as their value would be represented by hidden form fields.

Data Binding Event Listeners

The [DataBindingListener](#) interface provides a mechanism for listeners to be notified of data binding events. The interface looks like this:

```
package grails.databinding.events;

import grails.databinding.errors.BindingError;

/**
 * A listener which will be notified of events generated during data binding.
 *
 * @author Jeff Brown
 * @since 3.0
 * @see DataBindingListenerAdapter
 */
public interface DataBindingListener {

    /**
     * @return true if the listener is interested in events for the specified type.
     */
    boolean supports(Class<?> clazz);

    /**
     * Called when data binding is about to start.
     *
     * @param target The object data binding is being imposed upon
     * @param errors the Spring Errors instance (a org.springframework.validation.BindingResult)
     * @return true if data binding should continue
     */
    Boolean beforeBinding(Object target, Object errors);

    /**
     * Called when data binding is about to imposed on a property
     *
     * @param target The object data binding is being imposed upon
     * @param propertyName The name of the property being bound to
     * @param value The value of the property being bound
     * @param errors the Spring Errors instance (a org.springframework.validation.BindingResult)
     * @return true if data binding should continue, otherwise return false
     */
    Boolean beforeBinding(Object target, String propertyName, Object value, Object errors);

    /**
     * Called after data binding has been imposed on a property
     *
     * @param target The object data binding is being imposed upon
     * @param propertyName The name of the property that was bound to
     * @param errors the Spring Errors instance (a org.springframework.validation.BindingResult)
     */
    void afterBinding(Object target, String propertyName, Object errors);

    /**
     * Called after data binding has finished.
     */
}
```

```

    *
    * @param target The object data binding is being imposed upon
    * @param errors the Spring Errors instance (a org.springframework.validation.BindingResult)
    */
    void afterBinding(Object target, Object errors);

    /**
    * Called when an error occurs binding to a property
    * @param error encapsulates information about the binding error
    * @param errors the Spring Errors instance (a org.springframework.validation.BindingResult)
    * @see BindingError
    */
    void bindingError(BindingError error, Object errors);
}

```

Any bean in the Spring application context which implements that interface will automatically be registered with the data binder. The [DataBindingListenerAdapter](#) class implements the `DataBindingListener` interface and provides default implementations for all of the methods in the interface so this class is well suited for subclassing so your listener class only needs to provide implementations for the methods your listener is interested in.

Using The Data Binder Directly

There are situations where an application may want to use the data binder directly. For example, to do binding in a Service on some arbitrary object which is not a domain class. The following will not work because the `properties` property is read only.

src/main/groovy/bindingdemo/Widget.groovy

```

package bindingdemo

class Widget {
    String name
    Integer size
}

```

grails-app/services/bindingdemo/WidgetService.groovy

```

package bindingdemo

class WidgetService {

    def updateWidget(Widget widget, Map data) {
        // this will throw an exception because
        // properties is read-only
        widget.properties = data
    }
}

```

An instance of the data binder is in the Spring application context with a bean name of `grailsWebDataBinder`. That bean implements the [DataBinder](#) interface. The following code demonstrates using the data binder directly.

grails-app/services/bindingdemo/WidgetService

```

package bindingdemo

import grails.databinding.SimpleMapDataBindingSource

class WidgetService {

    // this bean will be autowired into the service
    def grailsWebDataBinder

    def updateWidget(Widget widget, Map data) {
        grailsWebDataBinder.bind widget, data as SimpleMapDataBindingSource
    }
}

```

See the [DataBinder](#) documentation for more information about overloaded versions of the `bind` method.

Data Binding and Security Concerns

When batch updating properties from request parameters you need to be careful not to allow

clients to bind malicious data to domain classes and be persisted in the database. You can limit what properties are bound to a given domain class using the subscript operator:

```
def p = Person.get(1)
p.properties['firstName', 'lastName'] = params
```

In this case only the `firstName` and `lastName` properties will be bound.

Another way to do this is to use [Command Objects](#) as the target of data binding instead of domain classes. Alternatively there is also the flexible [bindData](#) method.

The `bindData` method allows the same data binding capability, but to arbitrary objects:

```
def p = new Person()
bindData(p, params)
```

The `bindData` method also lets you exclude certain parameters that you don't want updated:

```
def p = new Person()
bindData(p, params, [exclude: 'dateOfBirth'])
```

Or include only certain properties:

```
def p = new Person()
bindData(p, params, [include: ['firstName', 'lastName']])
```

If an empty List is provided as a value for the `include` parameter then all fields will be subject to binding if they are not explicitly excluded.

The [bindable](#) constraint can be used to globally prevent data binding for certain properties.

7.1.6 Responding with JSON

Using the `respond` method to output JSON

The `respond` method is the preferred way to return JSON and integrates with [Content Negotiation](#) and [JSON Views](#).

The `respond` method provides content negotiation strategies to intelligently produce an appropriate response for the given client.

For example given the following controller and action:

grails-app/controllers/example/BookController.groovy

```
package example

class BookController {
    def index() {
        respond Book.list()
    }
}
```

The `respond` method will take the followings steps:

1. If the client `Accept` header specifies a media type (for example `application/json`) use that
2. If the file extension of the URI (for example `/books.json`) includes a format defined in the `grails.mime.types` property of `grails-app/conf/application.yml` use the media type defined in the configuration

The `respond` method will then look for an appropriate [Renderer](#) for the object and the

calculated media type from the [RendererRegistry](#).

Grails includes a number of pre-configured `Renderer` implementations that will produce default representations of JSON responses for the argument passed to `respond`. For example going to the `/book.json` URI will produce JSON such as:

```
[
  {id:1,"title":"The Stand"},
  {id:2,"title":"Shining"}
]
```

Controlling the Priority of Media Types

By default if you define a controller there is no priority in terms of which format is sent back to the client and Grails assumes you wish to serve HTML as a response type.

However if your application is primarily an API, then you can specify the priority using the `responseFormats` property:

`grails-app/controllers/example/BookController.groovy`

```
package example

class BookController {
    static responseFormats = ['json', 'html']
    def index() {
        respond Book.list()
    }
}
```

In the above example Grails will respond by default with `json` if the media type to respond with cannot be calculated from the `Accept` header or file extension.

Using Views to Output JSON Responses

If you define a view (either a GSP or a [JSON View](#)) then Grails will render the view when using the `respond` method by calculating a model from the argument passed to `respond`.

For example, in the previous listing, if you were to define `grails-app/views/index.json` and `grails-app/views/index.gsp` views, these would be used if the client requested `application/json` or `text/html` media types respectively. Thus allowing you to define a single backend capable of serving responses to a web browser or representing your application's API.

When rendering the view, Grails will calculate a model to pass to the view based on the type of the value passed to the `respond` method.

The following table summarizes this convention:

Example	Argument Type	Calculated Model Variable
<code>respond Book.list()</code>	<code>java.util.List</code>	<code>bookList</code>
<code>respond([])</code>	<code>java.util.List</code>	<code>emptyList</code>
<code>respond Book.get(1)</code>	<code>example.Book</code>	<code>book</code>
<code>respond([1,2])</code>	<code>java.util.List</code>	<code>integerList</code>

```
respond( [1,2] as Set )           java.util.Set           integerSet
```

```
respond( [1,2] as Integer[] )     Integer[]          integerArray
```

Using this convention you can reference the argument passed to `respond` from within your view:

grails-app/views/book/index.gson

```
@Field List<Book> bookList = []

json bookList, { Book book ->
    title book.title
}
```

You will notice that if `Book.list()` returns an empty list then the model variable name is translated to `emptyList`. This is by design and you should provide a default value in the view if no model variable is specified, such as the `List` in the example above:

grails-app/views/book/index.gson

```
// defaults to an empty list
@Field List<Book> bookList = []
...
```

There are cases where you may wish to be more explicit and control the name of the model variable. For example if you have a domain inheritance hierarchy where a call to `list()` may return different child classes relying on automatic calculation may not be reliable.

In this case you should pass the model directly using `respond` and a `map` argument:

```
respond bookList: Book.list()
```

When responding with any kind of mixed argument types in a collection, always use an explicit model name.

If you simply wish to augment the calculated model then you can do so by passing a `model` argument:

```
respond Book.list(), [model: [bookCount: Book.count()]]
```

The above example will produce a model like `[bookList:books, bookCount:totalBooks]`, where the calculated model is combined with the model passed in the `model` argument.

Using the render method to output JSON

The `render` method can also be used to output JSON, but should only be used for simple cases that don't warrant the creation of a JSON view:

```
def list() {
    def results = Book.list()
    render(contentType: "application/json") {
        books(results) { Book b ->
            title b.title
        }
    }
}
```

In this case the result would be something along the lines of:

```
[
```

```

    {"title":"The Stand"},
    {"title":"Shining"}
]

```

This technique for rendering JSON may be ok for very simple responses, but in general you should favour the use of [JSON Views](#) and use the view layer rather than embedding logic in your application.

The same dangers with naming conflicts described above for XML also apply to JSON building.

7.1.7 More on JSONBuilder

The previous section on XML and JSON responses covered simplistic examples of rendering XML and JSON responses. Whilst the XML builder used by Grails is the standard [XmlSlurper](#) found in Groovy.

For JSON, since Grails 3.1, Grails uses Groovy's [StreamingJsonBuilder](#) by default and you can refer to the [Groovy documentation](#) and [StreamingJsonBuilder](#) API documentation on how to use it.

7.1.8 Responding with XML

7.1.9 Uploading Files

Programmatic File Uploads

Grails supports file uploads using Spring's [MultipartHttpServletRequest](#) interface. The first step for file uploading is to create a multipart form like this:

```

Upload Form: <br />
<g:uploadForm action="upload">
  <input type="file" name="myFile" />
  <input type="submit" />
</g:uploadForm>

```

The `uploadForm` tag conveniently adds the `enctype="multipart/form-data"` attribute to the standard `<g:form>` tag.

There are then a number of ways to handle the file upload. One is to work with the Spring [MultipartFile](#) instance directly:

```

def upload() {
  def f = request.getFile('myFile')
  if (f.empty) {
    flash.message = 'file cannot be empty'
    render(view: 'uploadForm')
    return
  }

  f.transferTo(new File('/some/local/dir/myfile.txt'))
  response.sendError(200, 'Done')
}

```

This is convenient for doing transfers to other destinations and manipulating the file directly as you can obtain an `InputStream` and so on with the [MultipartFile](#) interface.

File Uploads through Data Binding

File uploads can also be performed using data binding. Consider this `Image` domain class:

```
class Image {
  byte[] myFile

  static constraints = {
    // Limit upload file size to 2MB
    myFile maxSize: 1024 * 1024 * 2
  }
}
```

If you create an image using the `params` object in the constructor as in the example below, Grails will automatically bind the file's contents as a `byte[]` to the `myFile` property:

```
def img = new Image(params)
```

It's important that you set the [size](#) or [maxSize](#) constraints, otherwise your database may be created with a small column size that can't handle reasonably sized files. For example, both H2 and MySQL default to a blob size of 255 bytes for `byte[]` properties.

It is also possible to set the contents of the file as a string by changing the type of the `myFile` property on the image to a `String` type:

```
class Image {
  String myFile
}
```

Increase Upload Max File Size

Grails default size for file uploads is 128000 (~128KB). When this limit is exceeded you'll see the following exception:

```
org.springframework.web.multipart.MultipartException: Could not parse multipart servlet request; nested exception i
```

You can configure the limit in your `application.yml` as follows:

`grails-app/conf/application.yml`

```
grails:
  controllers:
    upload:
      maxFileSize: 2000000
      maxRequestSize: 2000000
```

`maxFileSize` = The maximum size allowed for uploaded files.

`maxRequestSize` = The maximum size allowed for multipart/form-data requests.

You should keep in mind [OWASP recommendations - Unrestricted File Upload](#)

Limit the file size to a maximum value in order to prevent denial of service attacks.

These limits exist to prevent DoS attacks and to enforce overall application performance

7.1.10 Command Objects

Grails controllers support the concept of command objects. A command object is a class that is used in conjunction with [data binding](#), usually to allow validation of data that may not fit into an existing domain class.

A class is only considered to be a command object when it is used as a parameter of an action.

Declaring Command Objects

Command object classes are defined just like any other class.

```
class LoginCommand implements grails.validation.Validateable {
    String username
    String password

    static constraints = {
        username(blank: false, minSize: 6)
        password(blank: false, minSize: 6)
    }
}
```

In this example, the command object class implements the `validateable` trait. The `validateable` trait allows the definition of [Constraints](#) just like in [domain classes](#). If the command object is defined in the same source file as the controller that is using it, Grails will automatically make it `validateable`. It is not required that command object classes be `validateable`.

By default, all `validateable` object properties which are not instances of `java.util.Collection` or `java.util.Map` are `nullable: false`. Instances of `java.util.Collection` and `java.util.Map` default to `nullable: true`. If you want a `Validateable` that has `nullable: true` properties by default, you can specify this by defining a `defaultNullable` method in the class:

```
class AuthorSearchCommand implements grails.validation.Validateable {
    String name
    Integer age

    static boolean defaultNullable() {
        true
    }
}
```

In this example, both `name` and `age` will allow null values during validation.

Using Command Objects

To use command objects, controller actions may optionally specify any number of command object parameters. The parameter types must be supplied so that Grails knows what objects to create and initialize.

Before the controller action is executed Grails will automatically create an instance of the command object class and populate its properties by binding the request parameters. If the command object class is marked with `validateable` then the command object will be validated. For example:

```
class LoginController {
    def login(LoginCommand cmd) {
        if (cmd.hasErrors()) {
            redirect(action: 'loginForm')
            return
        }

        // work with the command object data
    }
}
```

If the command object's type is that of a domain class and there is an `id` request parameter then instead of invoking the domain class constructor to create a new instance a call will be made to the static `get` method on the domain class and the value of the `id` parameter will be passed as an argument.

Whatever is returned from that call to `get` is what will be passed into the controller action. This means that if there is an `id` request parameter and no corresponding record is found in the database then the value of the command object will be `null`. If an error occurs retrieving the instance from the database then `null` will be passed as an argument to the controller action and an error will be added the controller's `errors` property.

If the command object's type is a domain class and there is no `id` request parameter or there is an `id` request parameter and its value is empty then `null` will be passed into the controller action unless the HTTP request method is "POST", in which case a new instance of the domain class will be created by invoking the domain class constructor. For all of the cases where the domain class instance is non-null, data binding is only performed if the HTTP request method is "POST", "PUT" or "PATCH".

Command Objects And Request Parameter Names

Normally request parameter names will be mapped directly to property names in the command object. Nested parameter names may be used to bind down the object graph in an intuitive way.

In the example below a request parameter named `name` will be bound to the `name` property of the `Person` instance and a request parameter named `address.city` will be bound to the `city` property of the `address` property in the `Person`.

```
class StoreController {
  def buy(Person buyer) {
    // ...
  }
}

class Person {
  String name
  Address address
}

class Address {
  String city
}
```

A problem may arise if a controller action accepts multiple command objects which happen to contain the same property name. Consider the following example.

```
class StoreController {
  def buy(Person buyer, Product product) {
    // ...
  }
}

class Person {
  String name
  Address address
}

class Address {
  String city
}

class Product {
  String name
}
```

If there is a request parameter named `name` it isn't clear if that should represent the name of the `Product` or the name of the `Person`. Another version of the problem can come up if a controller action accepts 2 command objects of the same type as shown below.

```
class StoreController {
  def buy(Person buyer, Person seller, Product product) {
    // ...
  }
}

class Person {
  String name
  Address address
}

class Address {
  String city
}

class Product {
  String name
}
```

To help deal with this the framework imposes special rules for mapping parameter names to command object types. The command object data binding will treat all parameters that begin with the controller action parameter name as belonging to the corresponding command object.

For example, the `product.name` request parameter will be bound to the `name` property in the `product` argument, the `buyer.name` request parameter will be bound to the `name` property in the `buyer` argument the `seller.address.city` request parameter will be bound to the `city` property of the `address` property of the `seller` argument, etc...

Command Objects and Dependency Injection

Command objects can participate in dependency injection. This is useful if your command object has some custom validation logic which uses a Grails [service](#):

```
class LoginCommand implements grails.validation.Validateable {
    def loginService

    String username
    String password

    static constraints = {
        username validator: { val, obj ->
            obj.loginService.canLogin(obj.username, obj.password)
        }
    }
}
```

In this example the command object interacts with the `loginService` bean which is injected by name from the Spring `ApplicationContext`.

Binding The Request Body To Command Objects

When a request is made to a controller action which accepts a command object and the request contains a body, Grails will attempt to parse the body of the request based on the request content type and use the body to do data binding on the command object. See the following example.

grails-app/controllers/bindingdemo/DemoController.groovy

```
package bindingdemo

class DemoController {
    def createWidget(Widget w) {
        render "Name: ${w?.name}, Size: ${w?.size}"
    }
}

class Widget {
    String name
    Integer size
}

$ curl -H "Content-Type: application/json" -d '{"name":"Some Widget","42"}'[size] localhost:8080/demo/createWidget
Name: Some Widget, Size: 42

$ curl -H "Content-Type: application/xml" -d '<widget><name>Some Other Widget</name><size>2112</size></widget>' loc
Name: Some Other Widget, Size: 2112
```

The request body will not be parsed under the following conditions:

- The request method is GET
- The request method is DELETE
- The content length is 0

Note that the body of the request is being parsed to make that work. Any attempt to read the body of the request after that will fail since the corresponding input stream will be empty. The controller action can either use a command object or it can parse the body of the request on its own (either directly, or by referring to something like `request.JSON`), but cannot do both.

`grails-app/controllers/bindingdemo/DemoController.groovy`

```
package bindingdemo

class DemoController {

    def createWidget(Widget w) {
        // this will fail because it requires reading the body,
        // which has already been read.
        def json = request.JSON

        // ...
    }
}
```

Working with Lists of Command Objects

A common use case for command objects is a Command Object that contains a collection of another:

```
class DemoController {

    def createAuthor(AuthorCommand command) {
        // ...
    }

    class AuthorCommand {
        String fullName
        List<BookCommand> books
    }

    class BookCommand {
        String title
        String isbn
    }
}
```

On this example, we want to create an Author with multiple Books.

In order to make this work from the UI layer, you can do the following in your GSP:

```
<g:form name="submit-author-books" controller="demo" action="createAuthor">
  <g:fieldValue name="fullName" value=""/>
  <ul>
    <li>
      <g:fieldValue name="books[0].title" value=""/>
      <g:fieldValue name="books[0].isbn" value=""/>
    </li>
    <li>
      <g:fieldValue name="books[1].title" value=""/>
      <g:fieldValue name="books[1].isbn" value=""/>
    </li>
  </ul>
</g:form>
```

There is also support for JSON, so you can submit the following with correct databinding

```
{
  "fullName": "Graeme Rocher",
  "books": [{
    "title": "The Definitive Guide to Grails",
    "isbn": "1111-343455-1111"
  }, {
    "title": "The Definitive Guide to Grails 2",
    "isbn": "1111-343455-1112"
  }],
}
```

7.1.11 Handling Duplicate Form Submissions

Grails has built-in support for handling duplicate form submissions using the "Synchronizer Token Pattern". To get started you define a token on the [form](#) tag:

```
<g:form useToken="true" ...>
```

Then in your controller code you can use the [withForm](#) method to handle valid and invalid requests:

```
withForm {  
    // good request  
}.invalidToken {  
    // bad request  
}
```

If you only provide the [withForm](#) method and not the chained `invalidToken` method then by default Grails will store the invalid token in a `flash.invalidToken` variable and redirect the request back to the original page. This can then be checked in the view:

```
<g:if test="${flash.invalidToken}">  
    Don't click the button twice!  
</g:if>
```

The [withForm](#) tag makes use of the [session](#) and hence requires session affinity or clustered sessions if used in a cluster.

7.1.12 Simple Type Converters

Type Conversion Methods

If you prefer to avoid the overhead of [data binding](#) and simply want to convert incoming parameters (typically Strings) into another more appropriate type the [params](#) object has a number of convenience methods for each type:

```
def total = params.int('total')
```

The above example uses the `int` method, and there are also methods for `boolean`, `long`, `char`, `short` and so on. Each of these methods is null-safe and safe from any parsing errors, so you don't have to perform any additional checks on the parameters.

Each of the conversion methods allows a default value to be passed as an optional second argument. The default value will be returned if a corresponding entry cannot be found in the map or if an error occurs during the conversion. Example:

```
def total = params.int('total', 42)
```

These same type conversion methods are also available on the `attrs` parameter of GSP tags.

Handling Multi Parameters

A common use case is dealing with multiple request parameters of the same name. For example you could get a query string such as `?name=Bob&name=Judy`.

In this case dealing with one parameter and dealing with many has different semantics since Groovy's iteration mechanics for `String` iterate over each character. To avoid this problem the [params](#) object provides a `list` method that always returns a list:

```
for (name in params.list('name')) {  
    println name  
}
```

7.1.13 Declarative Controller Exception Handling

Grails controllers support a simple mechanism for declarative exception handling. If a controller declares a method that accepts a single argument and the argument type is `java.lang.Exception` or some subclass of `java.lang.Exception`, that method will be invoked any time an action in that controller throws an exception of that type. See the following example.

grails-app/controllers/demo/DemoController.groovy

```
package demo

class DemoController {

    def someAction() {
        // do some work
    }

    def handleSQLException(SQLException e) {
        render 'A SQLException Was Handled'
    }

    def handleBatchUpdateException(BatchUpdateException e) {
        redirect controller: 'logging', action: 'batchProblem'
    }

    def handleNumberFormatException(NumberFormatException nfe) {
        [problemDescription: 'A Number Was Invalid']
    }
}
```

That controller will behave as if it were written something like this...

grails-app/controllers/demo/DemoController.groovy

```
package demo

class DemoController {

    def someAction() {
        try {
            // do some work
        } catch (BatchUpdateException e) {
            return handleBatchUpdateException(e)
        } catch (SQLException e) {
            return handleSQLException(e)
        } catch (NumberFormatException e) {
            return handleNumberFormatException(e)
        }
    }

    def handleSQLException(SQLException e) {
        render 'A SQLException Was Handled'
    }

    def handleBatchUpdateException(BatchUpdateException e) {
        redirect controller: 'logging', action: 'batchProblem'
    }

    def handleNumberFormatException(NumberFormatException nfe) {
        [problemDescription: 'A Number Was Invalid']
    }
}
```

The exception handler method names can be any valid method name. The name is not what makes the method an exception handler, the `Exception` argument type is the important part.

The exception handler methods can do anything that a controller action can do including invoking `render`, `redirect`, returning a model, etc.

One way to share exception handler methods across multiple controllers is to use inheritance. Exception handler methods are inherited into subclasses so an application could define the exception handlers in an abstract class that multiple controllers extend from. Another way to share exception handler methods across multiple controllers is to use a trait, as shown below...

src/main/groovy/com/demo/DatabaseExceptionHandler.groovy

```
package com.demo

trait DatabaseExceptionHandler {
    def handleSQLException(SQLException e) {
        // handle SQLException
    }

    def handleBatchUpdateException(BatchUpdateException e) {
        // handle BatchUpdateException
    }
}
```

grails-app/controllers/com/demo/DemoController.groovy

```
package com.demo

class DemoController implements DatabaseExceptionHandler {

    // all of the exception handler methods defined
    // in DatabaseExceptionHandler will be added to
    // this class at compile time
}
```

Exception handler methods must be present at compile time. Specifically, exception handler methods which are runtime metaprogrammed onto a controller class are not supported.

7.2 Groovy Server Pages

[Groovy Servers Pages](#) (or GSP for short) is Grails' view technology. It is designed to be familiar for users of technologies such as ASP and JSP, but to be far more flexible and intuitive.

Although GSP can render any format, not just HTML, it is more designed around rendering markup. If you are looking for a way to simplify JSON responses take a look at [JSON Views](#).

GSPs live in the `grails-app/views` directory and are typically rendered automatically (by convention) or with the [render](#) method such as:

```
render(view: "index")
```

A GSP is typically a mix of mark-up and GSP tags which aid in view rendering.

Although it is possible to have Groovy logic embedded in your GSP and doing this will be covered in this document, the practice is strongly discouraged. Mixing mark-up and code is a **bad** thing and most GSP pages contain no code and needn't do so.

A GSP typically has a "model" which is a set of variables that are used for view rendering. The model is passed to the GSP view from a controller. For example consider the following controller action:

```
def show() {
    [book: Book.get(params.id)]
}
```

This action will look up a `Book` instance and create a model that contains a key called `book`. This key can then be referenced within the GSP view using the name `book`:

```
${book.title}
```

Embedding data received from user input has the risk of making your application vulnerable to an Cross Site Scripting (XSS) attack. Please read the documentation on [XSS prevention](#) for information on how to prevent XSS attacks.

For more information on using GSP please refer to the [dedicated GSP documentation](#).

7.3 URL Mappings

Throughout the documentation so far the convention used for URLs has been the default of `/controller/action/id`. However, this convention is not hard wired into Grails and is in fact controlled by a URL Mappings class located at `grails-app/controllers/mypackage/UrlMappings.groovy`.

The `UrlMappings` class contains a single property called `mappings` that has been assigned a block of code:

```
package mypackage

class UrlMappings {
    static mappings = {
    }
}
```

7.3.1 Mapping to Controllers and Actions

To create a simple mapping simply use a relative URL as the method name and specify named parameters for the controller and action to map to:

```
"/product"(controller: "product", action: "list")
```

In this case we've mapped the URL `/product` to the `list` action of the `ProductController`. Omit the action definition to map to the default action of the controller:

```
"/product"(controller: "product")
```

An alternative syntax is to assign the controller and action to use within a block passed to the method:

```
"/product" {
    controller = "product"
    action = "list"
}
```

Which syntax you use is largely dependent on personal preference.

If you have mappings that all fall under a particular path you can group mappings with the `group` method:

```
group "/product", {
    "/apple"(controller:"product", id:"apple")
    "/htc"(controller:"product", id:"htc")
}
```

You can also create nested `group` url mappings:

```
group "/store", {
    group "/product", {
        "$id"(controller:"product")
    }
}
```

To rewrite one URI onto another explicit URI (rather than a controller/action pair) do something like this:

```
"/hello"(uri: "/hello.dispatch")
```

Rewriting specific URIs is often useful when integrating with other frameworks.

7.3.2 Mapping to REST resources

Since Grails 2.3, it is possible to create RESTful URL mappings that map onto controllers by convention. The syntax to do so is as follows:

```
"/books"(resources: 'book')
```

You define a base URI and the name of the controller to map to using the `resources` parameter. The above mapping will result in the following URLs:

HTTP Method	URI	Grails Action
GET	/books	index
GET	/books/create	create
POST	/books	save
GET	/books/\${id}	show
GET	/books/\${id}/edit	edit
PUT	/books/\${id}	update
DELETE	/books/\${id}	delete

If you are not sure which mapping will be generated for your case just run the command `url-mappings-report` in your grails console. It will give you a really neat report for all the url mappings.

If you wish to include or exclude any of the generated URL mappings you can do so with the `includes` or `excludes` parameter, which accepts the name of the Grails action to include or exclude:

```
"/books"(resources: 'book', excludes: ['delete', 'update'])
```

or

```
"/books"(resources: 'book', includes: ['index', 'show'])
```

Explicit REST Mappings

As of Grails 3.1, if you prefer not to rely on a `resources` mapping to define your mappings then you can prefix any URL mapping with the HTTP method name (in lower case) to indicate the HTTP method it applies to. The following URL mapping:

```
"/books"(resources: 'book')
```


Is equivalent to:

```
get "/books"(controller:"book", action:"index")
get "/books/create"(controller:"book", action:"create")
post "/books"(controller:"book", action:"save")
get "/books/$id"(controller:"book", action:"show")
get "/books/$id/edit"(controller:"book", action:"edit")
put "/books/$id"(controller:"book", action:"update")
delete "/books/$id"(controller:"book", action:"delete")
```

Notice how the HTTP method name is prefixed prior to each URL mapping definition.

Single resources

A single resource is a resource for which there is only one (possibly per user) in the system. You can create a single resource using the `single` parameter (as opposed to `resources`):

```
"/book"(single:'book')
```

This results in the following URL mappings:

HTTP Method	URI	Grails Action
GET	/book/create	create
POST	/book	save
GET	/book	show
GET	/book/edit	edit
PUT	/book	update
DELETE	/book	delete

The main difference is that the id is not included in the URL mapping.

Nested Resources

You can nest resource mappings to generate child resources. For example:

```
"/books"(resources:'book') {
  "/authors"(resources:"author")
}
```

The above will result in the following URL mappings:

HTTP Method	URL	Grails Action
GET	/books/\${bookId}/authors	index
GET	/books/\${bookId}/authors/create	create

POST	/books/\${bookId}/authors	save
GET	/books/\${bookId}/authors/\${id}	show
GET	/books/\${bookId}/authors/edit/\${id}	edit
PUT	/books/\${bookId}/authors/\${id}	update
DELETE	/books/\${bookId}/authors/\${id}	delete

You can also nest regular URL mappings within a resource mapping:

```
"/books"(resources: "book") {
  "/publisher"(controller:"publisher")
}
```

This will result in the following URL being available:

HTTP Method	URL	Grails Action
GET	/books/\${bookId}/publisher	index

To map a URI directly below a resource then use a collection block:

```
"/books"(resources: "book") {
  collection {
    "/publisher"(controller:"publisher")
  }
}
```

This will result in the following URL being available (without the ID):

HTTP Method	URL	Grails Action
GET	/books/publisher	index

Linking to RESTful Mappings

You can link to any URL mapping created with the `g:link` tag provided by Grails simply by referencing the controller and action to link to:

```
<g:link controller="book" action="index">My Link</g:link>
```

As a convenience you can also pass a domain instance to the `resource` attribute of the `link` tag:

```
<g:link resource="${book}">My Link</g:link>
```

This will automatically produce the correct link (in this case `/books/1` for an id of `"1"`).

The case of nested resources is a little different as they typically required two identifiers (the id of the resource and the one it is nested within). For example given the nested resources:

```
"/books"(resources: 'book') {  
  "/authors"(resources: "author")  
}
```

If you wished to link to the `show` action of the `author` controller, you would write:

```
// Results in /books/1/authors/2  
<g:link controller="author" action="show" method="GET" params="[bookId:1]" id="2">The Author</g:link>
```

However, to make this more concise there is a `resource` attribute to the link tag which can be used instead:

```
// Results in /books/1/authors/2  
<g:link resource="book/author" action="show" bookId="1" id="2">My Link</g:link>
```

The resource attribute accepts a path to the resource separated by a slash (in this case "book/author"). The attributes of the tag can be used to specify the necessary `bookId` parameter.

7.3.3 Redirects In URL Mappings

Since Grails 2.3, it is possible to define URL mappings which specify a redirect. When a URL mapping specifies a redirect, any time that mapping matches an incoming request, a redirect is initiated with information provided by the mapping.

When a URL mapping specifies a redirect the mapping must either supply a String representing a URI to redirect to or must provide a Map representing the target of the redirect. That Map is structured just like the Map that may be passed as an argument to the `redirect` method in a controller.

```
"/viewBooks"(redirect: [uri: '/books/list'])  
"/viewAuthors"(redirect: [controller: 'author', action: 'list'])  
"/viewPublishers"(redirect: [controller: 'publisher', action: 'list', permanent: true])
```

Request parameters that were part of the original request will not be included in the redirect by default. To include them it is necessary to add the parameter `keepParamsWhenRedirect: true`.

```
"/viewBooks"(redirect: [uri: '/books/list', keepParamsWhenRedirect: true])  
"/viewAuthors"(redirect: [controller: 'author', action: 'list', keepParamsWhenRedirect: true])  
"/viewPublishers"(redirect: [controller: 'publisher', action: 'list', permanent: true, keepParamsWhenRedirect: true])
```

7.3.4 Embedded Variables

Simple Variables

The previous section demonstrated how to map simple URLs with concrete "tokens". In URL mapping speak tokens are the sequence of characters between each slash, '/'. A concrete token is one which is well defined such as `/product`. However, in many circumstances you don't know what the value of a particular token will be until runtime. In this case you can use variable placeholders within the URL for example:

```
static mappings = {  
  "/product/$id"(controller: "product")  
}
```

In this case by embedding a `$id` variable as the second token Grails will automatically map the second token into a parameter (available via the [params](#) object) called `id`. For example

given the URL `/product/MacBook`, the following code will render "MacBook" to the response:

```
class ProductController {  
    def index() { render params.id }  
}
```

You can of course construct more complex examples of mappings. For example the traditional blog URL format could be mapped as follows:

```
static mappings = {  
    "$blog/$year/$month/$day/$id"(controller: "blog", action: "show")  
}
```

The above mapping would let you do things like:

```
/graemerocher/2007/01/10/my_funky_blog_entry
```

The individual tokens in the URL would again be mapped into the [params](#) object with values available for `year`, `month`, `day`, `id` and so on.

Dynamic Controller and Action Names

Variables can also be used to dynamically construct the controller and action name. In fact the default Grails URL mappings use this technique:

```
static mappings = {  
    "$controller/$action?/$id?"()  
}
```

Here the name of the controller, action and id are implicitly obtained from the variables `controller`, `action` and `id` embedded within the URL.

You can also resolve the controller name and action name to execute dynamically using a closure:

```
static mappings = {  
    "$controller" {  
        action = { params.goHere }  
    }  
}
```

Optional Variables

Another characteristic of the default mapping is the ability to append a `?` at the end of a variable to make it an optional token. In a further example this technique could be applied to the blog URL mapping to have more flexible linking:

```
static mappings = {  
    "$blog/$year?/$month?/$day?/$id?"(controller:"blog", action:"show")  
}
```

With this mapping all of these URLs would match with only the relevant parameters being populated in the [params](#) object:

```
/graemerocher/2007/01/10/my_funky_blog_entry  
/graemerocher/2007/01/10  
/graemerocher/2007/01  
/graemerocher/2007  
/graemerocher
```

Optional File Extensions

If you wish to capture the extension of a particular path, then a special case mapping exists:

```
"$controller/$action?/$id?($format)?"()
```

By adding the `(.format)?` mapping you can access the file extension using the `response.format` property in a controller:

```
def index() {  
    render "extension is ${response.format}"  
}
```

Arbitrary Variables

You can also pass arbitrary parameters from the URL mapping into the controller by just setting them in the block passed to the mapping:

```
"/holiday/win" {  
    id = "Marrakech"  
    year = 2007  
}
```

This variables will be available within the [params](#) object passed to the controller.

Dynamically Resolved Variables

The hard coded arbitrary variables are useful, but sometimes you need to calculate the name of the variable based on runtime factors. This is also possible by assigning a block to the variable name:

```
"/holiday/win" {  
    id = { params.id }  
    isEligible = { session.user != null } // must be logged in  
}
```

In the above case the code within the blocks is resolved when the URL is actually matched and hence can be used in combination with all sorts of logic.

7.3.5 Mapping to Views

You can resolve a URL to a view without a controller or action involved. For example to map the root URL `/` to a GSP at the location `grails-app/views/index.gsp` you could use:

```
static mappings = {  
    "/"(view: "/index") // map the root URL  
}
```

Alternatively if you need a view that is specific to a given controller you could use:

```
static mappings = {  
    "/help"(controller: "site", view: "help") // to a view for a controller  
}
```

7.3.6 Mapping to Response Codes

Grails also lets you map HTTP response codes to controllers, actions or views. Just use a method name that matches the response code you are interested in:

```
static mappings = {  
    "403"(controller: "errors", action: "forbidden")  
    "404"(controller: "errors", action: "notFound")  
    "500"(controller: "errors", action: "serverError")  
}
```

Or you can specify custom error pages:

```
static mappings = {  
    "403"(view: "/errors/forbidden")  
    "404"(view: "/errors/notFound")  
}
```

```

    "500"(view: "/errors/serverError")
}

```

Declarative Error Handling

In addition you can configure handlers for individual exceptions:

```

static mappings = {
    "403"(view: "/errors/forbidden")
    "404"(view: "/errors/notFound")
    "500"(controller: "errors", action: "illegalArgument",
        exception: IllegalArgumentException)
    "500"(controller: "errors", action: "nullPointer",
        exception: NullPointerException)
    "500"(controller: "errors", action: "customException",
        exception: MyException)
    "500"(view: "/errors/serverError")
}

```

With this configuration, an `IllegalArgumentException` will be handled by the `illegalArgument` action in `ErrorsController`, a `NullPointerException` will be handled by the `nullPointer` action, and a `MyException` will be handled by the `customException` action. Other exceptions will be handled by the catch-all rule and use the `/errors/serverError` view.

You can access the exception from your custom error handling view or controller action using the request's `exception` attribute like so:

```

class ErrorsController {
    def handleError() {
        def exception = request.exception
        // perform desired processing to handle the exception
    }
}

```

If your error-handling controller action throws an exception as well, you'll end up with a `StackOverflowException`.

7.3.7 Mapping to HTTP methods

URL mappings can also be configured to map based on the HTTP method (GET, POST, PUT or DELETE). This is very useful for RESTful APIs and for restricting mappings based on HTTP method.

As an example the following mappings provide a RESTful API URL mappings for the `ProductController`:

```

static mappings = {
    "/product/$id"(controller:"product", action: "update", method: "PUT")
}

```

Note that if you specify a HTTP method other than GET in your URL mapping, you also have to specify it when creating the corresponding link by passing the `method` argument to `g:link` OR `g:createLink` to get a link of the desired format.

7.3.8 Mapping Wildcards

Grails' URL mappings mechanism also supports wildcard mappings. For example consider the following mapping:

```

static mappings = {
    "/images/*.jpg"(controller: "image")
}

```

This mapping will match all paths to images such as `/image/logo.jpg`. Of course you can

achieve the same effect with a variable:

```
static mappings = {
    "/images/$name.jpg"(controller: "image")
}
```

However, you can also use double wildcards to match more than one level below:

```
static mappings = {
    "/images/**/*.jpg"(controller: "image")
}
```

In this cases the mapping will match `/image/logo.jpg` as well as `/image/other/logo.jpg`. Even better you can use a double wildcard variable:

```
static mappings = {
    // will match /image/logo.jpg and /image/other/logo.jpg
    "/images/$name**.jpg"(controller: "image")
}
```

In this case it will store the path matched by the wildcard inside a `name` parameter obtainable from the [params](#) object:

```
def name = params.name
println name // prints "logo" or "other/logo"
```

If you use wildcard URL mappings then you may want to exclude certain URIs from Grails' URL mapping process. To do this you can provide an `excludes` setting inside the `UrlMappings.groovy` class:

```
class UrlMappings {
    static excludes = ["/images/**", "/css/**"]
    static mappings = {
        ...
    }
}
```

In this case Grails won't attempt to match any URIs that start with `/images` or `/css`.

7.3.9 Automatic Link Re-Writing

Another great feature of URL mappings is that they automatically customize the behaviour of the [link](#) tag so that changing the mappings don't require you to go and change all of your links.

This is done through a URL re-writing technique that reverse engineers the links from the URL mappings. So given a mapping such as the blog one from an earlier section:

```
static mappings = {
    "/$blog/$year?/$month?/$day?/$id?"(controller:"blog", action:"show")
}
```

If you use the link tag as follows:

```
<g:link controller="blog" action="show"
    params="[blog:'fred', year:2007]">
    My Blog
</g:link>

<g:link controller="blog" action="show"
    params="[blog:'fred', year:2007, month:10]">
    My Blog - October 2007 Posts
</g:link>
```

Grails will automatically re-write the URL in the correct format:

```
<a href="/fred/2007">My Blog</a>
<a href="/fred/2007/10">My Blog - October 2007 Posts</a>
```

7.3.10 Applying Constraints

URL Mappings also support Grails' unified [validation constraints](#) mechanism, which lets you further "constrain" how a URL is matched. For example, if we revisit the blog sample code from earlier, the mapping currently looks like this:

```
static mappings = {
    "$blog/$year?/$month?/$day?/$id?"(controller:"blog", action:"show")
}
```

This allows URLs such as:

```
/graemerocher/2007/01/10/my_funky_blog_entry
```

However, it would also allow:

```
/graemerocher/not_a_year/not_a_month/not_a_day/my_funky_blog_entry
```

This is problematic as it forces you to do some clever parsing in the controller code. Luckily, URL Mappings can be constrained to further validate the URL tokens:

```
"/$blog/$year?/$month?/$day?/$id?" {
    controller = "blog"
    action = "show"
    constraints {
        year(matches:/\\d{4}/)
        month(matches:/\\d{2}/)
        day(matches:/\\d{2}/)
    }
}
```

In this case the constraints ensure that the `year`, `month` and `day` parameters match a particular valid pattern thus relieving you of that burden later on.

7.3.11 Named URL Mappings

URL Mappings also support named mappings, that is mappings which have a name associated with them. The name may be used to refer to a specific mapping when links are generated.

The syntax for defining a named mapping is as follows:

```
static mappings = {
    name <mapping name>: <url pattern> {
        // ...
    }
}
```

For example:

```
static mappings = {
    name personList: "/showPeople" {
        controller = 'person'
        action = 'list'
    }
    name accountDetails: "/details/$acctNumber" {
        controller = 'product'
        action = 'accountDetails'
    }
}
```

The mapping may be referenced in a link tag in a GSP.

```
<g:link mapping="personList">List People</g:link>
```


That would result in:

```
<a href="/showPeople">List People</a>
```

Parameters may be specified using the params attribute.

```
<g:link mapping="accountDetails" params="[acctNumber:'8675309']">
  Show Account
</g:link>
```

That would result in:

```
<a href="/details/8675309">Show Account</a>
```

Alternatively you may reference a named mapping using the link namespace.

```
<link:personList>List People</link:personList>
```

That would result in:

```
<a href="/showPeople">List People</a>
```

The link namespace approach allows parameters to be specified as attributes.

```
<link:accountDetails acctNumber="8675309">Show Account</link:accountDetails>
```

That would result in:

```
<a href="/details/8675309">Show Account</a>
```

To specify attributes that should be applied to the generated href, specify a Map value to the attrs attribute. These attributes will be applied directly to the href, not passed through to be used as request parameters.

```
<link:accountDetails attrs="[class: 'fancy']" acctNumber="8675309">
  Show Account
</link:accountDetails>
```

That would result in:

```
<a href="/details/8675309" class="fancy">Show Account</a>
```

7.3.12 Customizing URL Formats

The default URL Mapping mechanism supports camel case names in the URLs. The default URL for accessing an action named `addNumbers` in a controller named `MathHelperController` would be something like `/mathHelper/addNumbers`. Grails allows for the customization of this pattern and provides an implementation which replaces the camel case convention with a hyphenated convention that would support URLs like `/math-helper/add-numbers`. To enable hyphenated URLs assign a value of "hyphenated" to the `grails.web.url.converter` property in `grails-app/conf/application.groovy`.

`grails-app/conf/application.groovy`

```
grails.web.url.converter = 'hyphenated'
```

Arbitrary strategies may be plugged in by providing a class which implements the [UrlConverter](#) interface and adding an instance of that class to the Spring application context with the bean name of `grails.web.UrlConverter.BEAN_NAME`. If Grails finds a bean in the context with that name, it will be used as the default converter and there is no need to assign a value to the `grails.web.url.converter` config property.

src/main/groovy/com/myapplication/MyUrlConverterImpl.groovy

```
package com.myapplication

class MyUrlConverterImpl implements grails.web.UrlConverter {

    String toUrlElement(String propertyOrClassName) {
        // return some representation of a property or class name that should be used in URLs...
    }
}
```

grails-app/conf/spring/resources.groovy

```
beans = {
    "${grails.web.UrlConverter.BEAN_NAME}"(com.myapplication.MyUrlConverterImpl)
}
```

7.3.13 Namespaced Controllers

If an application defines multiple controllers with the same name in different packages, the controllers must be defined in a namespace. The way to define a namespace for a controller is to define a static property named `namespace` in the controller and assign a `String` to the property that represents the namespace.

grails-app/controllers/com/app/reporting/AdminController.groovy

```
package com.app.reporting

class AdminController {

    static namespace = 'reports'

    // ...
}
```

grails-app/controllers/com/app/security/AdminController.groovy

```
package com.app.security

class AdminController {

    static namespace = 'users'

    // ...
}
```

When defining url mappings which should be associated with a namespaced controller, the `namespace` variable needs to be part of the URL mapping.

grails-app/controllers/UrlMappings.groovy

```
class UrlMappings {

    static mappings = {
        '/userAdmin' {
            controller = 'admin'
            namespace = 'users'
        }

        '/reportAdmin' {
            controller = 'admin'
            namespace = 'reports'
        }

        "$namespace/$controller/$action?"()
    }
}
```

Reverse URL mappings also require that the `namespace` be specified.

```
<g:link controller="admin" namespace="reports">Click For Report Admin</g:link>
<g:link controller="admin" namespace="users">Click For User Admin</g:link>
```

When resolving a URL mapping (forward or reverse) to a namespaced controller, a mapping will only match if the `namespace` has been provided. If the application provides several controllers with the same name in different packages, at most 1 of them may be defined without a `namespace` property. If there are multiple controllers with the same name that do not

define a `namespace` property, the framework will not know how to distinguish between them for forward or reverse mapping resolutions.

It is allowed for an application to use a plugin which provides a controller with the same name as a controller provided by the application and for neither of the controllers to define a `namespace` property as long as the controllers are in separate packages. For example, an application may include a controller named `com.accounting.ReportingController` and the application may use a plugin which provides a controller named `com.humanresources.ReportingController`. The only issue with that is the URL mapping for the controller provided by the plugin needs to be explicit in specifying that the mapping applies to the `ReportingController` which is provided by the plugin.

See the following example.

```
static mappings = {
    "/accountingReports" {
        controller = "reporting"
    }
    "/humanResourceReports" {
        controller = "reporting"
        plugin = "humanResources"
    }
}
```

With that mapping in place, a request to `/accountingReports` will be handled by the `ReportingController` which is defined in the application. A request to `/humanResourceReports` will be handled by the `ReportingController` which is provided by the `humanResources` plugin.

There could be any number of `ReportingController` controllers provided by any number of plugins but no plugin may provide more than one `ReportingController` even if they are defined in separate packages.

Assigning a value to the `plugin` variable in the mapping is only required if there are multiple controllers with the same name available at runtime provided by the application and/or plugins. If the `humanResources` plugin provides a `ReportingController` and there is no other `ReportingController` available at runtime, the following mapping would work.

```
static mappings = {
    "/humanResourceReports" {
        controller = "reporting"
    }
}
```

It is best practice to be explicit about the fact that the controller is being provided by a plugin.

7.4 CORS

Spring Boot provides CORS support out of the box, but it is difficult to configure in a Grails application due to the way `UrlMappings` are used instead of annotations that define URLs. Starting with Grails 3.2.1, we have added a way to configure CORS that makes sense in a Grails application.

Once enabled, the default setting is "wide open".

`application.yml`

```
grails:
  cors:
    enabled: true
```

That will produce a mapping to all urls `/**` with:

allowedOrigins	['*']
allowedMethods	['*']
allowedHeaders	['*']
exposedHeaders	null
maxAge	1800
allowCredentials	false

Some of these settings come directly from Spring Boot and can change in future versions. See [Spring CORS Configuration Documentation](#)

All of those settings can be easily overridden.

application.yml

```
grails:
  cors:
    enabled: true
    allowedOrigins:
      - http://localhost:5000
```

In the example above, the `allowedOrigins` setting will replace `['*']`.

You can also configure different URLs.

application.yml

```
grails:
  cors:
    enabled: true
    allowedHeaders:
      - Content-Type
    mappings:
      '["api/**"]':
        allowedOrigins:
          - http://localhost:5000
      # Other configurations not specified default to the global config
```

Note that the mapping key must be made with bracket notation (see <https://github.com/spring-projects/spring-boot/wiki/Spring-Boot-Configuration-Binding#map-based-binding>), which is a breaking change between Spring Boot 1.5 (Grails 3) and Spring Boot 2 (Grails 4).

Specifying at least one mapping will disable the creation of the global mapping (`/**`). If you wish to keep that setting, you should specify it along with your other mappings.

The settings above will produce a single mapping of `/api/**` with the following settings:

allowedOrigins	['http://localhost:5000']
----------------	---------------------------

allowedMethods	['*']
allowedHeaders	['Content-Type']
exposedHeaders	null
maxAge	1800
allowCredentials	false

If you don't wish to override any of the default settings, but only want to specify URLs, you can do so like this example:

application.yml

```
grails:
  cors:
    enabled: true
    mappings:
      '[/api/**]': inherit
```

7.5 Interceptors

Grails provides standalone Interceptors using the [create-interceptor](#) command:

```
$ grails create-interceptor MyInterceptor
```

The above command will create an Interceptor in the `grails-app/controllers` directory with the following default contents:

```
class MyInterceptor {
    boolean before() { true }
    boolean after() { true }
    void afterView() {
        // no-op
    }
}
```

Interceptors vs Filters

In versions of Grails prior to Grails 3.0, Grails supported the notion of filters. These are still supported for backwards compatibility but are considered deprecated.

The new interceptors concept in Grails 3.0 is superior in a number of ways, most significantly interceptors can use Groovy's `CompileStatic` annotation to optimize performance (something which is often critical as interceptors can be executed for every request.)

7.5.1 Defining Interceptors

By default interceptors will match the controllers with the same name. For example if you have an interceptor called `BookInterceptor` then all requests to the actions of the `BookController` will trigger the interceptor.

An `Interceptor` implements the [Interceptor](#) trait and provides 3 methods that can be used to intercept requests:

```
/**
 * Executed before a matched action
 *
 * @return Whether the action should continue and execute
 */
boolean before() { true }

/**
 * Executed after the action executes but prior to view rendering
 *
 * @return True if view rendering should continue, false otherwise
 */
boolean after() { true }

/**
 * Executed after view rendering completes
 */
void afterView() {}
```

As described above the `before` method is executed prior to an action and can cancel the execution of the action by returning `false`.

The `after` method is executed after an action executes and can halt view rendering if it returns `false`. The `after` method can also modify the view or model using the `view` and `model` properties respectively:

```
boolean after() {
    model.foo = "bar" // add a new model attribute called 'foo'
    view = 'alternate' // render a different view called 'alternate'
    true
}
```

The `afterView` method is executed after view rendering completes. If an exception occurs, the exception is available using the `throwable` property of the [Interceptor](#) trait.

7.5.2 Matching Requests with Interceptors

As mention in the previous section, by default an interceptor will match only requests to the associated controller by convention. However you can configure the interceptor to match any request using the `match` or `matchAll` methods defined in the [Interceptor API](#).

The matching methods return a [Matcher](#) instance which can be used to configure how the interceptor matches the request.

For example the following interceptor will match all requests except those to the `login` controller:

```
class AuthInterceptor {
  AuthInterceptor() {
    matchAll()
    .excludes(controller:"login")
  }

  boolean before() {
    // perform authentication
  }
}
```

You can also perform matching using named argument:

```
class LoggingInterceptor {
  LoggingInterceptor() {
    match(controller:"book", action:"show") // using strings
    match(controller: ~/ (author|publisher) /) // using regex
  }

  boolean before() {
```

```
} ...
}
```

You can use any number of matchers defined in your interceptor. They will be executed in the order in which they have been defined. For example the above interceptor will match for all of the following:

- when the `show` action of `BookController` is called
- when `AuthorController` OR `PublisherController` is called

All named arguments except for `uri` accept either a `String` or a `Regex` expression. The `uri` argument supports a `String` path that is compatible with Spring's [AntPathMatcher](#). The possible named arguments are:

- `namespace` - The namespace of the controller
- `controller` - The name of the controller
- `action` - The name of the action
- `method` - The HTTP method
- `uri` - The URI of the request. If this argument is used then all other arguments will be ignored and only this will be used.

7.5.3 Ordering Interceptor Execution

Interceptors can be ordered by defining an `order` property that defines a priority.

For example:

```
class AuthInterceptor {
    int order = HIGHEST_PRECEDENCE
    ...
}
```

The default value of the `order` property is 0. Interceptor execution order is determined by sorting the `order` property in an ascending direction and executing the lowest numerically ordered interceptor first.

The values `HIGHEST_PRECEDENCE` and `LOWEST_PRECEDENCE` can be used to define filters that should run first or last respectively.

Note that if you write an interceptor that is to be used by others it is better increment or decrement the `HIGHEST_PRECEDENCE` and `LOWEST_PRECEDENCE` to allow other interceptors to be inserted before or after the interceptor you are authoring:

```
int order = HIGHEST_PRECEDENCE + 50
// or
int order = LOWEST_PRECEDENCE - 50
```

To find out the computed order of interceptors you can add a debug logger to `logback.groovy` as follows:

```
logger 'grails.artefact.Interceptor', DEBUG, ['STDOUT'], false
```

You can override any interceptors default order by using bean override configuration in `grails-app/conf/application.yml`:

```
beans:
  authInterceptor:
    order: 50
```

Or in `grails-app/conf/application.groovy`:

```
beans {
  authInterceptor {
    order = 50
  }
}
```

Thus giving you complete control over interceptor execution order.

7.6 Content Negotiation

Grails has built in support for [Content negotiation](#) using either the HTTP `Accept` header, an explicit format request parameter or the extension of a mapped URI.

Configuring Mime Types

Before you can start dealing with content negotiation you need to tell Grails what content types you wish to support. By default Grails comes configured with a number of different content types within `grails-app/conf/application.yml` using the `grails.mime.types` setting:

```
grails:
  mime:
    types:
      all: '*'
      atom: application/atom+xml
      css: text/css
      csv: text/csv
      form: application/x-www-form-urlencoded
      html:
        - text/html
        - application/xhtml+xml
      js: text/javascript
      json:
        - application/json
        - text/json
      multipartForm: multipart/form-data
      rss: application/rss+xml
      text: text/plain
      hal:
        - application/hal+json
        - application/hal+xml
      xml:
        - text/xml
        - application/xml
```

The setting can also be done in `grails-app/conf/application.groovy` as shown below:

```
grails.mime.types = [ // the first one is the default format
  all:      '*', // 'all' maps to '*' or the first available format in withFormat
  atom:     'application/atom+xml',
  css:      'text/css',
  csv:      'text/csv',
  form:     'application/x-www-form-urlencoded',
  html:     ['text/html', 'application/xhtml+xml'],
  js:       'text/javascript',
  json:     ['application/json', 'text/json'],
  multipartForm: 'multipart/form-data',
  rss:      'application/rss+xml',
  text:     'text/plain',
  hal:      ['application/hal+json', 'application/hal+xml'],
  xml:      ['text/xml', 'application/xml']
]
```

The above bit of configuration allows Grails to detect to format of a request containing either the 'text/xml' or 'application/xml' media types as simply 'xml'. You can add your own

types by simply adding new entries into the map. The first one is the default format.

Content Negotiation using the format Request Parameter

Let's say a controller action can return a resource in a variety of formats: HTML, XML, and JSON. What format will the client get? The easiest and most reliable way for the client to control this is through a `format` URL parameter.

So if you, as a browser or some other client, want a resource as XML, you can use a URL like this:

```
http://my.domain.org/books.xml
```

The request parameters `format` is allowed as well <http://my.domain.org/books?format=xml>, but the default Grails URL Mapping `get "/$controller(.$format)?"(action:"index")` will override the `format` parameter with `null`. So the default mapping should be updated to `get "$controller"(action:"index")`.

The result of this on the server side is a `format` property on the `response` object with the value `xml`.

You can also define this parameter in the [URL Mappings](#) definition:

```
"/book/list"(controller:"book", action:"list") {  
    format = "xml"  
}
```

You could code your controller action to return XML based on this property, but you can also make use of the controller-specific `withFormat()` method:

This example requires the addition of the `org.grails.plugins:grails-plugin-converters` plugin

```
import grails.converters.JSON  
import grails.converters.XML  
  
class BookController {  
    def list() {  
        def books = Book.list()  
  
        withFormat {  
            html bookList: books  
            json { render books as JSON }  
            xml { render books as XML }  
            '*' { render books as JSON }  
        }  
    }  
}
```

In this example, Grails will only execute the block inside `withFormat()` that matches the requested content type. So if the preferred format is `html` then Grails will execute the `html()` call only. Each 'block' can either be a map model for the corresponding view (as we are doing for 'html' in the above example) or a closure. The closure can contain any standard action code, for example it can return a model or render content directly.

When no format matches explicitly, a `*` (wildcard) block can be used to handle all other formats.

There is a special format, "all", that is handled differently from the explicit formats. If "all" is specified (normally this happens through the Accept header - see below), then the first block of `withFormat()` is executed when there isn't a `*` (wildcard) block available.

You should not add an explicit "all" block. In this example, a format of "all" will trigger the `html` handler (`html` is the first block and there is no `*` block).

```
withFormat {
  html bookList: books
  json { render books as JSON }
  xml { render books as XML }
}
```

When using [withFormat](#) make sure it is the last call in your controller action as the return value of the `withFormat` method is used by the action to dictate what happens next.

Using the Accept header

Every incoming HTTP request has a special [Accept](#) header that defines what media types (or mime types) a client can "accept". In older browsers this is typically:

```
*/*
```

which simply means anything. However, newer browsers send more interesting values such as this one sent by Firefox:

```
text/xml, application/xml, application/xhtml+xml, text/html;q=0.9, \
text/plain;q=0.8, image/png, */*;q=0.5
```

This particular accept header is unhelpful because it indicates that XML is the preferred response format whereas the user is really expecting HTML. That's why Grails ignores the accept header by default for browsers. However, non-browser clients are typically more specific in their requirements and can send accept headers such as

```
application/json
```

As mentioned the default configuration in Grails is to ignore the accept header for browsers. This is done by the configuration setting `grails.mime.disable.accept.header.userAgents`, which is configured to detect the major rendering engines and ignore their `ACCEPT` headers. This allows Grails' content negotiation to continue to work for non-browser clients:

```
grails.mime.disable.accept.header.userAgents = ['Gecko', 'WebKit', 'Presto', 'Trident']
```

For example, if it sees the accept header above ('application/json') it will set `format` to `json` as you'd expect. And of course this works with the `withFormat()` method in just the same way as when the `format` URL parameter is set (although the URL parameter takes precedence).

An accept header of `'*/*'` results in a value of `all` for the `format` property.

If the accept header is used but contains no registered content types, Grails will assume a broken browser is making the request and will set the HTML format - note that this is different from how the other content negotiation modes work as those would activate the "all" format!

Request format vs. Response format

As of Grails 2.0, there is a separate notion of the *request* format and the *response* format. The request format is dictated by the `CONTENT_TYPE` header and is typically used to detect if the incoming request can be parsed into XML or JSON, whilst the response format uses the file extension, format parameter or `ACCEPT` header to attempt to deliver an appropriate response to the client.

The [withFormat](#) available on controllers deals specifically with the response format. If you wish to add logic that deals with the request format then you can do so using a separate `withFormat` method available on the request:

```
request.withFormat {
  xml {
    // read XML
  }
}
```

```

    }
    json {
        // read JSON
    }
}

```

Content Negotiation with URI Extensions

Grails also supports content negotiation using URI extensions. For example given the following URI:

```
/book/list.xml
```

This works as a result of the default URL Mapping definition which is:

```
"/$controller/$action?/$id?(.$format)?"{
```

Note the inclusion of the `format` variable in the path. If you do not wish to use content negotiation via the file extension then simply remove this part of the URL mapping:

```
"/$controller/$action?/$id?"{
```

Testing Content Negotiation

To test content negotiation in a unit or integration test (see the section on [Testing](#)) you can either manipulate the incoming request headers:

```

void testJavascriptOutput() {
    def controller = new TestController()
    controller.request.addHeader "Accept",
        "text/javascript, text/html, application/xml, text/xml, */*"

    controller.testAction()
    assertEquals "alert('hello')", controller.response.contentAsString
}

```

Or you can set the format parameter to achieve a similar effect:

```

void testJavascriptOutput() {
    def controller = new TestController()
    controller.params.format = 'js'

    controller.testAction()
    assertEquals "alert('hello')", controller.response.contentAsString
}

```

8 Traits

Overview

Grails provides a number of traits which provide access to properties and behavior that may be accessed from various Grails artefacts as well as arbitrary Groovy classes which are part of a Grails project. Many of these traits are automatically added to Grails artefact classes (like controllers and taglibs, for example) and are easy to add to other classes.

8.1 Traits Provided by Grails

Grails artefacts are automatically augmented with certain traits at compile time.

Domain Class Traits

- [grails.artefact.DomainClass](#)

- [grails.web.databinding.WebDataBinding](#)
- [org.grails.datastore.gorm.GormEntity](#)
- [org.grails.datastore.gorm.GormValidateable](#)

Controller Traits

- [grails.artefact.gsp.TagLibraryInvoker](#)
- [grails.artefact.AsyncController](#)
- [grails.artefact.controller.RestResponder](#)
- [grails.artefact.Controller](#)

Interceptor Trait

- [grails.artefact.Interceptor](#)

Tag Library Trait

- [grails.artefact.TagLibrary](#)

Below is a list of other traits provided by the framework. The javadocs provide more detail about methods and properties related to each trait.

Trait	Brief Description
grails.web.api.WebAttributes	Common Web Attributes
grails.web.api.ServletAttributes	Servlet API Attributes
grails.web.databinding.DataBinder	Data Binding API
grails.artefact.controller.support.RequestForwarder	Request Forwarding API
grails.artefact.controller.support.ResponseRedirector	Response Redirecting API
grails.artefact.controller.support.ResponseRenderer	Response Rendering API
grails.validation.Validateable	Validation API

8.1.1 WebAttributes Trait Example

[WebAttributes](#) is one of the traits provided by the framework. Any Groovy class may implement this trait to inherit all of the properties and behaviors provided by the trait.

```
src/main/groovy/demo/Helper.groovy

package demo

import grails.web.api.WebAttributes

class Helper implements WebAttributes {

    List<String> getControllerNames() {
        // There is no need to pass grailsApplication as an argument
        // or otherwise inject the grailsApplication property. The
        // WebAttributes trait provides access to grailsApplication.
        grailsApplication.getArtefacts('Controller')*.name
    }
}
```

The traits are compatible with static compilation...

```
src/main/groovy/demo/Helper.groovy

package demo

import grails.web.api.WebAttributes
import groovy.transform.CompileStatic

@CompileStatic
class Helper implements WebAttributes {

    List<String> getControllerNames() {
        // There is no need to pass grailsApplication as an argument
        // or otherwise inject the grailsApplication property. The
        // WebAttributes trait provides access to grailsApplication.
        grailsApplication.getArtefacts('Controller')*.name
    }
}
```

9 REST

REST is not really a technology in itself, but more an architectural pattern. REST is very simple and just involves using plain XML or JSON as a communication medium, combined with URL patterns that are "representational" of the underlying system, and HTTP methods such as GET, PUT, POST and DELETE.

Each HTTP method maps to an action type. For example GET for retrieving data, POST for creating data, PUT for updating and so on.

Grails includes flexible features that make it easy to create RESTful APIs. Creating a RESTful resource can be as simple as one line of code, as demonstrated in the next section.

9.1 Domain classes as REST resources

The easiest way to create a RESTful API in Grails is to expose a domain class as a REST resource. This can be done by adding the `grails.rest.Resource` transformation to any domain class:

```
import grails.rest.*

@Resource(uri='/books')
class Book {

    String title

    static constraints = {
        title blank:false
    }
}
```

Simply by adding the `Resource` transformation and specifying a URI, your domain class will automatically be available as a REST resource in either XML or JSON formats. The transformation will automatically register the necessary [RESTful URL mapping](#) and create a

controller called `BookController`.

You can try it out by adding some test data to `Bootstrap.groovy`:

```
def init = { servletContext ->
    new Book(title:"The Stand").save()
    new Book(title:"The Shining").save()
}
```

And then hitting the URL <http://localhost:8080/books/1>, which will render the response like:

```
<?xml version="1.0" encoding="UTF-8"?>
<book id="1">
  <title>The Stand</title>
</book>
```

If you change the URL to <http://localhost:8080/books/1.json> you will get a JSON response such as:

```
{"id":1,"title":"The Stand"}
```

If you wish to change the default to return JSON instead of XML, you can do this by setting the `formats` attribute of the `Resource` transformation:

```
import grails.rest.*

@Resource(uri='/books', formats=['json', 'xml'])
class Book {
    ...
}
```

With the above example JSON will be prioritized. The list that is passed should contain the names of the formats that the resource should expose. The names of formats are defined in the `grails.mime.types` setting of `application.groovy`:

```
grails.mime.types = [
    ...
    json:          ['application/json', 'text/json'],
    ...
    xml:           ['text/xml', 'application/xml']
]
```

See the section on [Configuring Mime Types](#) in the user guide for more information.

Instead of using the file extension in the URI, you can also obtain a JSON response using the `ACCEPT` header. Here's an example using the Unix `curl` tool:

```
$ curl -i -H "Accept: application/json" localhost:8080/books/1
{"id":1,"title":"The Stand"}
```

This works thanks to Grails' [Content Negotiation](#) features.

You can create a new resource by issuing a `POST` request:

```
$ curl -i -X POST -H "Content-Type: application/json" -d '{"title":"Along Came A Spider"}' localhost:8080/books
HTTP/1.1 201 Created
Server: Apache-Coyote/1.1
...
```

Updating can be done with a `PUT` request:

```
$ curl -i -X PUT -H "Content-Type: application/json" -d '{"title":"Along Came A Spider"}' localhost:8080/books/1
HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
...
```

Finally a resource can be deleted with `DELETE` request:

```
$ curl -i -X DELETE localhost:8080/books/1
HTTP/1.1 204 No Content
```

Server: Apache-Coyote/1.1

...

As you can see, the `Resource` transformation enables all of the HTTP method verbs on the resource. You can enable only read-only capabilities by setting the `readOnly` attribute to `true`:

```
import grails.rest.*

@Resource(uri='/books', readOnly=true)
class Book {
    ...
}
```

In this case `POST`, `PUT` and `DELETE` requests will be forbidden.

9.2 Mapping to REST resources

If you prefer to keep the declaration of the URL mapping in your `UrlMappings.groovy` file then simply removing the `uri` attribute of the `Resource` transformation and adding the following line to `UrlMappings.groovy` will suffice:

```
"/books"(resources:"book")
```

Extending your API to include more end points then becomes trivial:

```
"/books"(resources:"book") {
    "/publisher"(controller:"publisher", method:"GET")
}
```

The above example will expose the URI `/books/1/publisher`.

A more detailed explanation on [creating RESTful URL mappings](#) can be found in the [URL Mappings section](#) of the user guide.

9.3 Linking to REST resources from GSP pages

The `link` tag offers an easy way to link to any domain class resource:

```
<g:link resource="${book}">My Link</g:link>
```

However, currently you cannot use `g:link` to link to the `DELETE` action and most browsers do not support sending the `DELETE` method directly.

The best way to accomplish this is to use a form submit:

```
<form action="/book/2" method="post">
    <input type="hidden" name="_method" value="DELETE"/>
</form>
```

Grails supports overriding the request method via the hidden `_method` parameter. This is for browser compatibility purposes. This is useful when using restful resource mappings to create powerful web interfaces. To make a link fire this type of event, perhaps capture all click events for links with a `data-method` attribute and issue a form submit via JavaScript.

9.4 Versioning REST resources

A common requirement with a REST API is to expose different versions at the same time. There are a few ways this can be achieved in Grails.

Versioning using the URI

A common approach is to use the URI to version APIs (although this approach is discouraged in favour of Hypermedia). For example, you can define the following URL mappings:

```
"/books/v1"(resources:"book", namespace:'v1')
"/books/v2"(resources:"book", namespace:'v2')
```

That will match the following controllers:

```
package myapp.v1

class BookController {
    static namespace = 'v1'
}

package myapp.v2

class BookController {
    static namespace = 'v2'
}
```

This approach has the disadvantage of requiring two different URI namespaces for your API.

Versioning with the Accept-Version header

As an alternative Grails supports the passing of an `Accept-Version` header from clients. For example you can define the following URL mappings:

```
"/books"(version:'1.0', resources:"book", namespace:'v1')
"/books"(version:'2.0', resources:"book", namespace:'v2')
```

Then in the client simply pass which version you need using the `Accept-Version` header:

```
$ curl -i -H "Accept-Version: 1.0" -X GET http://localhost:8080/books
```

Versioning using Hypermedia / Mime Types

Another approach to versioning is to use Mime Type definitions to declare the version of your custom media types (see the section on "Hypermedia as the Engine of Application State" for more information about Hypermedia concepts). For example, in `application.groovy` you can declare a custom Mime Type for your resource that includes a version parameter (the 'v' parameter):

```
grails.mime.types = [
    all: '*/*',
    book: "application/vnd.books.org.book+json;v=1.0",
    bookv2: "application/vnd.books.org.book+json;v=2.0",
    ...
]
```

It is critical that place your new mime types after the 'all' Mime Type because if the Content Type of the request cannot be established then the first entry in the map is used for the response. If you have your new Mime Type at the top then Grails will always try and send back your new Mime Type if the requested Mime Type cannot be established.

Then override the renderer (see the section on "Customizing Response Rendering" for more information on custom renderers) to send back the custom Mime Type in

`grails-app/conf/spring/resources.groovy`:

```
import grails.rest.render.json.*
import grails.web.mime.*

beans = {
    bookRendererV1(JsonRenderer, myapp.v1.Book, new MimeType("application/vnd.books.org.book+json", [v:"1.0"]))
    bookRendererV2(JsonRenderer, myapp.v2.Book, new MimeType("application/vnd.books.org.book+json", [v:"2.0"]))
}
```


Then update the list of acceptable response formats in your controller:

```
class BookController extends RestfulController {
  static responseFormats = ['json', 'xml', 'book', 'bookv2']
  // ...
}
```

Then using the `Accept` header you can specify which version you need using the Mime Type:

```
$ curl -i -H "Accept: application/vnd.books.org.book+json;v=1.0" -X GET http://localhost:8080/books
```

9.5 Implementing REST controllers

The `Resource` transformation is a quick way to get started, but typically you'll want to customize the controller logic, the rendering of the response or extend the API to include additional actions.

9.5.1 Extending the `RestfulController` super class

The easiest way to get started doing so is to create a new controller for your resource that extends the `grails.rest.RestfulController` super class. For example:

```
class BookController extends RestfulController<Book> {
  static responseFormats = ['json', 'xml']
  BookController() {
    super(Book)
  }
}
```

To customize any logic you can just override the appropriate action. The following table provides the names of the action names and the URIs they map to:

HTTP Method	URI	Controller Action
GET	/books	index
GET	/books/create	create
POST	/books	save
GET	/books/\${id}	show
GET	/books/\${id}/edit	edit
PUT	/books/\${id}	update
DELETE	/books/\${id}	delete

The `create` and `edit` actions are only needed if the controller exposes an HTML interface.

As an example, if you have a [nested resource](#) then you would typically want to query both

the parent and the child identifiers. For example, given the following URL mapping:

```
"/authors"(resources:'author') {  
  "/books"(resources:'book')  
}
```

You could implement the nested controller as follows:

```
class BookController extends RestfulController {  
  static responseFormats = ['json', 'xml']  
  BookController() {  
    super(Book)  
  }  
  
  @Override  
  protected Book queryForResource(Serializable id) {  
    Book.where {  
      id == id && author.id == params.authorId  
    }.find()  
  }  
}
```

The example above subclasses `RestfulController` and overrides the protected `queryForResource` method to customize the query for the resource to take into account the parent resource.

Customizing Data Binding In A RestfulController Subclass

The `RestfulController` class contains code which does data binding for actions like `save` and `update`. The class defines a `getObjectToBind()` method which returns a value which will be used as the source for data binding. For example, the `update` action does something like this...

```
class RestfulController<T> {  
  def update() {  
    T instance = // retrieve instance from the database...  
  
    instance.properties = getObjectToBind()  
  
    // ...  
  }  
  // ...  
}
```

By default the `getObjectToBind()` method returns the [request](#) object. When the `request` object is used as the binding source, if the request has a body then the body will be parsed and its contents will be used to do the data binding, otherwise the request parameters will be used to do the data binding. Subclasses of `RestfulController` may override the `getObjectToBind()` method and return anything that is a valid binding source, including a [Map](#) or a [DataBindingSource](#). For most use cases binding the request is appropriate but the `getObjectToBind()` method allows for changing that behavior where desired.

Using custom subclass of RestfulController with Resource annotation

You can also customize the behaviour of the controller that backs the `Resource` annotation.

The class must provide a constructor that takes a domain class as its argument. The second constructor is required for supporting `Resource` annotation with `readOnly=true`.

This is a template that can be used for subclassed `RestfulController` classes used in `Resource` annotations:

```
class SubclassRestfulController<T> extends RestfulController<T> {  
  SubclassRestfulController(Class<T> domainClass) {  
    this(domainClass, false)  
  }  
  
  SubclassRestfulController(Class<T> domainClass, boolean readOnly) {
```

```

        super(domainClass, readOnly)
    }
}

```

You can specify the super class of the controller that backs the Resource annotation with the `superClass` attribute.

```

import grails.rest.*

@Resource(uri='/books', superClass=SubclassRestController)
class Book {

    String title

    static constraints = {
        title blank:false
    }
}

```

9.5.2 Implementing REST Controllers Step by Step

If you don't want to take advantage of the features provided by the `RestController` super class, then you can implement each HTTP verb yourself manually. The first step is to create a controller:

```
$ grails create-controller book
```

Then add some useful imports and enable `readOnly` by default:

```

import grails.gorm.transactions.*
import static org.springframework.http.HttpStatus.*
import static org.springframework.http.HttpMethod.*

@Transactional(readOnly = true)
class BookController {
    ...
}

```

Recall that each HTTP verb matches a particular Grails action according to the following conventions:

HTTP Method	URI	Controller Action
GET	/books	index
GET	/books/\${id}	show
GET	/books/create	create
GET	/books/\${id}/edit	edit
POST	/books	save
PUT	/books/\${id}	update
DELETE	/books/\${id}	delete

The `create` and `edit` actions are already required if you plan to implement an HTML interface for the REST resource. They are there in order to render appropriate HTML forms to create and edit a resource. They can be discarded if that is not a requirement.

The key to implementing REST actions is the [respond](#) method introduced in Grails 2.3. The `respond` method tries to produce the most appropriate response for the requested content type (JSON, XML, HTML etc.)

Implementing the 'index' action

For example, to implement the `index` action, simply call the `respond` method passing the list of objects to respond with:

```
def index(Integer max) {
    params.max = Math.min(max ?: 10, 100)
    respond Book.list(params), model:[bookCount: Book.count()]
}
```

Note that in the above example we also use the `model` argument of the `respond` method to supply the total count. This is only required if you plan to support pagination via some user interface.

The `respond` method will, using [Content Negotiation](#), attempt to reply with the most appropriate response given the content type requested by the client (via the `ACCEPT` header or file extension).

If the content type is established to be HTML then a model will be produced such that the action above would be the equivalent of writing:

```
def index(Integer max) {
    params.max = Math.min(max ?: 10, 100)
    [bookList: Book.list(params), bookCount: Book.count()]
}
```

By providing an `index.gsp` file you can render an appropriate view for the given model. If the content type is something other than HTML then the `respond` method will attempt to lookup an appropriate `grails.rest.render.Renderer` instance that is capable of rendering the passed object. This is done by inspecting the `grails.rest.render.RendererRegistry`.

By default there are already renderers configured for JSON and XML, to find out how to register a custom renderer see the section on "Customizing Response Rendering".

Implementing the 'show' action

The `show` action, which is used to display an individual resource by id, can be implemented in one line of Groovy code (excluding the method signature):

```
def show(Book book) {
    respond book
}
```

By specifying the domain instance as a parameter to the action Grails will automatically attempt to lookup the domain instance using the `id` parameter of the request. If the domain instance doesn't exist, then `null` will be passed into the action. The `respond` method will return a 404 error if `null` is passed otherwise once again it will attempt to render an appropriate response. If the format is HTML then an appropriate model will be produced. The following action is functionally equivalent to the above action:

```
def show(Book book) {
    if(book == null) {
        render status:404
    }
    else {

```

```

    return [book: book]
  }
}

```

Implementing the 'save' action

The `save` action creates new resource representations. To start off, simply define an action that accepts a resource as the first argument and mark it as `Transactional` with the `grails.gorm.transactions.Transactional` transform:

```

@Transactional
def save(Book book) {
    ...
}

```

Then the first thing to do is check whether the resource has any [validation errors](#) and if so respond with the errors:

```

if(book.hasErrors()) {
    respond book.errors, view:'create'
}
else {
    ...
}

```

In the case of HTML the 'create' view will be rendered again so the user can correct the invalid input. In the case of other formats (JSON, XML etc.), the errors object itself will be rendered in the appropriate format and a status code of 422 (UNPROCESSABLE_ENTITY) returned.

If there are no errors then the resource can be saved and an appropriate response sent:

```

book.save flush:true
withFormat {
    html {
        flash.message = message(code: 'default.created.message', args: [message(code: 'book.label', default: 'B
        redirect book
    }
    '*' { render status: CREATED }
}

```

In the case of HTML a redirect is issued to the originating resource and for other formats a status code of 201 (CREATED) is returned.

Implementing the 'update' action

The `update` action updates an existing resource representation and is largely similar to the `save` action. First define the method signature:

```

@Transactional
def update(Book book) {
    ...
}

```

If the resource exists then Grails will load the resource, otherwise null is passed. In the case of null, you should return a 404:

```

if(book == null) {
    render status: NOT_FOUND
}
else {
    ...
}

```

Then once again check for errors [validation errors](#) and if so respond with the errors:

```

if(book.hasErrors()) {
    respond book.errors, view:'edit'
}

```

```
else {
    ...
}
```

In the case of HTML the 'edit' view will be rendered again so the user can correct the invalid input. In the case of other formats (JSON, XML etc.) the errors object itself will be rendered in the appropriate format and a status code of 422 (UNPROCESSABLE_ENTITY) returned.

If there are no errors then the resource can be saved and an appropriate response sent:

```
book.save flush:true
withFormat {
    html {
        flash.message = message(code: 'default.updated.message', args: [message(code: 'book.label', default: 'Book')])
        redirect book
    }
    '*' { render status: OK }
}
```

In the case of HTML a redirect is issued to the originating resource and for other formats a status code of 200 (OK) is returned.

Implementing the 'delete' action

The `delete` action deletes an existing resource. The implementation is largely similar to the `update` action, except the `delete()` method is called instead:

```
book.delete flush:true
withFormat {
    html {
        flash.message = message(code: 'default.deleted.message', args: [message(code: 'Book.label', default: 'Book')])
        redirect action:"index", method:"GET"
    }
    '*' { render status: NO_CONTENT }
}
```

Notice that for an HTML response a redirect is issued back to the `index` action, whilst for other content types a response code 204 (NO_CONTENT) is returned.

9.5.3 Generating a REST controller using scaffolding

To see some of these concepts in action and help you get going, the [Scaffolding plugin](#), version 2.0 and above, can generate a REST ready controller for you, simply run the command:

```
$ grails generate-controller <<Domain Class Name>>
```

9.6 Calling REST Services with HttpClient

Calling Grails REST services - as well as third-party services - is very straightforward using the [Micronaut HTTP Client](#). This HTTP client has both a low-level API and a higher level AOP-driven API, making it useful for both simple requests as well as building declarative, type-safe API layers.

To use the Micronaut HTTP client you must have the `micronaut-http-client` dependency on your classpath. Add the following dependency to your `build.gradle` file.

```
build.gradle

implementation 'io.micronaut:micronaut-http-client'
```

Low-level API

The [HttpClient](#) interface forms the basis for the low-level API. This interface declares

methods to help ease executing HTTP requests and receive responses.

The majority of the methods in the `HttpClient` interface returns Reactive Streams Publisher instances, and a sub-interface called `RxHttpClient` is included that provides a variation of the `HttpClient` interface that returns RxJava Flowable types. When using `HttpClient` in a blocking flow, you may wish to call `toBlocking()` to return an instance of [BlockingHttpClient](#).

There are a few ways by which you can obtain a reference to a [HttpClient](#). The most simple way is using the [create](#) method

Creating an HTTP client

```
List<Album> searchWithApi(String searchTerm) {
    String baseUrl = "https://itunes.apple.com/"

    HttpClient client = HttpClient.create(baseUrl.toURL()).toBlocking() (1)

    HttpRequest request = HttpRequest.GET("/search?limit=25&media=music&entity=album&term=${searchTerm}")
    HttpResponse<String> resp = client.exchange(request, String)
    client.close() (2)

    String json = resp.body()
    ObjectMapper objectMapper = new ObjectMapper() (3)
    objectMapper.configure(DeserializationFeature.FAIL_ON_UNKNOWN_PROPERTIES, false)
    SearchResult searchResult = objectMapper.readValue(json, SearchResult)
    searchResult.results
}
```

- 1 Create a new instance of `HttpClient` with the `create` method, and convert to an instance of `BlockingHttpClient` with `toBlocking()`,
- 2 The client should be closed using the `close` method to prevent thread leaking.
- 3 Jackson's [ObjectMapper](#) API can be used to map the raw JSON to POGOs, in this case `SearchResult`

Consult the [Http Client section](#) of the [Micronaut user guide](#) for more information on using the `HttpClient` low-level API.

Declarative API

A declarative HTTP client can be written by adding the [@Client](#) annotation to any interface or abstract class. Using Micronaut's AOP support (see the Micronaut user guide section on [Introduction Advice](#)), the abstract or interface methods will be implemented for you at compilation time as HTTP calls. Declarative clients can return data-bound POGOs (or POJOs) without requiring special handling from the calling code.

```
package example.grails

import io.micronaut.http.annotation.Get
import io.micronaut.http.client.annotation.Client

@Client("https://start.grails.org")
interface GrailsAppForgeClient {

    @Get("/{version}/profiles")
    List<Map> profiles(String version)
}
```

Note that HTTP client methods are annotated with the appropriate HTTP method, such as `@Get` or `@Post`.

To use a client like the one in the above example, simply inject an instance of the client into any bean using the `@Autowired` annotation.

```
@Autowired GrailsAppForgeClient appForgeClient

List<Map> profiles(String grailsVersion) {
    respond appForgeClient.profiles(grailsVersion)
}
```

For more details on writing and using declarative clients, consult the [Http Client section](#) of the [Micronaut user guide](#).

9.7 The REST Profile

Since Grails 3.1, Grails supports a tailored profile for creating REST applications that provides a more focused set of dependencies and commands.

To get started with a REST API-type application:

```
$ grails create-restapi my-api
```

This will create a new REST application that provides the following features:

- Default set of commands for creating and generating REST endpoints
- Defaults to using JSON views for rendering responses (see the next section)
- Fewer plugins than a default Grails Web-style application (no GSP, no Asset Pipeline, nothing HTML related)

You will notice for example in the `grails-app/views` directory that there are `*.gson` files for rendering the default index page and as well as any 404 and 500 errors.

If you issue the following set of commands:

```
$ grails create-domain-class my.api.Book  
$ ./gradlew runCommand -Pargs="generate-all my.api.Book"
```

The `generate-*` commands are only available after adding the `org.grails.plugins:scaffolding` dependency to your project. They are not available by default in a REST application. Also, they will no longer produce `*.gson` files as that was a feature of the REST API-profile. Profiles were removed in Grails 6.

Instead of CRUD HTML interface a REST endpoint is generated that produces JSON responses. In addition, the generated functional and unit tests by default test the REST endpoint.

9.8 JSON Views

As mentioned in the previous section the REST profile by default uses JSON views to render JSON responses. These play a similar role to GSP, but instead are optimized for outputting JSON responses instead of HTML.

You can continue to separate your application in terms of MVC, with the logic of your application residing in controllers and services, whilst view related matters are handled by JSON views.

JSON views also provide the flexibility to easily customize the JSON presented to clients without having to resort to relatively complex marshalling libraries like Jackson or Grails' marshaller API.

Since Grails 3.1, JSON views are considered by the Grails team the best way to present JSON output for the client, the section on writing custommarshallers has been removed from the user guide. If you are looking for information on that topic, see [the Grails 3.0.x guide](#).

9.8.1 Getting Started

If you are using the REST application, then the JSON views plugin will already be included and you can skip the remainder of this section. Otherwise you will need to modify your `build.gradle` to include the necessary plugin to activate JSON views:

```
implementation 'org.grails.plugins:views-json:1.0.0' // or whatever is the latest version
```

The [source code repository for JSON views](#) can be found on Github if you are looking for more documentation and contributions

In order to compile JSON views for production deployment you should also activate the Gradle plugin by first modifying the `buildscript` block:

```
buildscript {  
    ...  
    dependencies {  
        ...  
        classpath "org.grails.plugins:views-gradle:1.0.0"  
    }  
}
```

Then apply the `org.grails.plugins.views-json` Gradle plugin after any Grails core gradle plugins:

```
...  
apply plugin: "org.grails.grails-web"  
apply plugin: "org.grails.plugins.views-json"
```

This will add a `compileGsonViews` task to Gradle, which is invoked prior to creating the production JAR or WAR file.

9.8.2 Creating JSON Views

JSON views go into the `grails-app/views` directory and end with the `.json` suffix. They are regular Groovy scripts and can be opened in any Groovy editor.

Example JSON view:

```
json.person {  
    name "bob"  
}
```

To open them in the Groovy editor in IntelliJ IDEA, double click on the file and when asked which file to associate it with, choose "Groovy"

The above JSON view produces:

```
{"person":{"name":"bob"}}
```

There is an implicit `json` variable which is an instance of [StreamingJsonBuilder](#).

Example usages:

```
json(1,2,3) == "[1,2,3]"  
json { name "Bob" } == '{"name":"Bob"}'  
json([1,2,3]) { n it } == ' [{"n":1}, {"n":2}, {"n":3}] '
```

Refer to the API documentation on [StreamingJsonBuilder](#) for more information about what is possible.

9.8.3 JSON View Templates

You can define templates starting with underscore `_`. For example given the following template called `_person.gson`:

```
model {
    Person person
}
json {
    name person.name
    age person.age
}
```

You can render it with a view as follows:

```
model {
    Family family
}
json {
    name family.father.name
    age family.father.age
    oldestChild g.render(template:"person", model:[person: family.children.max { Person p -> p.age } ])
    children g.render(template:"person", collection: family.children, var:'person')
}
```

Alternatively for a more concise way to invoke templates, using the `tmpl` variable:

```
model {
    Family family
}
json {
    name family.father.name
    age family.father.age
    oldestChild tmpl.person( family.children.max { Person p -> p.age } )
    children tmpl.person( family.children )
}
```

9.8.4 Rendering Domain Classes with JSON Views

Typically your model may involve one or many domain instances. JSON views provide a render method for rendering these.

For example given the following domain class:

```
class Book {
    String title
}
```

And the following template:

```
model {
    Book book
}

json g.render(book)
```

The resulting output is:

```
{id:1, title:"The Stand"}
```

You can customize the rendering by including or excluding properties:

```
json g.render(book, [includes:['title']])
```

Or by providing a closure to add additional JSON output:

```
json g.render(book) {
    pages 1000
}
```

9.8.5 JSON Views by Convention

There are a few useful conventions you can follow when creating JSON views. For example if you have a domain class called `Book`, then creating a template located at `grails-app/views/book/_book.gson` and using the [respond](#) method will result in rendering the template:

```
def show(Long id) {
    respond Book.get(id)
}
```

In addition if an error occurs during validation by default Grails will try to render a template called `grails-app/views/book/_errors.gson`, otherwise it will try to render `grails-app/views/errors/_errors.gson` if the former doesn't exist.

This is useful because when persisting objects you can `respond` with validation errors to render these aforementioned templates:

```
@Transactional
def save(Book book) {
    if (book.hasErrors()) {
        transactionStatus.setRollbackOnly()
        respond book.errors
    }
    else {
        // valid object
    }
}
```

If a validation error occurs in the above example the `grails-app/views/book/_errors.gson` template will be rendered.

For more information on JSON views (and Markup views), see the [JSON Views user guide](#).

9.9 Customizing Response Rendering

If you are looking for a more low-level API and JSON or Markup views don't suite your needs then you may want to consider implementing a custom renderer.

9.9.1 Customizing the Default Renderers

The default renderers for XML and JSON can be found in the `grails.rest.render.xml` and `grails.rest.render.json` packages respectively. These use the Grails converters (`grails.converters.XML` and `grails.converters.JSON`) by default for response rendering.

You can easily customize response rendering using these default renderers. A common change you may want to make is to include or exclude certain properties from rendering.

Including or Excluding Properties from Rendering

As mentioned previously, Grails maintains a registry of `grails.rest.render.Renderer` instances. There are some default configured renderers and the ability to register or override renderers for a given domain class or even for a collection of domain classes. To include a particular property from rendering you need to register a custom renderer by defining a bean in `grails-app/conf/spring/resources.groovy`:

```
import grails.rest.render.xml.*

beans = {
    bookRenderer(XmlRenderer, Book) {
        includes = ['title']
    }
}
```

The bean name is not important (Grails will scan the application context for all registered renderer beans), but for organizational and readability purposes it is recommended you

name it something meaningful.

To exclude a property, the `excludes` property of the `XmlRenderer` class can be used:

```
import grails.rest.render.xml.*

beans = {
    bookRenderer(XmlRenderer, Book) {
        excludes = ['isbn']
    }
}
```

Customizing the Converters

As mentioned previously, the default renders use the `grails.converters` package under the covers. In other words, under the covers they essentially do the following:

```
import grails.converters.*

...
render book as XML
// or render book as JSON
```

Why the separation between converters and renderers? Well a renderer has more flexibility to use whatever rendering technology you chose. When implementing a custom renderer you could use [Jackson](#), [Gson](#) or any Java library to implement the renderer. Converters on the other hand are very much tied to Grails' own marshalling implementation.

9.9.2 Implementing a Custom Renderer

If you want even more control of the rendering or prefer to use your own marshalling techniques then you can implement your own `Renderer` instance. For example below is a simple implementation that customizes the rendering of the `Book` class:

```
package myapp
import grails.rest.render.*
import grails.web.mime.MimeType

class BookXmlRenderer extends AbstractRenderer<Book> {
    BookXmlRenderer() {
        super(Book, [MimeType.XML, MimeType.TEXT_XML] as MimeType[])
    }

    void render(Book object, RenderContext context) {
        context.contentType = MimeType.XML.name

        def xml = new groovy.xml.MarkupBuilder(context.writer)
        xml.book(id: object.id, title:object.title)
    }
}
```

The `AbstractRenderer` super class has a constructor that takes the class that it renders and the `MimeType(s)` that are accepted (via the `ACCEPT` header or file extension) for the renderer.

To configure this renderer, simply add it is a bean to `grails-app/conf/spring/resources.groovy`:

```
beans = {
    bookRenderer(myapp.BookXmlRenderer)
}
```

The result will be that all `Book` instances will be rendered in the following format:

```
<book id="1" title="The Stand"/>
```

If you change the rendering to a completely different format like the above, then you also need to change the binding if you plan to support POST and PUT requests. Grails will not automatically know how to bind data from a custom XML format to a domain class otherwise. See the section on "Customizing Binding of Resources" for further information.

Container Renderers

A `grails.rest.render.ContainerRenderer` is a renderer that renders responses for containers of objects (lists, maps, collections etc.). The interface is largely the same as the `Renderer` interface except for the addition of the `getComponentType()` method, which should return the "contained" type. For example:

```
class BookListRenderer implements ContainerRenderer<List, Book> {
    Class<List> getTargetType() { List }
    Class<Book> getComponentType() { Book }
    MimeTypes[] getMimeTypes() { [ MimeTypes.XML ] as MimeTypes[] }
    void render(List object, RenderContext context) {
        ....
    }
}
```

9.9.3 Using GSP to Customize Rendering

You can also customize rendering on a per action basis using Groovy Server Pages (GSP). For example given the `show` action mentioned previously:

```
def show(Book book) {
    respond book
}
```

You could supply a `show.xml.gsp` file to customize the rendering of the XML:

```
<%@page contentType="application/xml"%>
<book id="${book.id}" title="${book.title}"/>
```

9.10 Hypermedia as the Engine of Application State

[HATEOAS](#), an abbreviation for Hypermedia as the Engine of Application State, is a common pattern applied to REST architectures that uses hypermedia and linking to define the REST API.

Hypermedia (also called Mime or Media Types) are used to describe the state of a REST resource, and links tell clients how to transition to the next state. The format of the response is typically JSON or XML, although standard formats such as [Atom](#) and/or [HAL](#) are frequently used.

9.10.1 HAL Support

[HAL](#) is a standard exchange format commonly used when developing REST APIs that follow HATEOAS principals. An example HAL document representing a list of orders can be seen below:

```
{
  "_links": {
    "self": { "href": "/orders" },
    "next": { "href": "/orders?page=2" },
    "find": {
      "href": "/orders/{id}",
      "templated": true
    },
    "admin": [{
      "href": "/admins/2",
      "title": "Fred"
    }, {
      "href": "/admins/5",
      "title": "Kate"
    }]
  },
  "currentlyProcessing": 14,
  "shippedToday": 20,
  "_embedded": {
    "order": [{
      "_links": {
```

```

        "self": { "href": "/orders/123" },
        "basket": { "href": "/baskets/98712" },
        "customer": { "href": "/customers/7809" }
    },
    "total": 30.00,
    "currency": "USD",
    "status": "shipped"
}, {
    "_links": {
        "self": { "href": "/orders/124" },
        "basket": { "href": "/baskets/97213" },
        "customer": { "href": "/customers/12369" }
    },
    "total": 20.00,
    "currency": "USD",
    "status": "processing"
}]
}
}
}

```

Exposing Resources Using HAL

To return HAL instead of regular JSON for a resource you can simply override the renderer in `grails-app/conf/spring/resources.groovy` with an instance of `grails.rest.render.hal.HalJsonRenderer` (or `HalXmlRenderer` for the XML variation):

```

import grails.rest.render.hal.*
beans = {
    halBookRenderer(HalJsonRenderer, rest.test.Book)
}

```

You will also need to update the acceptable response formats for the resource so that the HAL format is included. Not doing so will result in a 406 - Not Acceptable response being returned from the server.

This can be done by setting the `formats` attribute of the `Resource` transformation:

```

import grails.rest.*

@Resource(uri='/books', formats=['json', 'xml', 'hal'])
class Book {
    ...
}

```

Or by updating the `responseFormats` in the controller:

```

class BookController extends RestfulController {
    static responseFormats = ['json', 'xml', 'hal']

    // ...
}

```

With the bean in place requesting the HAL content type will return HAL:

```

$ curl -i -H "Accept: application/hal+json" http://localhost:8080/books/1

HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Content-Type: application/hal+json;charset=ISO-8859-1

{
  "_links": {
    "self": {
      "href": "http://localhost:8080/books/1",
      "hreflang": "en",
      "type": "application/hal+json"
    }
  },
  "title": "\"The Stand\""
}

```

To use HAL XML format simply change the renderer:

```

import grails.rest.render.hal.*
beans = {
    halBookRenderer(HalXmlRenderer, rest.test.Book)
}

```

Rendering Collections Using HAL

To return HAL instead of regular JSON for a list of resources you can simply override the renderer in `grails-app/conf/spring/resources.groovy` with an instance of

`grails.rest.render.hal.HalJsonCollectionRenderer`:

```
import grails.rest.render.hal.*
beans = {
    halBookCollectionRenderer(HalJsonCollectionRenderer, rest.test.Book)
}
```

With the bean in place requesting the HAL content type will return HAL:

```
$ curl -i -H "Accept: application/hal+json" http://localhost:8080/books
HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Content-Type: application/hal+json;charset=UTF-8
Transfer-Encoding: chunked
Date: Thu, 17 Oct 2013 02:34:14 GMT
```

```
{
  "_links": {
    "self": {
      "href": "http://localhost:8080/books",
      "hreflang": "en",
      "type": "application/hal+json"
    }
  },
  "_embedded": {
    "book": [
      {
        "_links": {
          "self": {
            "href": "http://localhost:8080/books/1",
            "hreflang": "en",
            "type": "application/hal+json"
          }
        },
        "title": "The Stand"
      },
      {
        "_links": {
          "self": {
            "href": "http://localhost:8080/books/2",
            "hreflang": "en",
            "type": "application/hal+json"
          }
        },
        "title": "Infinite Jest"
      },
      {
        "_links": {
          "self": {
            "href": "http://localhost:8080/books/3",
            "hreflang": "en",
            "type": "application/hal+json"
          }
        },
        "title": "Walden"
      }
    ]
  }
}
```

Notice that the key associated with the list of `Book` objects in the rendered JSON is `book` which is derived from the type of objects in the collection, namely `Book`. In order to customize the value of this key assign a value to the `collectionName` property on the `HalJsonCollectionRenderer` bean as shown below:

```
import grails.rest.render.hal.*
beans = {
    halBookCollectionRenderer(HalCollectionJsonRenderer, rest.test.Book) {
        collectionName = 'publications'
    }
}
```

With that in place the rendered HAL will look like the following:

```
$ curl -i -H "Accept: application/hal+json" http://localhost:8080/books
HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Content-Type: application/hal+json;charset=UTF-8
```

Transfer-Encoding: chunked
Date: Thu, 17 Oct 2013 02:34:14 GMT

```
{
  "_links": {
    "self": {
      "href": "http://localhost:8080/books",
      "hreflang": "en",
      "type": "application/hal+json"
    }
  },
  "_embedded": {
    "publications": [
      {
        "_links": {
          "self": {
            "href": "http://localhost:8080/books/1",
            "hreflang": "en",
            "type": "application/hal+json"
          }
        },
        "title": "The Stand"
      },
      {
        "_links": {
          "self": {
            "href": "http://localhost:8080/books/2",
            "hreflang": "en",
            "type": "application/hal+json"
          }
        },
        "title": "Infinite Jest"
      },
      {
        "_links": {
          "self": {
            "href": "http://localhost:8080/books/3",
            "hreflang": "en",
            "type": "application/hal+json"
          }
        },
        "title": "Walden"
      }
    ]
  }
}
```

Using Custom Media / Mime Types

If you wish to use a custom Mime Type then you first need to declare the Mime Types in `grails-app/conf/application.groovy`:

```
grails.mime.types = [
  all:      "**/*",
  book:     "application/vnd.books.org.book+json",
  bookList: "application/vnd.books.org.booklist+json",
  ...
]
```

It is critical that place your new mime types after the 'all' Mime Type because if the Content Type of the request cannot be established then the first entry in the map is used for the response. If you have your new Mime Type at the top then Grails will always try and send back your new Mime Type if the requested Mime Type cannot be established.

Then override the renderer to return HAL using the custom Mime Types:

```
import grails.rest.render.hal.*
import grails.web.mime.*

beans = {
  halBookRenderer(HalJsonRenderer, rest.test.Book, new MimeType("application/vnd.books.org.book+json", [v:"1.0"]))
  halBookListRenderer(HalJsonCollectionRenderer, rest.test.Book, new MimeType("application/vnd.books.org.booklist+json"))
}
```

In the above example the first bean defines a HAL renderer for a single book instance that returns a Mime Type of `application/vnd.books.org.book+json`. The second bean defines the Mime Type used to render a collection of books (in this case `application/vnd.books.org.booklist+json`).

`application/vnd.books.org.booklist+json` is an example of a media-range (

<http://www.w3.org/Protocols/rfc2616/rfc2616.html> - Header Field Definitions). This example uses entity (book) and operation (list) to form the media-range values but in reality, it may not be necessary to create a separate Mime type for each operation. Further, it may not be necessary to create Mime types at the entity level. See the section on "Versioning REST resources" for further information about how to define your own Mime types.

With this in place issuing a request for the new Mime Type returns the necessary HAL:

```
$ curl -i -H "Accept: application/vnd.books.org.book+json" http://localhost:8080/books/1

HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Content-Type: application/vnd.books.org.book+json;charset=ISO-8859-1

{
  "_links": {
    "self": {
      "href": "http://localhost:8080/books/1",
      "hreflang": "en",
      "type": "application/vnd.books.org.book+json"
    }
  },
  "title": "\"The Stand\""
}
```

Customizing Link Rendering

An important aspect of HATEOAS is the usage of links that describe the transitions the client can use to interact with the REST API. By default the `HalJsonRenderer` will automatically create links for you for associations and to the resource itself (using the "self" relationship).

However you can customize link rendering using the `link` method that is added to all domain classes annotated with `grails.rest.Resource` or any class annotated with `grails.rest.Linkable`. For example, the `show` action can be modified as follows to provide a new link in the resulting output:

```
def show(Book book) {
    book.link rel:'publisher', href: g.createLink(absolute: true, resource:"publisher", params:[bookId: book.id])
    respond book
}
```

Which will result in output such as:

```
{
  "_links": {
    "self": {
      "href": "http://localhost:8080/books/1",
      "hreflang": "en",
      "type": "application/vnd.books.org.book+json"
    }
  },
  "publisher": {
    "href": "http://localhost:8080/books/1/publisher",
    "hreflang": "en"
  }
},
"title": "\"The Stand\""
}
```

The `link` method can be passed named arguments that match the properties of the `grails.rest.Link` class.

9.10.2 Atom Support

[Atom](#) is another standard interchange format used to implement REST APIs. An example of Atom output can be seen below:

```
<?xml version="1.0" encoding="utf-8"?>
<feed xmlns="http://www.w3.org/2005/Atom">
```

```

<title>Example Feed</title>
<link href="http://example.org/" />
<updated>2003-12-13T18:30:02Z</updated>
<author>
  <name>John Doe</name>
</author>
<id>urn:uuid:60a76c80-d399-11d9-b93C-0003939e0af6</id>

<entry>
  <title>Atom-Powered Robots Run Amok</title>
  <link href="http://example.org/2003/12/13/atom03" />
  <id>urn:uuid:1225c695-cfb8-4ebb-aaaa-80da344efa6a</id>
  <updated>2003-12-13T18:30:02Z</updated>
  <summary>Some text.</summary>
</entry>
</feed>

```

To use Atom rendering again simply define a custom renderer:

```

import grails.rest.render.atom.*
beans = {
    halBookRenderer(AtomRenderer, rest.test.Book)
    halBookListRenderer(AtomCollectionRenderer, rest.test.Book)
}

```

9.10.3 Vnd.Error Support

[Vnd.Error](#) is a standardised way of expressing an error response.

By default when a validation error occurs when attempting to POST new resources then the errors object will be sent back allow with a 422 respond code:

```
$ curl -i -H "Accept: application/json" -H "Content-Type: application/json" -X POST -d "" http://localhost:8080/bo
```

```

HTTP/1.1 422 Unprocessable Entity
Server: Apache-Coyote/1.1
Content-Type: application/json; charset=ISO-8859-1

```

```

{
  "errors": [
    {
      "object": "rest.test.Book",
      "field": "title",
      "rejected-value": null,
      "message": "Property [title] of class [class rest.test.Book] cannot be null"
    }
  ]
}

```

If you wish to change the format to Vnd.Error then simply register

grails.rest.render.errors.VndErrorJsonRenderer bean in grails-app/conf/spring/resources.groovy:

```

beans = {
    vndJsonErrorRenderer(grails.rest.render.errors.VndErrorJsonRenderer)
    // for Vnd.Error XML format
    vndXmlErrorRenderer(grails.rest.render.errors.VndErrorXmlRenderer)
}

```

Then if you alter the client request to accept Vnd.Error you get an appropriate response:

```
$ curl -i -H "Accept: application/vnd.error+json,application/json" -H "Content-Type: application/json" -X POST -d "
HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Content-Type: application/vnd.error+json; charset=ISO-8859-1

```

```

[
  {
    "logref": "book.nullable",
    "message": "Property [title] of class [class rest.test.Book] cannot be null",
    "_links": {
      "resource": {
        "href": "http://localhost:8080/rest-test/books"
      }
    }
  }
]

```

9.11 Customizing Binding of Resources

The framework provides a sophisticated but simple mechanism for binding REST requests to domain objects and command objects. One way to take advantage of this is to bind the request property in a controller the properties of a domain class. Given the following XML as the body of the request, the `createBook` action will create a new `Book` and assign "The Stand" to the `title` property and "Stephen King" to the `authorName` property.

```
<?xml version="1.0" encoding="UTF-8"?>
<book>
  <title>The Stand</title>
  <authorName>Stephen King</authorName>
</book>
```

```
class BookController {
  def createBook() {
    def book = new Book()
    book.properties = request

    // ...
  }
}
```

Command objects will automatically be bound with the body of the request:

```
class BookController {
  def createBook(BookCommand book) {

    // ...
  }
}

class BookCommand {
  String title
  String authorName
}
```

If the command object type is a domain class and the root element of the XML document contains an `id` attribute, the `id` value will be used to retrieve the corresponding persistent instance from the database and then the rest of the document will be bound to the instance. If no corresponding record is found in the database, the command object reference will be null.

```
<?xml version="1.0" encoding="UTF-8"?>
<book id="42">
  <title>Walden</title>
  <authorName>Henry David Thoreau</authorName>
</book>

class BookController {
  def updateBook(Book book) {
    // The book will have been retrieved from the database and updated
    // by doing something like this:
    //
    // book == Book.get('42')
    // if(book != null) {
    //   book.properties = request
    // }
    //
    // the code above represents what the framework will
    // have done. There is no need to write that code.

    // ...
  }
}
```

The data binding depends on an instance of the [DataBindingSource](#) interface created by an instance of the [DataBindingSourceCreator](#) interface. The specific implementation of `DataBindingSourceCreator` will be selected based on the `contentType` of the request. Several implementations are provided to handle common content types. The default implementations will be fine for most use cases. The following table lists the content types

which are supported by the core framework and which `DataBindingSourceCreator` implementations are used for each. All of the implementation classes are in the `org.grails.databinding.bindingsource` package.

Content Type(s)	Bean Name	<code>DataBindingSourceCreator</code> Impl.
application/xml, text/xml	xmlDataBindingSourceCreator	XmlDataBindingSourceCreator
application/json, text/json	jsonDataBindingSourceCreator	JsonDataBindingSourceCreator
application/hal+json	halJsonDataBindingSourceCreator	HalJsonDataBindingSourceCreator
application/hal+xml	halXmlDataBindingSourceCreator	HalXmlDataBindingSourceCreator

In order to provide your own `DataBindingSourceCreator` for any of those content types, write a class which implements `DataBindingSourceCreator` and register an instance of that class in the Spring application context. If you are replacing one of the existing helpers, use the corresponding bean name from above. If you are providing a helper for a content type other than those accounted for by the core framework, the bean name may be anything that you like but you should take care not to conflict with one of the bean names above.

The `DataBindingSourceCreator` interface defines just 2 methods:

```
package org.grails.databinding.bindingsource

import grails.web.mime.MimeType
import grails.databinding.DataBindingSource

/**
 * A factory for DataBindingSource instances
 *
 * @since 2.3
 * @see DataBindingSourceRegistry
 * @see DataBindingSource
 */
interface DataBindingSourceCreator {

    /**
     * `return All of the {@link MimeType} supported by this helper
     */
    MimeType[] getMimeTypeTypes()

    /**
     * Creates a DataBindingSource suitable for binding bindingSource to bindingTarget
     *
     * @param mimeType a mime type
     * @param bindingTarget the target of the data binding
     * @param bindingSource the value being bound
     * @return a DataBindingSource
     */
    DataBindingSource createDataBindingSource(MimeType mimeType, Object bindingTarget, Object bindingSource)
}
```

[AbstractRequestBodyDataBindingSourceCreator](#) is an abstract class designed to be extended to simplify writing custom `DataBindingSourceCreator` classes. Classes which extend `AbstractRequestBodyDataBindingSourceCreator` need to implement a method named `createBindingSource` which accepts an `InputStream` as an argument and returns a `DataBindingSource`

as well as implementing the `getMimeTypes` method described in the `DataBindingSourceCreator` interface above. The `InputStream` argument to `createBindingSource` provides access to the body of the request.

The code below shows a simple implementation.

`src/main/groovy/com/demo/myapp/databinding/MyCustomDataBindingSourceCreator.groovy`

```
package com.demo.myapp.databinding

import grails.web.mime.MimeType
import grails.databinding.DataBindingSource
import org...databinding.SimpleMapDataBindingSource
import org...databinding.bindingsource.AbstractRequestBodyDataBindingSourceCreator

/**
 * A custom DataBindingSourceCreator capable of parsing key value pairs out of
 * a request body containing a comma separated list of key:value pairs like:
 *
 * name:Herman,age:99,town:STL
 *
 */
class MyCustomDataBindingSourceCreator extends AbstractRequestBodyDataBindingSourceCreator {

    @Override
    public MimeType[] getMimeTypes() {
        [new MimeType('text/custom+demo+csv')] as MimeType[]
    }

    @Override
    protected DataBindingSource createBindingSource(InputStream inputStream) {
        def map = [:]

        def reader = new InputStreamReader(inputStream)

        // this is an obviously naive parser and is intended
        // for demonstration purposes only.

        reader.eachLine { line ->
            def keyValuePairs = line.split(',')
            keyValuePairs.each { keyValuePair ->
                if(keyValuePair?.trim()) {
                    def keyValuePieces = keyValuePair.split(':')
                    def key = keyValuePieces[0].trim()
                    def value = keyValuePieces[1].trim()
                    map<<key>> = value
                }
            }
        }

        // create and return a DataBindingSource which contains the parsed data
        new SimpleMapDataBindingSource(map)
    }
}
```

An instance of `MyCustomDataSourceCreator` needs to be registered in the spring application context.

`grails-app/conf/spring/resources.groovy`

```
beans = {

    myCustomCreator com.demo.myapp.databinding.MyCustomDataBindingSourceCreator

    // ...
}
```

With that in place the framework will use the `myCustomCreator` bean any time a `DataBindingSourceCreator` is needed to deal with a request which has a `contentType` of `"text/custom+demo+csv"`.

9.12 RSS and Atom

No direct support is provided for RSS or Atom within Grails. You could construct RSS or ATOM feeds with the [render](#) method's XML capability.

10 Asynchronous Programming

With modern hardware featuring multiple cores, many programming languages have been adding asynchronous, parallel programming APIs, Groovy being no exception.

Popular asynchronous libraries include:

- RxJava - <http://reactivex.io>
- GPars - <http://gpars.org>
- Reactor - <https://projectreactor.io>

By building on top of these various libraries the [Async features of Grails](#) aim to simplify concurrent programming within the framework, include the concept of Promises, and a unified event model.

In general, since the Reactive programming model is an evolving space, Grails tries to provide generic support for integrating a range of asynchronous libraries and doesn't recommend any single library as they all have various advantages and disadvantages.

For more information on Asynchronous programming with Grails see the user guide for the [Grails Asynchronous Framework](#).

11 Validation

Grails validation capability is built on [Spring's Validator API](#) and data binding capabilities. However Grails takes this further and provides a unified way to define validation "constraints" with its constraints mechanism.

Constraints in Grails are a way to declaratively specify validation rules. Most commonly they are applied to [domain classes](#), however [URL Mappings](#) and [Command Objects](#) also support constraints.

11.1 Declaring Constraints

Within a domain class [constraints](#) are defined with the constraints property that is assigned a code block:

```
class User {
    String login
    String password
    String email
    Integer age

    static constraints = {
        ...
    }
}
```

You then use method calls that match the property name for which the constraint applies in combination with named parameters to specify constraints:

```
class User {
    ...

    static constraints = {
```

```

        login size: 5..15, blank: false, unique: true
        password size: 5..15, blank: false
        email email: true, blank: false
        age min: 18
    }
}

```

In this example we've declared that the `login` property must be between 5 and 15 characters long, it cannot be blank and must be unique. We've also applied other constraints to the `password`, `email` and `age` properties.

By default, all domain class properties are not nullable (i.e. they have an implicit `nullable: false` constraint).

A complete reference for the available constraints can be found in the Quick Reference section under the Constraints heading.

Note that constraints are only evaluated once which may be relevant for a constraint that relies on a value like an instance of `java.util.Date`.

```

class User {
    ...

    static constraints = {
        // this Date object is created when the constraints are evaluated, not
        // each time an instance of the User class is validated.
        birthDate max: new Date()
    }
}

```

A word of warning - referencing domain class properties from constraints

It's very easy to attempt to reference instance variables from the static constraints block, but this isn't legal in Groovy (or Java). If you do so, you will get a `MissingPropertyException` for your trouble. For example, you may try

```

class Response {
    Survey survey
    Answer answer

    static constraints = {
        survey blank: false
        answer blank: false, inList: survey.answers
    }
}

```

See how the `inList` constraint references the instance property `survey`? That won't work. Instead, use a custom [validator](#):

```

class Response {
    ...
    static constraints = {
        survey blank: false
        answer blank: false, validator: { val, obj -> val in obj.survey.answers }
    }
}

```

In this example, the `obj` argument to the custom validator is the domain *instance* that is being validated, so we can access its `survey` property and return a boolean to indicate whether the new value for the `answer` property, `val`, is valid.

11.2 Validating Constraints

Validation Basics

Call the [validate](#) method to validate a domain class instance:

```
def user = new User(params)

if (user.validate()) {
    // do something with user
}
else {
    user.errors.allErrors.each {
        println it
    }
}
```

The `errors` property on domain classes is an instance of the Spring [Errors](#) interface. The `Errors` interface provides methods to navigate the validation errors and also retrieve the original values.

Validation Phases

Within Grails there are two phases of validation, the first one being [data binding](#) which occurs when you bind request parameters onto an instance such as:

```
def user = new User(params)
```

At this point you may already have errors in the `errors` property due to type conversion (such as converting Strings to Dates). You can check these and obtain the original input value using the `Errors` API:

```
if (user.hasErrors()) {
    if (user.errors.hasFieldErrors("login")) {
        println user.errors.getFieldError("login").rejectedValue
    }
}
```

The second phase of validation happens when you call [validate](#) or [save](#). This is when Grails will validate the bound values against the [constraints](#) you defined. For example, by default the [save](#) method calls `validate` before executing, allowing you to write code like:

```
if (user.save()) {
    return user
}
else {
    user.errors.allErrors.each {
        println it
    }
}
```

11.3 Sharing Constraints Between Classes

A common pattern in Grails is to use [Command Objects](#) for validating user-submitted data and then copy the properties of the command object to the relevant domain classes. This often means that your command objects and domain classes share properties and their constraints. You could manually copy and paste the constraints between the two, but that's a very error-prone approach. Instead, make use of Grails' global constraints and import mechanism.

Global Constraints

In addition to defining constraints in domain classes, command objects and [other validateable classes](#), you can also define them in `grails-app/conf/runtime.groovy`:

```
grails.gorm.default.constraints = {
    '*'(nullable: true, size: 1..20)
    myShared(nullable: false, blank: false)
}
```

These constraints are not attached to any particular classes, but they can be easily referenced from any validateable class:


```
class User {
    ...

    static constraints = {
        login shared: "myShared"
    }
}
```

Note the use of the `shared` argument, whose value is the name of one of the constraints defined in `grails.gorm.default.constraints`. Despite the name of the configuration setting, you can reference these shared constraints from any validateable class, such as command objects.

The `'*` constraint is a special case: it means that the associated constraints (`'nullable'` and `'size'` in the above example) will be applied to all properties in all validateable classes. These defaults can be overridden by the constraints declared in a validateable class.

Importing Constraints

Grails 2 introduced an alternative approach to sharing constraints that allows you to import a set of constraints from one class into another.

Let's say you have a domain class like so:

```
class User {
    String firstName
    String lastName
    String passwordHash

    static constraints = {
        firstName blank: false, nullable: false
        lastName blank: false, nullable: false
        passwordHash blank: false, nullable: false
    }
}
```

You then want to create a command object, `UserCommand`, that shares some of the properties of the domain class and the corresponding constraints. You do this with the `importFrom()` method:

```
class UserCommand {
    String firstName
    String lastName
    String password
    String confirmPassword

    static constraints = {
        importFrom User

        password blank: false, nullable: false
        confirmPassword blank: false, nullable: false
    }
}
```

This will import all the constraints from the `User` domain class and apply them to `UserCommand`. The import will ignore any constraints in the source class (`User`) that don't have corresponding properties in the importing class (`UserCommand`). In the above example, only the `'firstName'` and `'lastName'` constraints will be imported into `UserCommand` because those are the only properties shared by the two classes.

If you want more control over which constraints are imported, use the `include` and `exclude` arguments. Both of these accept a list of simple or regular expression strings that are matched against the property names in the source constraints. So for example, if you only wanted to import the `'lastName'` constraint you would use:

```
...
static constraints = {
    importFrom User, include: ["lastName"]
    ...
}
```

or if you wanted all constraints that ended with 'Name':

```
...
static constraints = {
    importFrom User, include: [/.*Name/]
    ...
}
```

Of course, `exclude` does the reverse, specifying which constraints should *not* be imported.

11.4 Validation on the Client

Displaying Errors

Typically if you get a validation error you redirect back to the view for rendering. Once there you need some way of displaying errors. Grails supports a rich set of tags for dealing with errors. To render the errors as a list you can use [renderErrors](#):

```
<g:renderErrors bean="${user}" />
```

If you need more control you can use [hasErrors](#) and [eachError](#):

```
<g:hasErrors bean="${user}">
  <ul>
    <g:eachError var="err" bean="${user}">
      <li>${err}</li>
    </g:eachError>
  </ul>
</g:hasErrors>
```

Highlighting Errors

It is often useful to highlight using a red box or some indicator when a field has been incorrectly input. This can also be done with the [hasErrors](#) by invoking it as a method. For example:

```
<div class='value ${hasErrors(bean:user, field:'login', 'errors')}'>
  <input type="text" name="login" value="${fieldValue(bean:user, field:'login')}" />
</div>
```

This code checks if the `login` field of the `user` bean has any errors and if so it adds an `errors` CSS class to the `div`, allowing you to use CSS rules to highlight the `div`.

Retrieving Input Values

Each error is actually an instance of the [FieldError](#) class in Spring, which retains the original input value within it. This is useful as you can use the error object to restore the value input by the user using the [fieldValue](#) tag:

```
<input type="text" name="login" value="${fieldValue(bean:user, field:'login')}" />
```

This code will check for an existing `FieldError` in the `User` bean and if there is obtain the originally input value for the `login` field.

11.5 Validation and Internationalization

Another important thing to note about errors in Grails is that error messages are not hard coded anywhere. The [FieldError](#) class in Spring resolves messages from message bundles using Grails' [i18n](#) support.

Constraints and Message Codes

The codes themselves are dictated by a convention. For example consider the constraints we looked at earlier:

```
package com.mycompany.myapp

class User {
    ...

    static constraints = {
        login size: 5..15, blank: false, unique: true
        password size: 5..15, blank: false
        email email: true, blank: false
        age min: 18
    }
}
```

If a constraint is violated, Grails looks by convention for a message code:

Constraint	Error Code
blank	className.propertyName.blank
creditCard	className.propertyName.creditCard.invalid
email	className.propertyName.email.invalid
inList	className.propertyName.not.inList
matches	className.propertyName.matches.invalid
max	className.propertyName.max.exceeded
maxSize	className.propertyName.maxSize.exceeded
min	className.propertyName.min.notmet
minSize	className.propertyName.minSize.notmet
notEqual	className.propertyName.notEqual
nullable	className.propertyName.nullable
range	className.propertyName.range.toosmall OR className.propertyName.range.toobig

[size](#)

```
className.propertyName.size.toosmall OR  
className.propertyName.size.toobig
```

[unique](#)

```
className.propertyName.unique
```

[url](#)

```
className.propertyName.url.invalid
```

[validator](#)

```
classname.propertyName. + String returned by  
Closure
```

In the case of the `blank` constraint this would be `user.login.blank` so you would need a message such as the following in your `grails-app/i18n/messages.properties` file:

```
user.login.blank=Your login name must be specified!
```

The class name is looked for both with and without a package, with the packaged version taking precedence. So for example, `com.mycompany.myapp.User.login.blank` will be used before `user.login.blank`. This allows for cases where your domain class message codes clash with a plugin's.

For a reference on what codes are for which constraints refer to the reference guide for each constraint (e.g. [blank](#)).

Displaying Messages

The [renderErrors](#) tag will automatically look up messages for you using the [message](#) tag. If you need more control of rendering you can handle this yourself:

```
<g:hasErrors bean="{user}">  
  <ul>  
    <g:eachError var="err" bean="{user}">  
      <li><g:message error="{err}" /></li>  
    </g:eachError>  
  </ul>  
</g:hasErrors>
```

In this example within the body of the [eachError](#) tag we use the [message](#) tag in combination with its `error` argument to read the message for the given error.

11.6 Applying Validation to Other Classes

[Domain classes](#) and [Command Objects](#) support validation by default. Other classes may be made validateable by defining the static `constraints` property in the class (as described above) and then telling the framework about them. It is important that the application register the validateable classes with the framework. Simply defining the `constraints` property is not sufficient.

The Validateable Trait

Classes which define the static `constraints` property and implement the [Validateable](#) trait will be validateable. Consider this example:

```
src/main/groovy/com/mycompany/myapp/User.groovy
```

```

package com.mycompany.myapp

import grails.validation.Validateable

class User implements Validateable {
    ...

    static constraints = {
        login size: 5..15, blank: false, unique: true
        password size: 5..15, blank: false
        email email: true, blank: false
        age min: 18
    }
}

```

Programmatic access

Accessing the constraints on a validateable object is slightly different. You can access a command object's constraints programmatically in another context by accessing the `constraintsMap` static property of the class. That property is an instance of `Map<String, ConstrainedProperty>`

In the example above, accessing `User.constraintsMap.login.blank` would yield `false`, while `User.constraintsMap.login.unique` would yield `true`.

12 The Service Layer

Grails defines the notion of a service layer. The Grails team discourages the embedding of core application logic inside controllers, as it does not promote reuse and a clean separation of concerns.

Services in Grails are the place to put the majority of the logic in your application, leaving controllers responsible for handling request flow with redirects and so on.

Creating a Service

You can create a Grails service by running the [create-service](#) command from the root of your project in a terminal window:

```
grails create-service helloworld.simple
```

If no package is specified with the create-service script, Grails automatically uses the `grails.defaultPackage` defined in `grails-app/conf/application.yml` as the package name.

The above example will create a service at the location `grails-app/services/helloworld/SimpleService.groovy`. A service's name ends with the convention `Service`, other than that a service is a plain Groovy class:

```

package helloworld

class SimpleService {
}

```

12.1 Declarative Transactions

Declarative Transactions

Services are typically involved with coordinating logic between [domain classes](#), and hence often involved with persistence that spans large operations. Given the nature of services, they frequently require transactional behaviour. You can use programmatic transactions with

the [withTransaction](#) method, however this is repetitive and doesn't fully leverage the power of Spring's underlying transaction abstraction.

Services enable transaction demarcation, which is a declarative way of defining which methods are to be made transactional. To enable transactions on a service use the Transactional transform:

```
import grails.gorm.transactions.*

@Transactional
class CountryService {

}
```

The result is that all methods are wrapped in a transaction and automatic rollback occurs if a method throws an exception (both Checked or Runtime exceptions) or an Error. The propagation level of the transaction is by default set to [PROPAGATION_REQUIRED](#).

Version Grails 3.2.0 was the first version to use GORM 6 by default. Checked exceptions did not roll back transactions before GORM 6. Only a method which threw a runtime exception (i.e. one that extends RuntimeException) rolled back a transaction.

Warning: [dependency injection](#) is the **only** way that declarative transactions work. You will not get a transactional service if you use the `new` operator such as `new BookService()`

The Transactional annotation vs the transactional property

In versions of Grails prior to Grails 3.1, Grails created Spring proxies and used the `transactional` property to enable and disable proxy creation. These proxies are disabled by default in applications created with Grails 3.1 and above in favor of the `@Transactional` transformation.

For versions of Grails 3.1.x and 3.2.x, if you wish to reenable this feature (not recommended) then you must set `grails.spring.transactionManagement` to `true` or remove the configuration in `grails-app/conf/application.yml` OR `grails-app/conf/application.groovy`.

In Grails 3.3.x Spring proxies for transaction management has been dropped completely, and you must use Grails' AST transforms. In Grails 3.3.x, if you wish to continue to use Spring proxies for transaction management you will have to configure them manually, using the appropriate Spring configuration.

In addition, prior to Grails 3.1 services were transactional by default, as of Grails 3.1 they are only transactional if the `@Transactional` transformation is applied.

Custom Transaction Configuration

Grails also provides `@Transactional` and `@NotTransactional` annotations for cases where you need more fine-grained control over transactions at a per-method level or need to specify an alternative propagation level. For example, the `@NotTransactional` annotation can be used to mark a particular method to be skipped when a class is annotated with `@Transactional`.

Annotating a service method with `Transactional` disables the default Grails transactional behavior for that service (in the same way that adding `transactional=false` does) so if you use any annotations you must annotate all methods that require transactions.

In this example `listBooks` uses a read-only transaction, `updateBook` uses a default read-write transaction, and `deleteBook` is not transactional (probably not a good idea given its name).

```
import grails.gorm.transactions.Transactional
```

```

class BookService {
    @Transactional(readOnly = true)
    def listBooks() {
        Book.list()
    }

    @Transactional
    def updateBook() {
        // ...
    }

    def deleteBook() {
        // ...
    }
}

```

You can also annotate the class to define the default transaction behavior for the whole service, and then override that default per-method:

```

import grails.gorm.transactions.Transactional

@Transactional
class BookService {
    def listBooks() {
        Book.list()
    }

    def updateBook() {
        // ...
    }

    def deleteBook() {
        // ...
    }
}

```

This version defaults to all methods being read-write transactional (due to the class-level annotation), but the `listBooks` method overrides this to use a read-only transaction:

```

import grails.gorm.transactions.Transactional

@Transactional
class BookService {
    @Transactional(readOnly = true)
    def listBooks() {
        Book.list()
    }

    def updateBook() {
        // ...
    }

    def deleteBook() {
        // ...
    }
}

```

Although `updateBook` and `deleteBook` aren't annotated in this example, they inherit the configuration from the class-level annotation.

For more information refer to the section of the Spring user guide on [Using @Transactional](#).

Unlike Spring you do not need any prior configuration to use `Transactional`; just specify the annotation as needed and Grails will detect them up automatically.

Transaction status

An instance of [TransactionStatus](#) is available by default in Grails transactional service methods.

Example:

```

import grails.gorm.transactions.Transactional

@Transactional

```

```

class BookService {
    def deleteBook() {
        transactionStatus.setRollbackOnly()
    }
}

```

12.1.1 Transactions and Multi-DataSources

Given two domain classes such as:

```

class Movie {
    String title
}

class Book {
    String title

    static mapping = {
        datasource 'books'
    }
}

```

You can supply the desired data source to `@Transactional` or `@ReadOnly` annotations.

```

import grails.gorm.transactions.ReadOnly
import grails.gorm.transactions.Transactional
import groovy.transform.CompileStatic

@CompileStatic
class BookService {

    @ReadOnly('books')
    List<Book> findAll() {
        Book.where {}.findAll()
    }

    @Transactional('books')
    Book save(String title) {
        Book book = new Book(title: title)
        book.save()
        book
    }
}

@CompileStatic
class MovieService {

    @ReadOnly
    List<Movie> findAll() {
        Movie.where {}.findAll()
    }
}

```

12.1.2 Transactions Rollback and the Session

Understanding Transactions and the Hibernate Session

When using transactions there are important considerations you must take into account with regards to how the underlying persistence session is handled by Hibernate. When a transaction is rolled back the Hibernate session used by GORM is cleared. This means any objects within the session become detached and accessing uninitialized lazy-loaded collections will lead to a `LazyInitializationException`.

To understand why it is important that the Hibernate session is cleared. Consider the following example:

```

class Author {
    String name
    Integer age

    static hasMany = [books: Book]
}

```


If you were to save two authors using consecutive transactions as follows:

```
Author.withTransaction { status ->
  new Author(name: "Stephen King", age: 40).save()
  status.setRollbackOnly()
}

Author.withTransaction { status ->
  new Author(name: "Stephen King", age: 40).save()
}
```

Only the second author would be saved since the first transaction rolls back the author `save()` by clearing the Hibernate session. If the Hibernate session were not cleared then both author instances would be persisted and it would lead to very unexpected results.

It can, however, be frustrating to get a `LazyInitializationException` due to the session being cleared.

For example, consider the following example:

```
class AuthorService {
  void updateAge(id, int age) {
    def author = Author.get(id)
    author.age = age
    if (author.isTooOld()) {
      throw new AuthorException("too old", author)
    }
  }
}

class AuthorController {
  def authorService

  def updateAge() {
    try {
      authorService.updateAge(params.id, params.int("age"))
    }
    catch(e) {
      render "Author books ${e.author.books}"
    }
  }
}
```

In the above example the transaction will be rolled back if the age of the `Author` age exceeds the maximum value defined in the `isTooOld()` method by throwing an `AuthorException`. The `AuthorException` references the author but when the `books` association is accessed a `LazyInitializationException` will be thrown because the underlying Hibernate session has been cleared.

To solve this problem you have a number of options. One is to ensure you query eagerly to get the data you will need:

```
class AuthorService {
  ...
  void updateAge(id, int age) {
    def author = Author.findById(id, [fetch:[books:"eager"]])
    ...
  }
}
```

In this example the `books` association will be queried when retrieving the `Author`.

This is the optimal solution as it requires fewer queries than the following suggested solutions.

Another solution is to redirect the request after a transaction rollback:

```
class AuthorController {
  AuthorService authorService

  def updateAge() {
    try {
      authorService.updateAge(params.id, params.int("age"))
    }
  }
}
```

```

    }
    catch(e) {
        flash.message = "Can't update age"
        redirect action:"show", id:params.id
    }
}

```

In this case a new request will deal with retrieving the `Author` again. And, finally a third solution is to retrieve the data for the `Author` again to make sure the session remains in the correct state:

```

class AuthorController {
    def authorService
    def updateAge() {
        try {
            authorService.updateAge(params.id, params.int("age"))
        }
        catch(e) {
            def author = Author.read(params.id)
            render "Author books ${author.books}"
        }
    }
}

```

Validation Errors and Rollback

A common use case is to rollback a transaction if there are validation errors. For example consider this service:

```

import grails.validation.ValidationException

class AuthorService {
    void updateAge(id, int age) {
        def author = Author.get(id)
        author.age = age
        if (!author.validate()) {
            throw new ValidationException("Author is not valid", author.errors)
        }
    }
}

```

To re-render the same view that a transaction was rolled back in you can re-associate the errors with a refreshed instance before rendering:

```

import grails.validation.ValidationException

class AuthorController {
    def authorService
    def updateAge() {
        try {
            authorService.updateAge(params.id, params.int("age"))
        }
        catch (ValidationException e) {
            def author = Author.read(params.id)
            author.errors = e.errors
            render view: "edit", model: [author:author]
        }
    }
}

```

12.2 Scoped Services

By default, access to service methods is not synchronised, so nothing prevents concurrent execution of those methods. In fact, because the service is a singleton and may be used concurrently, you should be very careful about storing state in a service. Or take the easy (and better) road and never store state in a service.

You can change this behaviour by placing a service in a particular scope. The supported scopes are:

- `prototype` - A new service is created every time it is injected into another class
- `request` - A new service will be created per request
- `flash` - A new service will be created for the current and next request only
- `flow` - In web flows the service will exist for the scope of the flow
- `conversation` - In web flows the service will exist for the scope of the conversation. ie a root flow and its sub flows
- `session` - A service is created for the scope of a user session
- `singleton` (default) - Only one instance of the service ever exists

If your service is `flash`, `flow` OR `conversation` scoped it must implement `java.io.Serializable` and can only be used in the context of a Web Flow.

To enable one of the scopes, add a static scope property to your class whose value is one of the above, for example

```
static scope = "flow"
```

Upgrading

Starting with Grails 2.3, new applications are generated with configuration that defaults the scope of controllers to `singleton`. If `singleton` controllers interact with `prototype` scoped services, the services effectively behave as per-controller singletons. If non-singleton services are required, controller scope should be changed as well.

See [Controllers and Scopes](#) in the user guide for more information.

Lazy initialization

You can also configure whether the service is lazily initialized. By default, this is set to `true`, but you can disable this and make initialization eager with the `lazyInit` property:

```
static lazyInit = false
```

12.3 Dependency Injection and Services

Dependency Injection Basics

A key aspect of Grails services is the ability to use [Spring Framework](#)'s dependency injection features. Grails supports "dependency injection by convention". In other words, you can use the property name representation of the class name of a service to automatically inject them into controllers, tag libraries, and so on.

As an example, given a service called `BookService`, if you define a property called `bookService` in a controller as follows:

```
class BookController {
    def bookService
    ...
}
```

In this case, the Spring container will automatically inject an instance of that service based on its configured scope. All dependency injection is done by name. You can also specify the type as follows:

```
class AuthorService {
    BookService bookService
}
```

NOTE: Normally the property name is generated by lower casing the first letter of the type. For example, an instance of the `BookService` class would map to a property named `bookService`.

To be consistent with standard JavaBean conventions, if the first 2 letters of the class name are upper case, the property name is the same as the class name. For example, the property name of the `JDBCHelperService` class would be `JDBCHelperService`, not `jdbchelperService` or `jdbchelperService`.

See section 8.8 of the JavaBean specification for more information on de-capitalization rules.

Only the top level object is subjected to injection as traversing all nested objects to perform injection would be a performance issue.

Be careful when injecting the non-default datasources. For example, using this config:

```
dataSources:
  dataSource:
    pooled: true
    jmxExport: true
    .....
  secondary:
    pooled: true
    jmxExport: true
    .....
```

You can inject the primary `dataSource` like you would expect:

```
class BookSqlService {
    def dataSource
}
```

But to inject the `secondary` datasource, you have to use Spring's `Autowired` injection or `resources.groovy`.

```
class BookSqlSecondaryService {
    @Autowired
    @Qualifier('dataSource_secondary')
    def dataSource2
}
```

Dependency Injection and Services

You can inject services in other services with the same technique. If you had an `AuthorService` that needed to use the `BookService`, declaring the `AuthorService` as follows would allow that:

```
class AuthorService {
    def bookService
}
```

Dependency Injection and Domain Classes / Tag Libraries

You can even inject services into domain classes and tag libraries, which can aid in the development of rich domain models and views:

```
class Book {
```

```

...
def bookService

def buyBook() {
    bookService.buyBook(this)
}
}

```

Since Grails 3.2.8 this is not enabled by default. If you want to enable it again, take a look at [Spring Autowiring of Domain Instance](#)

Service Bean Names

The default bean name which is associated with a service can be problematic if there are multiple services with the same name defined in different packages. For example consider the situation where an application defines a service class named `com.demo.ReportingService` and the application uses a plugin named `ReportingUtilities` and that plugin provides a service class named `com.reporting.util.ReportingService`.

The default bean name for each of those would be `reportingService` so they would conflict with each other. Grails manages this by changing the default bean name for services provided by plugins by prefixing the bean name with the plugin name.

In the scenario described above the `reportingService` bean would be an instance of the `com.demo.ReportingService` class defined in the application and the `reportingUtilitiesReportingService` bean would be an instance of the `com.reporting.util.ReportingService` class provided by the `ReportingUtilities` plugin.

For all service beans provided by plugins, if there are no other services with the same name within the application or other plugins in the application then a bean alias will be created which does not include the plugin name and that alias points to the bean referred to by the name that does include the plugin name prefix.

For example, if the `ReportingUtilities` plugin provides a service named `com.reporting.util.AuthorService` and there is no other `AuthorService` in the application or in any of the plugins that the application is using then there will be a bean named `reportingUtilitiesAuthorService` which is an instance of this `com.reporting.util.AuthorService` class and there will be a bean alias defined in the context named `authorService` which points to that same bean.

13 Static Type Checking And Compilation

Groovy is a dynamic language and by default Groovy uses a dynamic dispatch mechanism to carry out method calls and property access. This dynamic dispatch mechanism provides a lot of flexibility and power to the language. For example, it is possible to dynamically add methods to classes at runtime and it is possible to dynamically replace existing methods at runtime. Features like these are important and provide a lot of power to the language. However, there are times when you may want to disable this dynamic dispatch in favor of a more static dispatch mechanism and Groovy provides a way to do that. The way to tell the Groovy compiler that a particular class should be compiled statically is to mark the class with the [groovy.transform.CompileStatic](#) annotation as shown below.

```

import groovy.transform.CompileStatic

@CompileStatic
class MyClass {

    // this class will be statically compiled...
}

```

See [these notes on Groovy static compilation](#) for more details on how `CompileStatic` works and why you might want to use it.

One limitation of using `CompileStatic` is that when you use it you give up access to the power and flexibility offered by dynamic dispatch. For example, in Grails you would not be able to invoke a GORM dynamic finder from a class that is marked with `CompileStatic` because the compiler cannot verify that the dynamic finder method exists, because it doesn't exist at compile time. It may be that you want to take advantage of Groovy's static compilation benefits without giving up access to dynamic dispatch for Grails specific things like dynamic finders and this is where [grails.compiler.GrailsCompileStatic](#) comes in.

`GrailsCompileStatic` behaves just like `CompileStatic` but is aware of certain Grails features and allows access to those specific features to be accessed dynamically.

13.1 The GrailsCompileStatic Annotation

GrailsCompileStatic

The `grailsCompileStatic` annotation may be applied to a class or methods within a class.

```
import grails.compiler.GrailsCompileStatic

@GrailsCompileStatic
class SomeClass {

    // all of the code in this class will be statically compiled

    def methodOne() {
        // ...
    }

    def methodTwo() {
        // ...
    }

    def methodThree() {
        // ...
    }
}

import grails.compiler.GrailsCompileStatic

class SomeClass {

    // methodOne and methodThree will be statically compiled
    // methodTwo will be dynamically compiled

    @GrailsCompileStatic
    def methodOne() {
        // ...
    }

    def methodTwo() {
        // ...
    }

    @GrailsCompileStatic
    def methodThree() {
        // ...
    }
}
```

It is possible to mark a class with `GrailsCompileStatic` and exclude specific methods by marking them with `GrailsCompileStatic` and specifying that the type checking should be skipped for that particular method as shown below.

```
import grails.compiler.GrailsCompileStatic
import groovy.transform.TypeCheckingMode

@GrailsCompileStatic
class SomeClass {

    // methodOne and methodThree will be statically compiled
    // methodTwo will be dynamically compiled

    def methodOne() {
        // ...
    }
}
```

```

@GrailsCompileStatic(TypeCheckingMode.SKIP)
def methodTwo() {
    // ...
}

def methodThree() {
    // ...
}
}

```

Code that is marked with `GrailsCompileStatic` will all be statically compiled except for Grails specific interactions that cannot be statically compiled but that `GrailsCompileStatic` can identify as permissible for dynamic dispatch. These include things like invoking dynamic finders and DSL code in configuration blocks like constraints and mapping closures in domain classes.

Care must be taken when deciding to statically compile code. There are benefits associated with static compilation but in order to take advantage of those benefits you are giving up the power and flexibility of dynamic dispatch. For example if code is statically compiled it cannot take advantage of runtime metaprogramming enhancements which may be provided by plugins.

13.2 The GrailsTypeChecked Annotation

GrailsTypeChecked

The [grails.compiler.GrailsTypeChecked](#) annotation works a lot like the `GrailsCompileStatic` annotation except that it only enables static type checking, not static compilation. This affords compile time feedback for expressions which cannot be validated statically at compile time while still leaving dynamic dispatch in place for the class.

```

import grails.compiler.GrailsTypeChecked

@GrailsTypeChecked
class SomeClass {

    // all of the code in this class will be statically type
    // checked and will be dynamically dispatched at runtime

    def methodOne() {
        // ...
    }

    def methodTwo() {
        // ...
    }

    def methodThree() {
        // ...
    }
}

```

14 Testing

Automated testing is a critical aspect of Grails development. Grails provides a rich set of testing capabilities, ranging from low-level unit testing to high-level functional tests. This comprehensive guide explores these diverse testing features in detail.

Automatic Test Generation

When you use the `create-` and `generate-` commands, Grails automatically generates unit or integration tests. For example, running the `create-controller` command as shown below:

```
grails create-controller com.example.simple
```

Grails generates a controller at `grails-app/controllers/com/example/SimpleController.groovy` and a corresponding unit test at `src/test/groovy/com/example/SimpleControllerSpec.groovy`. It's important to note that Grails only creates the test structure; you need to implement the test logic.

Running Tests

To execute tests, you can use the Gradle check task:

```
./gradlew check
```

This command will execute all the Unit tests in `src/main/groovy/com/example/` directory.

Targeting Tests

To selectively target tests for execution, you have several options:

1. To run all tests for a controller named `SimpleController`, use this command:

```
./gradlew check --tests SimpleController
```

2. To test all classes ending in `Controller`, you can employ wildcards:

```
./gradlew check --tests *Controller
```

3. To specify package names:

```
./gradlew check --tests some.org.*Controller
```

4. To run all tests in a package:

```
./gradlew check --tests some.org.*
```

5. To run all tests in a package, including subpackages:

```
./gradlew check --tests some.org.**.*
```

6. To target specific test methods:

```
./gradlew check --tests SimpleController.testLogin
```

You can combine multiple patterns as needed:

```
./gradlew check --tests some.org.* SimpleController.testLogin BookController
```

You might need to specify the package name before the class name and append "Spec" to it. For instance, to run the test for the `ProductController`, use `./gradlew test *.ProductControllerSpec`. You can also use the star wildcard if you want to avoid typing the entire package hierarchy.

Debugging

To debug your tests using a remote debugger, you can add `--debug-jvm` after `./gradlew` in any commands, like so:

```
./gradlew check --debug-jvm
```

This will open the default Java remote debugging port, 5005, allowing you to attach a remote debugger from your code editor or integrated development environment.

Targeting Test Phases / Running Unit & Integration Separately

To execute "unit" tests, use this command:

```
./gradlew test
```

For "integration" tests, you would run:

```
./gradlew integrationTest
```

Targeting Tests When Using Phases

You can combine test and phase targeting:

```
./gradlew test some.org.**.*
```

This command will run all tests in the unit phase within the some.org package or its subpackages. For more detailed information, it's recommended to consult the Gradle documentation on [Testing in Java & JVM projects](#).

14.1 Unit Testing

Unit testing are tests at the "unit" level. In other words you are testing individual methods or blocks of code without consideration for surrounding infrastructure. Unit tests are typically run without the presence of physical resources that involve I/O such as databases, socket connections or files. This is to ensure they run as quick as possible since quick feedback is important.

Since Grails 3.3, the [Grails Testing Support Framework](#) is used for all unit tests. This support provides a set of traits. An example hello world test can be seen below:

```
import spock.lang.Specification
import grails.testing.web.controllers.ControllerUnitTest

class HelloControllerTests extends Specification implements ControllerUnitTest<HelloController> {

    void "Test message action"() {
        when:"The message action is invoked"
        controller.message()

        then:"Hello is returned"
        response.text == 'Hello'
    }
}
```

For more information on writing tests with Grails Testing Support see the [dedicated documentation](#).

Versions of Grails below 3.2 used the [Grails Test Mixin Framework](#) which was based on the `@TestMixin` AST transformation. This library has been superceded by the simpler and more IDE friendly trait based implementation.

14.2 Integration Testing

Integration tests differ from unit tests in that you have full access to the Grails environment within the test. You can create an integration test using the [create-integration-test](#) command:

```
$ grails create-integration-test Example
```

The above command will create a new integration test at the location

```
src/integration-test/groovy/<PACKAGE>/ExampleSpec.groovy.
```

Grails uses the test environment for integration tests and loads the application prior to the first test run. All tests use the same application state.

Transactions

Integration test methods run inside their own database transaction by default, which is rolled back at the end of each test method. This means that data saved during a test is not persisted to the database (which is shared across all tests). The default generated integration test template includes the [Rollback](#) annotation:

```
import grails.testing.mixin.integration.Integration
import grails.gorm.transactions.*
import spock.lang.*

@Integration
@Rollback
class ExampleSpec extends Specification {

    ...

    void "test something"() {
        expect:"fix me"
        true == false
    }
}
```

The `Rollback` annotation ensures that each test method runs in a transaction that is rolled back. Generally this is desirable because you do not want your tests depending on order or application state.

In Grails 3.0 tests rely on `grails.gorm.transactions.Rollback` annotation to bind the session in integration tests. Though each test method transaction is rolled back, the `setup()` method uses a separate transaction that is not rolled back. Data will persist to the database and will need to be cleaned up manually if `setup()` sets up data and persists them as shown in the below sample:

```
import grails.testing.mixin.integration.Integration
import grails.gorm.transactions.*
import spock.lang.*

@Integration
@Rollback
class BookSpec extends Specification {

    void setup() {
        // Below line would persist and not roll back
        new Book(name: 'Grails in Action').save(flush: true)
    }

    void "test something"() {
        expect:
        Book.count() == 1
    }
}
```

To preload the database and automatically roll back setup logic, any persistence operations need to be called from the test method itself so that they can run within the test method's rolled back transaction. Similar to usage of the `setupData()` method shown below which creates a record in database and after running other test will be rolled back:

```
import grails.testing.mixin.integration.Integration
import grails.gorm.transactions.*
import spock.lang.*

@Integration
@Rollback
class BookSpec extends Specification {

    void setupData() {
        // Below line would roll back
        new Book(name: 'Grails in Action').save(flush: true)
    }
}
```

```

    void "test something"() {
        given:
            setupData()

        expect:
            Book.count() == 1
    }
}

```

Using Spring's Rollback annotation

Another transactional approach could be to use Spring's [@Rollback](#) instead.

```

import grails.testing.mixin.integration.Integration
import org.springframework.test.annotation.Rollback
import spock.lang.*

@Integration
@Rollback
class BookSpec extends Specification {

    void setup() {
        new Book(name: 'Grails in Action').save(flush: true)
    }

    void "test something"() {
        expect:
            Book.count() == 1
    }
}

```

It isn't possible to make `grails.gorm.transactions.Rollback` behave the same way as Spring's `Rollback` annotation because `grails.gorm.transactions.Rollback` transforms the byte code of the class, eliminating the need for a proxy (which Spring's version requires). This has the downside that you cannot implement it differently for different cases (as Spring does for testing).

DirtyContext

If you do have a series of tests that will share state you can remove the `Rollback` and the last test in the suite should feature the [DirtyContext](#) annotation which will shutdown the environment and restart it fresh (note that this will have an impact on test run times).

Autowiring

To obtain a reference to a bean you can use the [Autowired](#) annotation. For example:

```

...
import org.springframework.beans.factory.annotation.*

@Integration
@Rollback
class ExampleServiceSpec extends Specification {

    @Autowired
    ExampleService exampleService
    ...

    void "Test example service"() {
        expect:
            exampleService.countExamples() == 0
    }
}

```

Testing Controllers

To integration test controllers it is recommended you use [create-functional-test](#) command to create a Geb functional test. See the following section on functional testing for more information.

14.3 Functional Testing

Functional tests involve making HTTP requests against the running application and verifying the resultant behaviour. This is useful for end-to-end testing scenarios, such as making REST calls against a JSON API.

Grails by default ships with support for writing functional tests using the [Geb framework](#). To create a functional test you can use the `create-functional-test` command which will create a new functional test:

```
$ grails create-functional-test MyFunctional
```

The above command will create a new Spock spec called `MyFunctionalSpec.groovy` in the `src/integration-test/groovy` directory. The test is annotated with the [Integration](#) annotation to indicate it is an integration test and extends the `GebSpec` super class:

```
@Integration
class HomeSpec extends GebSpec {

    def setup() {
    }

    def cleanup() {
    }

    void "Test the home page renders correctly"() {
        when:"The home page is visited"
        go '/'

        then:"The title is correct"
        $('title').text() == "Welcome to Grails"
    }
}
```

When the test is run the application container will be loaded up in the background and you can send requests to the running application using the Geb API.

Note that the application is only loaded once for the entire test run, so functional tests share the state of the application across the whole suite.

In addition the application is loaded in the JVM as the test, this means that the test has full access to the application state and can interact directly with data services such as GORM to setup and cleanup test data.

The `Integration` annotation supports an optional `applicationClass` attribute which may be used to specify the application class to use for the functional test. The class must extend [GrailsAutoConfiguration](#).

```
@Integration(applicationClass=com.demo.Application)
class HomeSpec extends GebSpec {

    // ...
}
```

If the `applicationClass` is not specified then the test runtime environment will attempt to locate the application class dynamically which can be problematic in multiproject builds where multiple application classes may be present.

When running the server port by default will be randomly assigned. The `Integration` annotation adds a property of `serverPort` to the test class that you can use if you want to know what port the application is running on this isn't needed if you are extending the `GebSpec` as shown above but can be useful information.

If you want to run the tests on a fixed port (defined by the `server.port` configuration property), you need to manually annotate your test with `@SpringBootTest`:

```
import grails.testing.mixin.integration.Integration
import org.springframework.boot.test.context.SpringBootTest
```

```
import spock.lang.Specification

@Integration
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.DEFINED_PORT)
class MySpec extends Specification {

    // ...

}
```

15 Internationalization

Grails supports Internationalization (i18n) out of the box by leveraging the underlying Spring MVC internationalization support. With Grails you are able to customize the text that appears in a view based on the user's Locale. To quote the javadoc for the [Locale](#) class:

A Locale object represents a specific geographical, political, or cultural region. An operation that requires a Locale to perform its task is called locale-sensitive and uses the Locale to tailor information for the user. For example, displaying a number is a locale-sensitive operation—the number should be formatted according to the customs/conventions of the user's native country, region, or culture.

A Locale is made up of a [language code](#) and a [country code](#). For example "en_US" is the code for US English, whilst "en_GB" is the code for British English.

15.1 Understanding Message Bundles

Now that you have an idea of locales, to use them in Grails you create message bundle file containing the different languages that you wish to render. Message bundles in Grails are located inside the `grails-app/i18n` directory and are simple Java properties files.

Each bundle starts with the name `messages` by convention and ends with the locale. Grails ships with several message bundles for a whole range of languages within the `grails-app/i18n` directory. For example:

- `messages.properties`
- `messages_da.properties`
- `messages_de.properties`
- `messages_es.properties`
- `messages_fr.properties`
- ...

By default Grails looks in `messages.properties` for messages unless the user has specified a locale. You can create your own message bundle by simply creating a new properties file that ends with the locale you are interested in. For example `messages_en_GB.properties` for British English.

15.2 Changing Locales

By default, the user locale is detected from the incoming `Accept-Language` header. You can

provide users the capability to switch locales by simply passing a parameter called `lang` to Grails as a request parameter:

```
/book/list?lang=es
```

Grails will automatically switch the user's locale and subsequent requests will use the switched locale.

By default, Grails uses [SessionLocaleResolver](#) as the `localeResolver` bean.

You can change the default locale easily:

```
grails-app/conf/spring/resources.groovy
```

```
import org.springframework.web.servlet.i18n.SessionLocaleResolver

beans = {
    localeResolver(SessionLocaleResolver) {
        defaultLocale= new Locale('es')
    }
}
```

Other `localeResolver` are available. For example, you could use save the switched locale in a Cookie:

```
grails-app/conf/spring/resources.groovy
```

```
import org.springframework.web.servlet.i18n.CookieLocaleResolver

beans = {
    localeResolver(CookieLocaleResolver) {
        defaultLocale= new Locale('es')
    }
}
```

Or fix the locale:

```
grails-app/conf/spring/resources.groovy
```

```
import org.springframework.web.servlet.i18n.FixedLocaleResolver

beans = {
    localeResolver(FixedLocaleResolver, new Locale('de'))
}
```

15.3 Reading Messages

Reading Messages in the View

The most common place that you need messages is inside the view. Use the [message](#) tag for this:

```
<g:message code="my.localized.content" />
```

As long as you have a key in your `messages.properties` (with appropriate locale suffix) such as the one below then Grails will look up the message:

```
my.localized.content=Hola, me llamo John. Hoy es domingo.
```

Messages can also include arguments, for example:

```
<g:message code="my.localized.content" args="${ ['Juan', 'lunes'] }" />
```

The message declaration specifies positional parameters which are dynamically specified:

```
my.localized.content=Hola, me llamo {0}. Hoy es {1}.
```

Reading Messages in Grails Artifacts with MessageSource

In a Grails artifact, you can inject `messageSource` and use the method `getMessage` with the arguments: message code, message arguments, default message and locale to retrieve a message.

```
import org.springframework.context.MessageSource
import org.springframework.context.i18n.LocaleContextHolder

class MyAppController {
    MessageSource messageSource

    def show() {
        def msg = messageSource.getMessage('my.localized.content', ['Juan', 'lunes'] as Object[], 'Default Message')
    }
}
```

Reading Messages in Controllers and Tag Libraries with the Message Tag

Additionally, you can read a message inside Controllers and Tag Libraries with the [Message Tag](#). However, using the message tag relies on GSP support which a Grails application may not necessarily have; e.g. a rest application.

In a controller, you can invoke tags as methods.

```
def show() {
    def msg = message(code: "my.localized.content", args: ['Juan', 'lunes'])
}
```

The same technique can be used in [tag libraries](#), but if your tag library uses a custom [namespace](#) then you must prefix the call with `g.:`

```
def myTag = { attrs, body ->
    def msg = g.message(code: "my.localized.content", args: ['Juan', 'lunes'])
}
```

15.4 Scaffolding and i18n

Grails [scaffolding](#) templates for controllers and views are fully i18n-aware. The GSPs use the [message](#) tag for labels, buttons etc. and controller `flash` messages use i18n to resolve locale-specific messages.

The scaffolding includes locale specific labels for domain classes and domain fields. For example, if you have a `Book` domain class with a `title` field:

```
class Book {
    String title
}
```

The scaffolding will use labels with the following keys:

```
book.label = Libro
book.title.label = Título del libro
```

You can use this property pattern if you'd like or come up with one of your own. There is nothing special about the use of the word `label` as part of the key other than it's the convention used by the scaffolding.

16 Security

Grails is no more or less secure than Java Servlets. However, Java servlets (and hence

Grails) are extremely secure and largely immune to common buffer overrun and malformed URL exploits due to the nature of the Java Virtual Machine underpinning the code.

Web security problems typically occur due to developer naivety or mistakes, and there is a little Grails can do to avoid common mistakes and make writing secure applications easier to write.

What Grails Automatically Does

Grails has a few built in safety mechanisms by default.

- All standard database access via [GORM](#) domain objects is automatically SQL escaped to prevent SQL injection attacks
- The default [scaffolding](#) templates HTML escape all data fields when displayed
- Grails link creating tags ([link](#), [form](#), [createLink](#), [createLinkTo](#) and others) all use appropriate escaping mechanisms to prevent code injection
- Grails provides [codecs](#) to let you trivially escape data when rendered as HTML, JavaScript and URLs to prevent injection attacks here.

16.1 Securing Against Attacks

SQL injection

Hibernate, which is the technology underlying GORM domain classes, automatically escapes data when committing to database so this is not an issue. However it is still possible to write bad dynamic HQL code that uses unchecked request parameters. For example doing the following is vulnerable to HQL injection attacks:

```
def vulnerable() {  
    def books = Book.find("from Book as b where b.title = '" + params.title + "'")  
}
```

or the analogous call using a GString:

```
def vulnerable() {  
    def books = Book.find("from Book as b where b.title = '${params.title}')"  
}
```

Do **not** do this. Use named or positional parameters instead to pass in parameters:

```
def safe() {  
    def books = Book.find("from Book as b where b.title = ?",  
                          [params.title])  
}
```

or

```
def safe() {  
    def books = Book.find("from Book as b where b.title = :title",  
                          [title: params.title])  
}
```

Phishing

This really a public relations issue in terms of avoiding hijacking of your branding and a declared communication policy with your customers. Customers need to know how to identify valid emails.

XSS - cross-site scripting injection

It is important that your application verifies as much as possible that incoming requests were originated from your application and not from another site. It is also important to ensure that all data values rendered into views are escaped correctly. For example when rendering to HTML or XHTML you must ensure that people cannot maliciously inject JavaScript or other HTML into data or tags viewed by others.

Grails 2.3 and above include special support for automatically encoded data placed into GSP pages. See the documentation on [Cross Site Scripting \(XSS\) prevention](#) for further information.

You must also avoid the use of request parameters or data fields for determining the next URL to redirect the user to. If you use a `successURL` parameter for example to determine where to redirect a user to after a successful login, attackers can imitate your login procedure using your own site, and then redirect the user back to their own site once logged in, potentially allowing JavaScript code to then exploit the logged-in account on the site.

Cross-site request forgery

CSRF involves unauthorized commands being transmitted from a user that a website trusts. A typical example would be another website embedding a link to perform an action on your website if the user is still authenticated.

The best way to decrease risk against these types of attacks is to use the `useToken` attribute on your forms. See [Handling Duplicate Form Submissions](#) for more information on how to use it. An additional measure would be to not use remember-me cookies.

HTML/URL injection

This is where bad data is supplied such that when it is later used to create a link in a page, clicking it will not cause the expected behaviour, and may redirect to another site or alter request parameters.

HTML/URL injection is easily handled with the [codecs](#) supplied by Grails, and the tag libraries supplied by Grails all use [encodeAsURL](#) where appropriate. If you create your own tags that generate URLs you will need to be mindful of doing this too.

Denial of service

Load balancers and other appliances are more likely to be useful here, but there are also issues relating to excessive queries for example where a link is created by an attacker to set the maximum value of a result set so that a query could exceed the memory limits of the server or slow the system down. The solution here is to always sanitize request parameters before passing them to dynamic finders or other GORM query methods:

```
int limit = 100
def safeMax = Math.min(params.max?.toInteger() ?: limit, limit) // limit to 100 results
return Book.list(max:safeMax)
```

Guessable IDs

Many applications use the last part of the URL as an "id" of some object to retrieve from GORM or elsewhere. Especially in the case of GORM these are easily guessable as they are typically sequential integers.

Therefore you must assert that the requesting user is allowed to view the object with the requested id before returning the response to the user.

Not doing this is "security through obscurity" which is inevitably breached, just like having a default password of "letmein" and so on.

You must assume that every unprotected URL is publicly accessible one way or another.

16.2 Cross Site Scripting (XSS) Prevention

Cross Site Scripting (XSS) attacks are a common attack vector for web applications. They typically involve submitting HTML or Javascript code in a form such that when that code is displayed, the browser does something nasty. It could be as simple as popping up an alert box, or it could be much worse like for example one could [access other users session cookies](#).

The solution is to escape all untrusted user input when it is displayed in a page. For example,

```
<script>alert('Got ya!');</script>
```

will become

```
&lt;script&gt;alert('Got ya!');&lt;/script&gt;
```

when rendered, nullifying the effects of the malicious input.

By default, Grails plays it safe and escapes all content in `${}` expressions in GSPs. All the standard GSP tags are also safe by default, escaping any relevant attribute values.

So what happens when you want to stop Grails from escaping some content? There are valid use cases for putting HTML into the database and rendering it as-is, as long as that content is **trusted**. In such cases, you can tell Grails that the content is safe as should be rendered raw, i.e. without any escaping:

```
<section>${raw(page.content)}</section>
```

The `raw()` method you see here is available from controllers, tag libraries and GSP pages.

XSS prevention is hard and requires a lot of developer attention

Although Grails plays it safe by default, that is no guarantee that your application will be invulnerable to an XSS-style attack. Such an attack is less likely to succeed than would otherwise be the case, but developers should always be conscious of potential attack vectors and attempt to uncover vulnerabilities in the application during testing. It's also easy to switch to an unsafe default, thereby increasing the risk of a vulnerability being introduced.

There are more details about the XSS in [OWASP - XSS prevention rules](#) and [OWASP - Types of Cross-Site Scripting](#). Types of XSS are: [Stored XSS](#), [Reflected XSS](#) and [DOM based XSS](#). [DOM based XSS prevention](#) is coming more important because of the popularity of Javascript client side templating and Single Page Apps.

Grails codecs are mainly for preventing stored and reflected XSS type of attacks. Grails 2.4 includes HTMLJS codec that assists in preventing some DOM based XSS attacks.

It's difficult to make a solution that works for everyone, and so Grails provides a lot of flexibility with regard to fine-tuning how escaping works, allowing you to keep most of your application safe while switching off default escaping or changing the codec used for

pages, tags, page fragments, and more.

Configuration

It is recommended that you review the configuration of a newly created Grails application to garner an understanding of XSS prevention works in Grails.

When you tag a cookie with the `HttpOnly` flag, it tells the browser that this particular cookie should only be accessed by the server. Any attempt to access the cookie from client script is strictly forbidden. This can be configured in the `application.yml` configuration file as seen below:

```
server:
  session:
    cookie:
      domain: example.org
      http-only: true
      path: /
      secure: true
```

GSP features the ability to automatically HTML encode GSP expressions, and as of Grails 2.3 this is the default configuration. The default configuration (found in `application.yml`) for a newly created Grails application can be seen below:

```
grails:
  views:
    gsp:
      encoding: UTF-8
      htmlcodec: xml # use xml escaping instead of HTML4 escaping
      codecs:
        expression: html # escapes values inside ${}
        scriptlets: html # escapes output from scriptlets in GSPs
        taglib: none # escapes output from taglibs
        staticparts: none # escapes output from static template parts
```

GSP features several codecs that it uses when writing the page to the response. The codecs are configured in the `codecs` block and are described below:

- `expression` - The expression codec is used to encode any code found within `${..}` expressions. The default for newly created application is `html` encoding.
- `scriptlet` - Used for output from GSP scriptlets (`<% %>`, `<%= %>` blocks). The default for newly created applications is `html` encoding
- `taglib` - Used to encode output from GSP tag libraries. The default is `none` for new applications, as typically it is the responsibility of the tag author to define the encoding of a given tag and by specifying `none` Grails remains backwards compatible with older tag libraries.
- `staticparts` - Used to encode the raw markup output by a GSP page. The default is `none`.

Double Encoding Prevention

Versions of Grails prior to 2.3, included the ability to set the default codec to `html`, however enabling this setting sometimes proved problematic when using existing plugins due to encoding being applied twice (once by the `html` codec and then again if the plugin manually called `encodeAsHTML`).

Grails 2.3 includes double encoding prevention so that when an expression is evaluated, it will not encode if the data has already been encoded (Example `${foo.encodeAsHTML()}`).

Raw Output

If you are 100% sure that the value you wish to present on the page has not been received from user input, and you do not wish the value to be encoded then you can use the `raw` method:

```
${raw(book.title)}
```

The 'raw' method is available in tag libraries, controllers and GSP pages.

Per Plugin Encoding

Grails also features the ability to control the codecs used on a per plugin basis. For example if you have a plugin named `foo` installed, then placing the following configuration in your `application.groovy` will disable encoding for only the `foo` plugin

```
foo.grails.views.gsp.codecs.expression = "none"
```

Per Page Encoding

You can also control the various codecs used to render a GSP page on a per page basis, using a page directive:

```
<%@page expressionCodec="none" %>
```

Per Tag Library Encoding

Each tag library created has the opportunity to specify a default codec used to encode output from the tag library using the "defaultEncodeAs" property:

```
static defaultEncodeAs = 'html'
```

Encoding can also be specified on a per tag basis using "encodeAsForTags":

```
static encodeAsForTags = [tagName: 'raw']
```

Context Sensitive Encoding Switching

Certain tags require certain encodings and Grails features the ability to enable a codec only a certain part of a tag's execution using the "withCodec" method. Consider for example the "`<g:javascript>`" tag which allows you to embed JavaScript code in the page. This tag requires JavaScript encoding, not HTML coding for the execution of the body of the tag (but not for the markup that is output):

```
out.println '<script type="text/javascript">'
    withCodec("JavaScript") {
        out << body()
    }
    out.println()
    out.println '</script>'
```

Forced Encoding for Tags

If a tag specifies a default encoding that differs from your requirements you can force the encoding for any tag by passing the optional 'encodeAs' attribute:

```
<g:message code="foo.bar" encodeAs="JavaScript" />
```

Default Encoding for All Output

The default configuration for new applications is fine for most use cases, and backwards compatible with existing plugins and tag libraries. However, you can also make your application even more secure by configuring Grails to always encode all output at the end of

a response. This is done using the `filteringCodecForContentType` configuration in `application.groovy`:

```
grails.views.gsp.filteringCodecForContentType.'text/html' = 'html'
```

Note that, if activated, the `staticparts` codec typically needs to be set to `raw` so that static markup is not encoded:

```
codecs {
    expression = 'html' // escapes values inside ${}
    scriptlet = 'html' // escapes output from scriptlets in GSPs
    taglib = 'none' // escapes output from taglibs
    staticparts = 'raw' // escapes output from static template parts
}
```

16.3 Encoding and Decoding Objects

Grails supports the concept of dynamic encode/decode methods. A set of standard codecs are bundled with Grails. Grails also supports a simple mechanism for developers to contribute their own codecs that will be recognized at runtime.

Codec Classes

A Grails codec class is one that may contain an encode closure, a decode closure or both. When a Grails application starts up the Grails framework dynamically loads codecs from the `grails-app/utils/` directory.

The framework looks under `grails-app/utils/` for class names that end with the convention `Codec`. For example one of the standard codecs that ships with Grails is `HTMLCodec`.

If a codec contains an `encode` closure Grails will create a dynamic `encode` method and add that method to the `Object` class with a name representing the codec that defined the `encode` closure. For example, the `HTMLCodec` class defines an `encode` closure, so Grails attaches it with the name `encodeAsHTML`.

The `HTMLCodec` and `URLCodec` classes also define a `decode` closure, so Grails attaches those with the names `decodeHTML` and `decodeURL` respectively. Dynamic codec methods may be invoked from anywhere in a Grails application. For example, consider a case where a report contains a property called 'description' which may contain special characters that must be escaped to be presented in an HTML document. One way to deal with that in a GSP is to encode the description property using the dynamic `encode` method as shown below:

```
${report.description.encodeAsHTML()}
```

Decoding is performed using `value.decodeHTML()` syntax.

Encoder and Decoder interfaces for statically compiled code

A preferred way to use codecs is to use the `codecLookup` bean to get hold of `Encoder` and `Decoder` instances .

```
package org.grails.encoder;

public interface CodecLookup {
    public Encoder lookupEncoder(String codecName);
    public Decoder lookupDecoder(String codecName);
}
```

example of using `CodecLookup` and `Encoder` interface

```
import org.grails.encoder.CodecLookup
```

```
class CustomTagLib {
  CodecLookup codecLookup

  def myTag = { Map attrs, body ->
    out << codecLookup.lookupEncoder('HTML').encode(attrs.something)
  }
}
```

Standard Codecs

HTMLCodec

This codec performs HTML escaping and unescaping, so that values can be rendered safely in an HTML page without creating any HTML tags or damaging the page layout. For example, given a value "Don't you know that 2 > 1?" you wouldn't be able to show this safely within an HTML page because the > will look like it closes a tag, which is especially bad if you render this data within an attribute, such as the value attribute of an input field.

Example of usage:

```
<input name="comment.message" value="${comment.message.encodeAsHTML()}" />
```

Note that the HTML encoding does not re-encode apostrophe/single quote so you must use double quotes on attribute values to avoid text with apostrophes affecting your page.

HTMLCodec defaults to HTML4 style escaping (legacy HTMLCodec implementation in Grails versions before 2.3.0) which escapes non-ascii characters.

You can use plain XML escaping instead of HTML4 escaping by setting this config property in application.groovy:

```
grails.views.gsp.htmlcodec = 'xml'
```

XMLCodec

This codec performs XML escaping and unescaping. It escapes & , < , > , " , ' , \\ , @ , ` , non breaking space (\\u00a0), line separator (\\u2028) and paragraph separator (\\u2029).

HTMLJSCodec

This codec performs HTML and JS encoding. It is used for preventing some DOM-XSS vulnerabilities. See [OWASP - DOM based XSS Prevention Cheat Sheet](#) for guidelines of preventing DOM based XSS attacks.

URLCodec

URL encoding is required when creating URLs in links or form actions, or any time data is used to create a URL. It prevents illegal characters from getting into the URL and changing its meaning, for example "Apple & Blackberry" is not going to work well as a parameter in a GET request as the ampersand will break parameter parsing.

Example of usage:

```
<a href="/mycontroller/find?searchKey=${lastSearch.encodeAsURL()}">
Repeat last search
</a>
```

Base64Codec

Performs Base64 encode/decode functions. Example of usage:

Your registration code is: `${user.registrationCode.encodeAsBase64()}`

JavaScriptCodec

Escapes Strings so they can be used as valid JavaScript strings. For example:

```
Element.update('${elementId}',  
    '${render(template: "/common/message").encodeAsJavaScript()}')
```

HexCodec

Encodes byte arrays or lists of integers to lowercase hexadecimal strings, and can decode hexadecimal strings into byte arrays. For example:

Selected colour: `#{[255,127,255].encodeAsHex()}`

MD5Codec

Uses the MD5 algorithm to digest byte arrays or lists of integers, or the bytes of a string (in default system encoding), as a lowercase hexadecimal string. Example of usage:

Your API Key: `${user.uniqueID.encodeAsMD5()}`

MD5BytesCodec

Uses the MD5 algorithm to digest byte arrays or lists of integers, or the bytes of a string (in default system encoding), as a byte array. Example of usage:

```
byte[] passwordHash = params.password.encodeAsMD5Bytes()
```

SHA1Codec

Uses the SHA1 algorithm to digest byte arrays or lists of integers, or the bytes of a string (in default system encoding), as a lowercase hexadecimal string. Example of usage:

Your API Key: `${user.uniqueID.encodeAsSHA1()}`

SHA1BytesCodec

Uses the SHA1 algorithm to digest byte arrays or lists of integers, or the bytes of a string (in default system encoding), as a byte array. Example of usage:

```
byte[] passwordHash = params.password.encodeAsSHA1Bytes()
```

SHA256Codec

Uses the SHA256 algorithm to digest byte arrays or lists of integers, or the bytes of a string (in default system encoding), as a lowercase hexadecimal string. Example of usage:

Your API Key: `${user.uniqueID.encodeAsSHA256()}`

SHA256BytesCodec

Uses the SHA256 algorithm to digest byte arrays or lists of integers, or the bytes of a string (in default system encoding), as a byte array. Example of usage:

```
byte[] passwordHash = params.password.encodeAsSHA256Bytes()
```

Custom Codecs

Applications may define their own codecs and Grails will load them along with the standard codecs. A custom codec class must be defined in the `grails-app/utils/` directory and the class name must end with `Codec`. The codec may contain a `static encode` closure, a `static decode` closure or both. The closure must accept a single argument which will be the object that the dynamic method was invoked on. For Example:

```
class PigLatinCodec {
    static encode = { str ->
        // convert the string to pig latin and return the result
    }
}
```

With the above codec in place an application could do something like this:

```
${lastName.encodeAsPigLatin()}}
```

16.4 Authentication

Grails has no default mechanism for authentication as it is possible to implement authentication in many different ways. It is however, easy to implement a simple authentication mechanism using [interceptors](#). This is sufficient for simple use cases but it's highly preferable to use an established security framework, for example by using the [Spring Security](#) or the [Shiro](#) plugin.

Interceptors let you apply authentication across all controllers or across a URI space. For example you can create a new set of filters in a class called `grails-app/controllers/SecurityInterceptor.groovy` by running:

```
grails create-interceptor security
```

and implement your interception logic there:

```
class SecurityInterceptor {
    SecurityInterceptor() {
        matchAll()
        .except(controller:'user', action:'login')
    }

    boolean before() {
        if (!session.user && actionName != "login") {
            redirect(controller: "user", action: "login")
            return false
        }
        return true
    }
}
```

Here the interceptor intercepts execution *before* all actions except `login` are executed, and if there is no user in the session then redirect to the `login` action.

The `login` action itself is simple too:

```
def login() {
    if (request.get) {
        return // render the login view
    }

    def u = User.findByLogin(params.login)
    if (u) {
        if (u.password == params.password) {
            session.user = u
            redirect(action: "home")
        }
        else {
            render(view: "login", model: [message: "Password incorrect"])
        }
    }
    else {

```



```
}  
    render(view: "login", model: [message: "User not found"])  
}
```

16.5 Security Plugins

If you need more advanced functionality beyond simple authentication such as authorization, roles etc. then you should consider using the spring security core plugin.

16.5.1 Spring Security

The Spring Security plugins are built on the [Spring Security](#) project which provides a flexible, extensible framework for building all sorts of authentication and authorization schemes. The plugins are modular so you can install just the functionality that you need for your application. The Spring Security plugins are the official security plugins for Grails and are actively maintained and supported.

There is a [Spring Security Core](#) plugin which supports form-based authentication, encrypted/salted passwords, HTTP Basic authentication, etc. and secondary dependent plugins provide alternate functionality such as [ACL support](#), [single sign-on with Jasig CAS](#), [LDAP authentication](#), [Kerberos authentication](#), and a plugin providing [user interface extensions](#) and security workflows.

See the [Spring Security Core](#) plugin page for basic information and [user guide](#) for detailed information.

17 Plugins

Grails is first and foremost a web application framework, but it is also a platform. By exposing a number of extension points that let you extend anything from the command line interface to the runtime configuration engine, Grails can be customised to suit almost any needs. To hook into this platform, all you need to do is create a plugin.

Extending the platform may sound complicated, but plugins can range from trivially simple to incredibly powerful. If you know how to build a Grails application, you'll know how to create a plugin for [sharing a data model](#) or some static resources.

17.1 Creating and Installing Plugins

Creating Plugins

Creating a Grails plugin is a simple matter of running the command:

```
grails create-plugin <<PLUGIN NAME>>
```

This will create a web-plugin project for the name you specify. For example running `grails create-plugin example` would create a new web-plugin project called `example`.

In Grails 3.0 you should consider whether the plugin you create requires a web environment or whether the plugin can be used with other profiles. If your plugin does not require a web environment then use the "plugin" profile instead of the default "web-plugin" profile:

```
grails create-plugin <<PLUGIN NAME>> --profile=plugin
```

Make sure the plugin name does not contain more than one capital letter in a row, or it won't work. Camel case is fine, though.

Being a regular Grails project has a number of benefits in that you can immediately test your plugin by running (if the plugin targets the "web" profile):

```
./gradlew bootRun
```

Plugin projects don't provide an `index.gsp` by default since most plugins don't need it. So, if you try to view the plugin running in a browser right after creating it, you will receive a page not found error. You can easily create a `grails-app/views/index.gsp` for your plugin if you'd like.

The structure of a Grails plugin is very nearly the same as a Grails application project's except that in the `src/main/groovy` directory under the plugin package structure you will find a plugin descriptor class (a class that ends in "GrailsPlugin"). For example:

```
import grails.plugins.*

class ExampleGrailsPlugin extends Plugin {
    ...
}
```

All plugins must have this class under the `src/main/groovy` directory, otherwise they are not regarded as a plugin. The plugin class defines metadata about the plugin, and optionally various hooks into plugin extension points (covered shortly).

You can also provide additional information about your plugin using several special properties:

- `title` - short one-sentence description of your plugin
- `grailsVersion` - The version range of Grails that the plugin supports. eg. `"1.2 > *"` (indicating 1.2 or higher)
- `author` - plugin author's name
- `authorEmail` - plugin author's contact e-mail
- `developers` - Any additional developers beyond the author specified above.
- `description` - full multi-line description of plugin's features
- `documentation` - URL of the plugin's documentation
- `license` - License of the plugin
- `issueManagement` - Issue Tracker of the plugin
- `scm` - Source code management location of the plugin

Here is a slimmed down example from the [Quartz Grails plugin](#):

```
package quartz

@Slf4j
class QuartzGrailsPlugin extends Plugin {
    // the version or versions of Grails the plugin is designed for
    def grailsVersion = "3.0.0.BUILD-SNAPSHOT > *"
    // resources that are excluded from plugin packaging
    def pluginExcludes = [
```

```

        "grails-app/views/error.gsp"
    ]
    def title = "Quartz" // Headline display name of the plugin
    def author = "Jeff Brown"
    def authorEmail = "zzz@yyy.com"
    def description = '''\
Adds Quartz job scheduling features
'''
    def profiles = ['web']
    List loadAfter = ['hibernate3', 'hibernate4', 'hibernate5', 'services']
    def documentation = "http://grails.org/plugin/quartz"
    def license = "APACHE"
    def issueManagement = [ system: "Github Issues", url: "http://github.com/grails3-plugins/quartz/issues" ]
    def developers = [
        [ name: "Joe Dev", email: "joedev@gmail.com" ]
    ]
    def scm = [ url: "https://github.com/grails3-plugins/quartz/" ]

    Closure doWithSpring().....

```

Plugin Configuration

Instead of directly accessing Grails configuration as

`grailsApplication.config.getProperty('mail.hostName', String)`, use a Spring Boot configuration bean (or a POJO) annotated with [ConfigurationProperties](#) annotation. Here is an example plugin configuration:

./src/main/groovy/example/MailPluginConfiguration.groovy

```

package example

import org.springframework.boot.context.properties.ConfigurationProperties

@ConfigurationProperties(prefix = "mail")
class MailPluginConfiguration {

    String hostName
    int port
    String from
}

```

You can inject the `MailPluginConfiguration` bean into your bean like any other bean.

./grails-app/services/example/MailService.groovy

```

package example

class MailService {

    MailPluginConfiguration mailPluginConfiguration

    void sendMail() {

    }

}

```

Please read the [Spring Boot Externalized Configuration](#) section for more information.

Installing Local Plugins

In order to install the Grails plugin to your local Maven, you could use Gradle [Maven Publish](#) plugin. You may also need to configure the publishing extension as:

```

publishing {
    publications {
        maven(MavenPublication) {
            versionMapping {
                usage('java-api') {
                    fromResolutionOf('runtimeClasspath')
                }
                usage('java-runtime') {
                    fromResolutionResult()
                }
            }
        }
    }
    from components.java
}

```

```

    }
  }
}

```

Please refer to the [Gradle Maven Publish plugin documentation](#) for up-to-date information.

To make your plugin available for use in a Grails application run the `./gradlew publishToMavenLocal` command:

```
./gradlew publishToMavenLocal
```

This will install the plugin into your local Maven cache. Then to use the plugin within an application declare a dependency on the plugin in your `build.gradle` file and include `mavenLocal()` in your repositories hash:

```

...
repositories {
    ...
    mavenLocal()
}
...
implementation "org.grails.plugins:quartz:0.1"

```

In Grails 2.x plugins were packaged as ZIP files, however in Grails 3.x plugins are simple JAR files that can be added to the classpath of the IDE.

Plugins and Multi-Project Builds

If you wish to setup a plugin as part of a multi project build then follow these steps.

Step 1: Create the application and the plugin

Using the `grails` command create an application and a plugin:

```

$ grails create-app myapp
$ grails create-plugin myplugin

```

Step 2: Create a settings.gradle file

In the same directory create a `settings.gradle` file with the following contents:

```
include "myapp", "myplugin"
```

The directory structure should be as follows:

```

PROJECT_DIR
- settings.gradle
- myapp
  - build.gradle
- myplugin
  - build.gradle

```

Step 3: Declare a project dependency on the plugin

Within the `build.gradle` of the application declare a dependency on the plugin within the `plugins` block:

```

grails {
    plugins {
        implementation project(':myplugin')
    }
}

```

You can also declare the dependency within the `dependencies` block, however you will not get subproject reloading if you do this!

Step 4: Configure the plugin to enable reloading

In the plugin directory, add or modify the `gradle.properties` file. A new property `exploded=true` needs to be set in order for the plugin to add the exploded directories to the classpath.

Step 5: Run the application

Now run the application using the `./gradlew bootRun` command from the root of the application directory, you can use the `verbose` flag to see the Gradle output:

```
$ cd myapp
$ ./gradlew bootRun --verbose
```

You will notice from the Gradle output that plugins sources are built and placed on the classpath of your application:

```
:myplugin:compileAstJava UP-TO-DATE
:myplugin:compileAstGroovy UP-TO-DATE
:myplugin:processAstResources UP-TO-DATE
:myplugin:astClasses UP-TO-DATE
:myplugin:compileJava UP-TO-DATE
:myplugin:configScript UP-TO-DATE
:myplugin:compileGroovy
:myplugin:copyAssets UP-TO-DATE
:myplugin:copyCommands UP-TO-DATE
:myplugin:copyTemplates UP-TO-DATE
:myplugin:processResources
:myapp:compileJava UP-TO-DATE
:myapp:compileGroovy
:myapp:processResources UP-TO-DATE
:myapp:classes
:myapp:findMainClass
:myapp:bootRun
Grails application running at http://localhost:8080 in environment: development
```

Notes on excluded Artefacts

Although the [create-plugin](#) command creates certain files for you so that the plugin can be run as a Grails application, not all of these files are included when packaging a plugin. The following is a list of artefacts created, but not included by [package-plugin](#):

- `grails-app/build.gradle` (although it is used to generate `dependencies.groovy`)
- `grails-app/conf/application.yml` (renamed to `plugin.yml`)
- `grails-app/conf/spring/resources.groovy`
- `grails-app/conf/logback.groovy`
- Everything within `/src/test/*\*`
- SCM management files within `*\*/.svn/*\*` and `*\*/cvs/*\*`

Customizing the plugin contents

When developing a plugin you may create test classes and sources that are used during the development and testing of the plugin but should not be exported to the application.

To exclude test sources you need to modify the `pluginExcludes` property of the plugin descriptor AND exclude the resources inside your `build.gradle` file. For example say you have some classes under the `com.demo` package that are in your plugin source tree but should not be packaged in the application. In your plugin descriptor you should exclude these:

```
// resources that should be loaded by the plugin once installed in the application
def pluginExcludes = [
    '**/com/demo/**'
]
```

And in your `build.gradle` you should exclude the compiled classes from the JAR file:

```
jar {  
    exclude "com/demo/**/**"  
}
```

Inline Plugins in Grails 3.0

In Grails 2.x it was possible to specify inline plugins in `BuildConfig`, in Grails 3.x this functionality has been replaced by Gradle's multi-project build feature.

To set up a multi project build create an application and a plugin in a parent directory:

```
$ grails create-app myapp  
$ grails create-plugin myplugin
```

Then create a `settings.gradle` file in the parent directory specifying the location of your application and plugin:

```
include 'myapp', 'myplugin'
```

Finally add a dependency in your application's `build.gradle` on the plugin:

```
implementation project(':myplugin')
```

Using this technique you have achieved the equivalent of inline plugins from Grails 2.x.

17.2 Plugin Repositories

Distributing Plugins in the Grails Central Plugin Repository

The preferred way to distribute plugin is to publish to the official Grails Central Plugin Repository. This will make your plugin visible to the [list-plugins](#) command:

```
grails list-plugins
```

which lists all plugins that are in the central repository. Your plugin will also be available to the [plugin-info](#) command:

```
grails plugin-info [plugin-name]
```

which prints extra information about it, such as its description, who wrote, etc.

If you have created a Grails plugin and want it to be hosted in the central repository, you'll find instructions for getting an account on the [plugin portal](#) website.

17.3 Providing Basic Artefacts

Add Command Line Commands

A plugin can add new commands to the Grails 3.0 interactive shell in one of two ways. First, using the [create-script](#) you can create a code generation script which will become available to the application. The `create-script` command will create the script in the `src/main/scripts` directory:

```
+ src/main/scripts      <-- additional scripts here  
+ grails-app  
  + controllers  
  + services  
  + etc.
```

Code generation scripts can be used to create artefacts within the project tree and automate interactions with Gradle.

If you want to create a new shell command that interacts with a loaded Grails application instance then you should use the `create-command` command:

```
$ grails create-command MyExampleCommand
```

This will create a file called `grails-app/commands/PACKAGE_PATH/MyExampleCommand.groovy` that extends [ApplicationCommand](#):

```
import grails.dev.commands.*

class MyExampleCommand implements ApplicationCommand {

    boolean handle(ExecutionContext ctx) {
        println "Hello World"
        return true
    }
}
```

An `ApplicationCommand` has access to the `GrailsApplication` instance and is subject to autowiring like any other Spring bean.

You can also inform Grails to skip the execution of `Bootstrap.groovy` files with a simple property in your command:

```
class MyExampleCommand implements ApplicationCommand {

    boolean skipBootstrap = true

    boolean handle(ExecutionContext ctx) {
        ...
    }
}
```

For each `ApplicationCommand` present Grails will create a shell command and a Gradle task to invoke the `ApplicationCommand`. In the above example you can invoke the `MyExampleCommand` class using either:

```
$ grails my-example
```

Or

```
$ gradle myExample
```

The Grails version is all lower case hyphen separated and excludes the "Command" suffix.

The main difference between code generation scripts and `ApplicationCommand` instances is that the latter has full access to the Grails application state and hence can be used to perform tasks that interactive with the database, call into GORM etc.

In Grails 2.x Gant scripts could be used to perform both these tasks, in Grails 3.x code generation and interacting with runtime application state has been cleanly separated.

Adding a new grails-app artifact (Controller, Tag Library, Service, etc.)

A plugin can add new artifacts by creating the relevant file within the `grails-app` tree.

```
+ grails-app
+ controllers <-- additional controllers here
+ services <-- additional services here
+ etc. <-- additional XXX here
```

Providing Views, Templates and View resolution

When a plugin provides a controller it may also provide default views to be rendered. This is an excellent way to modularize your application through plugins. Grails' view resolution mechanism will first look for the view in the application it is installed into and if that fails will attempt to look for the view within the plugin. This means that you can override views provided by a plugin by creating corresponding GSPs in the application's `grails-app/views` directory.

For example, consider a controller called `BookController` that's provided by an 'amazon' plugin. If the action being executed is `list`, Grails will first look for a view called `grails-app/views/book/list.gsp` then if that fails it will look for the same view relative to the plugin.

However if the view uses templates that are also provided by the plugin then the following syntax may be necessary:

```
<g:render template="fooTemplate" plugin="amazon"/>
```

Note the usage of the `plugin` attribute, which contains the name of the plugin where the template resides. If this is not specified then Grails will look for the template relative to the application.

Excluded Artefacts

By default Grails excludes the following files during the packaging process:

- `grails-app/conf/logback.groovy`
- `grails-app/conf/application.yml` (renamed to `plugin.yml`)
- `grails-app/conf/spring/resources.groovy`
- Everything within `/src/test/*\*`
- SCM management files within `*\*/.svn/*\*` and `*\*/cvs/*\*`

The default `UrlMappings.groovy` file is not excluded, so remove any mappings that are not required for the plugin to work. You are also free to add a `UrlMappings` definition under a different name which **will** be included. For example a file called `grails-app/controllers/BlogUrlMappings.groovy` is fine.

The list of excludes is extensible with the `pluginExcludes` property:

```
// resources that are excluded from plugin packaging
def pluginExcludes = [
    "grails-app/views/error.gsp"
]
```

This is useful for example to include demo or test resources in the plugin repository, but not include them in the final distribution.

17.4 Evaluating Conventions

Before looking at providing runtime configuration based on conventions you first need to understand how to evaluate those conventions from a plugin. Every plugin has an implicit application variable which is an instance of the [GrailsApplication](#) interface.

The `GrailsApplication` interface provides methods to evaluate the conventions within the

project and internally stores references to all artifact classes within your application.

Artifacts implement the [GrailsClass](#) interface, which represents a Grails resource such as a controller or a tag library. For example to get all `GrailsClass` instances you can do:

```
for (grailsClass in application.allClasses) {  
    println grailsClass.name  
}
```

`GrailsApplication` has a few "magic" properties to narrow the type of artefact you are interested in. For example to access controllers you can use:

```
for (controllerClass in application.controllerClasses) {  
    println controllerClass.name  
}
```

The dynamic method conventions are as follows:

- `*Classes` - Retrieves all the classes for a particular artefact name. For example `application.controllerClasses`.
- `get*Class` - Retrieves a named class for a particular artefact. For example `application.getControllerClass("PersonController")`
- `is*Class` - Returns `true` if the given class is of the given artefact type. For example `application.isControllerClass(PersonController)`

The `GrailsClass` interface has a number of useful methods that let you further evaluate and work with the conventions. These include:

- `getPropertyValue` - Gets the initial value of the given property on the class
- `hasProperty` - Returns `true` if the class has the specified property
- `newInstance` - Creates a new instance of this class.
- `getName` - Returns the logical name of the class in the application without the trailing convention part if applicable
- `getShortName` - Returns the short name of the class without package prefix
- `getFullName` - Returns the full name of the class in the application with the trailing convention part and with the package name
- `getPropertyName` - Returns the name of the class as a property name
- `getLogicalPropertyName` - Returns the logical property name of the class in the application without the trailing convention part if applicable
- `getNaturalName` - Returns the name of the property in natural terms (e.g. 'lastName' becomes 'Last Name')
- `getPackageName` - Returns the package name

For a full reference refer to the [javadoc API](#).

17.5 Hooking into Runtime Configuration

Grails provides a number of hooks to leverage the different parts of the system and perform runtime configuration by convention.

Hooking into the Grails Spring configuration

First, you can hook in Grails runtime configuration overriding the `doWithSpring` method from the [Plugin](#) class and returning a closure that defines additional beans. For example the following snippet is from one of the core Grails plugins that provides [i18n](#) support:

```
import org.springframework.web.servlet.i18n.CookieLocaleResolver
import org.springframework.web.servlet.i18n.LocaleChangeInterceptor
import org.springframework.context.support.ReloadableResourceBundleMessageSource
import grails.plugins.*

class I18nGrailsPlugin extends Plugin {

    def version = "0.1"

    Closure doWithSpring() {{->
        messageSource(ReloadableResourceBundleMessageSource) {
            basename = "WEB-INF/grails-app/i18n/messages"
        }
        localeChangeInterceptor(LocaleChangeInterceptor) {
            paramName = "lang"
        }
        localeResolver(CookieLocaleResolver)
    }}
}
```

This plugin configures the Grails `messageSource` bean and a couple of other beans to manage Locale resolution and switching. It using the [Spring Bean Builder](#) syntax to do so.

Customizing the Servlet Environment

In previous versions of Grails it was possible to dynamically modify the generated `web.xml`. In Grails 3.x there is no `web.xml` file and it is not possible to programmatically modify the `web.xml` file anymore.

However, it is possible to perform the most commons tasks of modifying the Servlet environment in Grails 3.x.

Adding New Servlets

If you want to add a new Servlet instance the simplest way is simply to define a new Spring bean in the `doWithSpring` method:

```
Closure doWithSpring() {{->
    myServlet(MyServlet)
}}
```

If you need to customize the servlet you can use Spring Boot's [ServletRegistrationBean](#):

```
Closure doWithSpring() {{->
    myServlet(ServletRegistrationBean, new MyServlet(), "/myServlet/*") {
        loadOnStartup = 2
    }
}}
```

Adding New Servlet Filters

Just like Servlets, the simplest way to configure a new filter is to simply define a Spring bean:

```
Closure doWithSpring() {{->
    myFilter(MyFilter)
}}
```

However, if you want to control the order of filter registrations you will need to use Spring

Boot's [FilterRegistrationBean](#):

```
myFilter(FilterRegistrationBean) {
    filter = bean(MyFilter)
    urlPatterns = ['//*']
    order = Ordered.HIGHEST_PRECEDENCE
}
```

Grails' internal registered filters (`GrailsWebRequestFilter`, `HiddenHttpMethodFilter` etc.) are defined by incrementing `HIGHEST_PRECEDENCE` by 10 thus allowing several filters to be inserted before or between Grails' filters.

Doing Post Initialisation Configuration

Sometimes it is useful to be able to do some runtime configuration after the Spring [ApplicationContext](#) has been built. In this case you can define a `doWithApplicationContext` closure property.

```
class SimplePlugin extends Plugin{

    def name = "simple"
    def version = "1.1"

    @Override
    void doWithApplicationContext() {
        def sessionFactory = applicationContext.sessionFactory
        // do something here with session factory
    }
}
```

17.6 Adding Methods at Compile Time

Grails 3.0 makes it easy to add new traits to existing artefact types from a plugin. For example say you wanted to add methods for manipulating dates to controllers. This can be done by defining a trait in `src/main/groovy`:

```
package myplugin

@Enhances("Controller")
trait DateTrait {
    Date currentDate() {
        return new Date()
    }
}
```

The `@Enhances` annotation defines the types of artefacts that the trait should be applied to.

As an alternative to using the `@Enhances` annotation above, you can implement a [TraitInjector](#) to tell Grails which artefacts you want to inject the trait into at compile time:

```
package myplugin

@CompileStatic
class ControllerTraitInjector implements TraitInjector {

    @Override
    Class getTrait() {
        SomeTrait
    }

    @Override
    String[] getArtefactTypes() {
        ['Controller'] as String[]
    }
}
```

The above `TraitInjector` will add the `SomeTrait` to all controllers. The `getArtefactTypes` method defines the types of artefacts that the trait should be applied to.

Applying traits conditionally

A `TraitInjector` implementation can also implement the [SupportsClassNode](#) interface to apply traits to only those artefacts which satisfy a custom requirement. For example, if a trait should only be applied if the target artefact class has a specific annotation, it can be done as below

```
package myplugin

@CompileStatic
class AnnotationBasedTraitInjector implements TraitInjector, SupportsClassNode {

    @Override
    Class getTrait() {
        SomeTrait
    }

    @Override
    String[] getArtefactTypes() {
        ['Controller'] as String[]
    }

    boolean supports(ClassNode classNode) {
        return GrailsASTUtils.hasAnnotation(classNode, SomeAnnotation)
    }
}
```

Above `TraitInjector` will add the `SomeTrait` to only those controllers which has the `SomeAnnotation` declared.

The framework discovers trait injectors by way of a `META-INF/grails.factories` descriptor that is in the `.jar` file. This descriptor is automatically generated. The descriptor generated for the code shown above would look like this:

```
#Grails Factories File
grails.compiler.traits.TraitInjector=
myplugin.ControllerTraitInjector,myplugin.DateTraitTraitInjector
```

Due to formatting issues, above code snippet includes a line break after equal sign.

That file is generated automatically and added to the `.jar` file at build time. If for any reason the application defines its own `grails.factories` file at

`src/main/resources/META-INF/grails.factories`, it is important that the trait injectors be explicitly defined in that file. The auto-generated metadata is only reliable if the application does not define its own `src/main/resources/META-INF/grails.factories` file.

17.7 Adding Dynamic Methods at Runtime

The Basics

Grails plugins let you register dynamic methods with any Grails-managed or other class at runtime. This work is done in a `doWithDynamicMethods` method.

Note that Grails 3.x features newer features such as traits that are usable from code compiled with `CompileStatic`. It is recommended that dynamic behavior is only added for cases that are not possible with traits.

```
class ExamplePlugin extends Plugin {
    void doWithDynamicMethods() {
        for (controllerClass in grailsApplication.controllerClasses) {
            controllerClass.metaClass.myNewMethod = {-> println "hello world" }
        }
    }
}
```

In this case we use the implicit application object to get a reference to all of the controller classes' `MetaClass` instances and add a new method called `myNewMethod` to each controller. If you know beforehand the class you wish to add a method to you can simply reference its `metaClass` property.

For example we can add a new method `swapCase` to `java.lang.String`:

```
class ExamplePlugin extends Plugin {

    @Override
    void doWithDynamicMethods() {
        String.metaClass.swapCase = {->
            def sb = new StringBuilder()
            delegate.each {
                sb << (Character.isUpperCase(it as char) ?
                    Character.toLowerCase(it as char) :
                    Character.toUpperCase(it as char))
            }
            sb.toString()
        }

        assert "UpAndDown" == "uPaNdDOWn".swapCase()
    }
}
```

Interacting with the ApplicationContext

The `doWithDynamicMethods` closure gets passed the Spring `ApplicationContext` instance. This is useful as it lets you interact with objects within it. For example if you were implementing a method to interact with Hibernate you could use the `SessionFactory` instance in combination with a `HibernateTemplate`:

```
import org.springframework.orm.hibernate3.HibernateTemplate

class ExampleHibernatePlugin extends Plugin{

    void doWithDynamicMethods() {

        for (domainClass in grailsApplication.domainClasses) {

            domainClass.metaClass.static.load = { Long id->
                def sf = applicationContext.sessionFactory
                def template = new HibernateTemplate(sf)
                template.load(delegate, id)
            }
        }
    }
}
```

Also because of the autowiring and dependency injection capability of the Spring container you can implement more powerful dynamic constructors that use the application context to wire dependencies into your object at runtime:

```
class MyConstructorPlugin {

    void doWithDynamicMethods()
        for (domainClass in grailsApplication.domainClasses) {
            domainClass.metaClass.constructor = {->
                return applicationContext.getBean(domainClass.name)
            }
        }
}
```

Here we actually replace the default constructor with one that looks up prototyped Spring beans instead!

17.8 Participating in Auto Reload Events

Monitoring Resources for Changes

Often it is valuable to monitor resources for changes and perform some action when they occur. This is how Grails implements advanced reloading of application state at runtime. For example, consider this simplified snippet from the Grails `ServicesPlugin`:

```
class ServicesGrailsPlugin extends Plugin {
    ...
}
```

```

def watchedResources = "file:./grails-app/services/**/*Service.groovy"

...
void onChange( Map<String, Object> event) {
    if (event.source) {
        def serviceClass = grailsApplication.addServiceClass(event.source)
        def serviceName = "${serviceClass.propertyName}"
        beans {
            "$serviceName"(serviceClass.getClass()) { bean ->
                bean.autowire = true
            }
        }
    }
}
}
}
}
}

```

First it defines `watchedResources` as either a String or a List of strings that contain either the references or patterns of the resources to watch. If the watched resources specify a Groovy file, when it is changed it will automatically be reloaded and passed into the `onChange` closure in the `event` object.

The `event` object defines a number of useful properties:

- `event.source` - The source of the event, either the reloaded `Class` or a `Spring Resource`
- `event.ctx` - The `Spring ApplicationContext` instance
- `event.plugin` - The plugin object that manages the resource (usually `this`)
- `event.application` - The `GrailsApplication` instance
- `event.manager` - The `GrailsPluginManager` instance

These objects are available to help you apply the appropriate changes based on what changed. In the "Services" example above, a new service bean is re-registered with the `ApplicationContext` when one of the service classes changes.

Influencing Other Plugins

In addition to reacting to changes, sometimes a plugin needs to "influence" another.

Take for example the `Services` and `Controllers` plugins. When a service is reloaded, unless you reload the controllers too, problems will occur when you try to auto-wire the reloaded service into an older controller `Class`.

To get around this, you can specify which plugins another plugin "influences". This means that when one plugin detects a change, it will reload itself and then reload its influenced plugins. For example consider this snippet from the `ServicesGrailsPlugin`:

```
def influences = ['controllers']
```

Observing other plugins

If there is a particular plugin that you would like to observe for changes but not necessary watch the resources that it monitors you can use the "observe" property:

```
def observe = ["controllers"]
```

In this case when a controller is changed you will also receive the event chained from the `controllers` plugin.

It is also possible for a plugin to observe all loaded plugins by using a wildcard:

```
def observe = ["*"]
```

The Logging plugin does exactly this so that it can add the `log` property back to *any* artefact that changes while the application is running.

17.9 Understanding Plugin Load Order

Controlling Plugin Dependencies

Plugins often depend on the presence of other plugins and can adapt depending on the presence of others. This is implemented with two properties. The first is called `dependsOn`. For example, take a look at this snippet from the Hibernate plugin:

```
class HibernateGrailsPlugin {  
    def version = "1.0"  
    def dependsOn = [dataSource: "1.0",  
                    domainClass: "1.0",  
                    i18n: "1.0",  
                    core: "1.0"]  
}
```

The Hibernate plugin is dependent on the presence of four plugins: the `dataSource`, `domainClass`, `i18n` and `core` plugins.

The dependencies will be loaded before the Hibernate plugin and if all dependencies do not load, then the plugin will not load.

The `dependsOn` property also supports a mini expression language for specifying version ranges. A few examples of the syntax can be seen below:

```
def dependsOn = [foo: "*" > 1.0]  
def dependsOn = [foo: "1.0 > 1.1"]  
def dependsOn = [foo: "1.0 > *"]
```

When the wildcard `*` character is used it denotes "any" version. The expression syntax also excludes any suffixes such as `-BETA`, `-ALPHA` etc. so for example the expression `"1.0 > 1.1"` would match any of the following versions:

- 1.1
- 1.0
- 1.0.1
- 1.0.3-SNAPSHOT
- 1.1-BETA2

Controlling Load Order

Using `dependsOn` establishes a "hard" dependency in that if the dependency is not resolved, the plugin will give up and won't load. It is possible though to have a weaker dependency using the `loadAfter` and `loadBefore` properties:

```
def loadAfter = ['controllers']
```

Here the plugin will be loaded after the `controllers` plugin if it exists, otherwise it will just be loaded. The plugin can then adapt to the presence of the other plugin, for example the Hibernate plugin has this code in its `dowithSpring` closure:

```

if (manager?.hasGrailsPlugin("controllers")) {
    openSessionInViewInterceptor(OpenSessionInViewInterceptor) {
        flushMode = HibernateAccessor.FLUSH_MANUAL
        sessionFactory = sessionFactory
    }
    grailsUrlHandlerMapping.interceptors << openSessionInViewInterceptor
}

```

Here the Hibernate plugin will only register an `openSessionInViewInterceptor` if the `controllers` plugin has been loaded. The `manager` variable is an instance of the [GrailsPluginManager](#) interface and it provides methods to interact with other plugins.

You can also use the `loadBefore` property to specify one or more plugins that your plugin should load before:

```

def loadBefore = ['rabbitmq']

```

Scopes and Environments

It's not only plugin load order that you can control. You can also specify which environments your plugin should be loaded in and which scopes (stages of a build). Simply declare one or both of these properties in your plugin descriptor:

```

def environments = ['development', 'test', 'myCustomEnv']
def scopes = [excludes: 'war']

```

In this example, the plugin will only load in the 'development' and 'test' environments. Nor will it be packaged into the WAR file, because it's excluded from the 'war' phase. This allows development-only plugins to not be packaged for production use.

The full list of available scopes are defined by the enum [BuildScope](#), but here's a summary:

- `test` - when running tests
- `functional-test` - when running functional tests
- `run` - for `run-app` and `run-war`
- `war` - when packaging the application as a WAR file
- `all` - plugin applies to all scopes (default)

Both properties can be one of:

- a string - a sole inclusion
- a list - a list of environments or scopes to include
- a map - for full control, with 'includes' and/or 'excludes' keys that can have string or list values

For example,

```

def environments = "test"

```

will only include the plugin in the test environment, whereas

```

def environments = ["development", "test"]

```

will include it in both the development *and* test environments. Finally,

```

def environments = [includes: ["development", "test"]]

```


will do the same thing.

17.10 The Artefact API

You should by now understand that Grails has the concept of artefacts: special types of classes that it knows about and can treat differently from normal Groovy and Java classes, for example by enhancing them with extra properties and methods. Examples of artefacts include domain classes and controllers. What you may not be aware of is that Grails allows application and plugin developers access to the underlying infrastructure for artefacts, which means you can find out what artefacts are available and even enhance them yourself. You can even provide your own custom artefact types.

17.10.1 Asking About Available Artefacts

As a plugin developer, it can be important for you to find out about what domain classes, controllers, or other types of artefact are available in an application. For example, the [Elasticsearch plugin](#) needs to know what domain classes exist so it can check them for any searchable properties and index the appropriate ones. So how does it do it? The answer lies with the `grailsApplication` object, and instance of [GrailsApplication](#) that's available automatically in controllers and GSPs and can be [injected](#) everywhere else.

The `grailsApplication` object has several important properties and methods for querying artefacts. Probably the most common is the one that gives you all the classes of a particular artefact type:

```
for (cls in grailsApplication.<artefactType>Classes) {  
    ...  
}
```

In this case, `artefactType` is the property name form of the artefact type. With core Grails you have:

- domain
- controller
- tagLib
- service
- codec
- bootstrap
- urlMappings

So for example, if you want to iterate over all the domain classes, you use:

```
for (cls in grailsApplication.domainClasses) {  
    ...  
}
```

and for URL mappings:

```
for (cls in grailsApplication.urlMappingsClasses) {  
    ...  
}
```

You need to be aware that the objects returned by these properties are not instances of [Class](#).

Instead, they are instances of [GrailsClass](#) that has some particularly useful properties and methods, including one for the underlying `class`:

- `shortName` - the class name of the artefact without the package (equivalent of `class.getSimpleName()`).
- `logicalPropertyName` - the artefact name in property form without the 'type' suffix. So `MyGreatController` becomes 'myGreat'.
- `isAbstract()` - a boolean indicating whether the artefact class is abstract or not.
- `getPropertyValue(name)` - returns the value of the given property, whether it's a static or an instance one. This works best if the property is initialised on declaration, e.g. `static transactional = true`.

The artefact API also allows you to fetch classes by name and check whether a class is an artefact:

- `get<type>Class(String name)`
- `is<type>Class(Class clazz)`

The first method will retrieve the `GrailsClass` instance for the given name, e.g. 'MyGreatController'. The second will check whether a class is a particular type of artefact. For example, you can use `grailsApplication.isControllerClass(org.example.MyGreatController)` to check whether `MyGreatController` is in fact a controller.

17.10.2 Adding Your Own Artefact Types

Plugins can easily provide their own artefacts so that they can easily find out what implementations are available and take part in reloading. All you need to do is create an `ArtefactHandler` implementation and register it in your main plugin class:

```
class MyGrailsPlugin {
    def artefacts = [ org.somewhere.MyArtefactHandler ]
    ...
}
```

The `artefacts` list can contain either handler classes (as above) or instances of handlers.

So, what does an artefact handler look like? Well, put simply it is an implementation of the [ArtefactHandler](#) interface. To make life a bit easier, there is a skeleton implementation that can readily be extended: [ArtefactHandlerAdapter](#).

In addition to the handler itself, every new artefact needs a corresponding wrapper class that implements [GrailsClass](#). Again, skeleton implementations are available such as [AbstractInjectableGrailsClass](#), which is particularly useful as it turns your artefact into a Spring bean that is auto-wired, just like controllers and services.

The best way to understand how both the handler and wrapper classes work is to look at the Quartz plugin:

- [GrailsJobClass](#)
- [DefaultGrailsJobClass](#)
- [JobArtefactHandler](#)

Another example is the [Shiro plugin](#) which adds a realm artefact.

18 Grails and Spring

This section is for advanced users and those who are interested in how Grails integrates with and builds on the [Spring Framework](#). It is also useful for [plugin developers](#) considering doing runtime configuration Grails.

18.1 Configuring Additional Beans

Using the Spring Bean DSL

You can easily register new (or override existing) beans by configuring them in `grails-app/conf/spring/resources.groovy` which uses the Grails [Spring DSL](#). Beans are defined inside a `beans` property (a Closure):

```
beans = {  
    // beans here  
}
```

As a simple example you can configure a bean with the following syntax:

```
import my.company.MyBeanImpl  
  
beans = {  
    myBean(MyBeanImpl) {  
        someProperty = 42  
        otherProperty = "blue"  
    }  
}
```

Once configured, the bean can be auto-wired into Grails artifacts and other classes that support dependency injection (for example `Bootstrap.groovy` and integration tests) by declaring a public field whose name is your bean's name (in this case `myBean`):

```
class ExampleController {  
    def myBean  
    ...  
}
```

Using the DSL has the advantage that you can mix bean declarations and logic, for example based on the [environment](#):

```
import grails.util.Environment  
import my.company.mock.MockImpl  
import my.company.MyBeanImpl  
  
beans = {  
    switch(Environment.current) {  
        case Environment.PRODUCTION:  
            myBean(MyBeanImpl) {  
                someProperty = 42  
                otherProperty = "blue"  
            }  
            break  
  
        case Environment.DEVELOPMENT:  
            myBean(MockImpl) {  
                someProperty = 42  
                otherProperty = "blue"  
            }  
            break  
    }  
}
```

The `grailsApplication` object can be accessed with the `application` variable and can be used to access the Grails configuration (amongst other things):

```

import grails.util.Environment
import my.company.mock.MockImpl
import my.company.MyBeanImpl

beans = {
    if (application.config.getProperty('my.company.mockService')) {
        myBean(MockImpl) {
            someProperty = 42
            otherProperty = "blue"
        }
    } else {
        myBean(MyBeanImpl) {
            someProperty = 42
            otherProperty = "blue"
        }
    }
}

```

If you define a bean in `resources.groovy` with the same name as one previously registered by Grails or an installed plugin, your bean will replace the previous registration. This is a convenient way to customize behavior without resorting to editing plugin code or other approaches that would affect maintainability.

Using XML

Beans can also be configured using a `grails-app/conf/spring/resources.xml`. In earlier versions of Grails this file was automatically generated for you by the `run-app` script, but the DSL in `resources.groovy` is the preferred approach now so it isn't automatically generated now. But it is still supported - you just need to create it yourself.

This file is typical Spring XML file and the Spring documentation has an [excellent reference](#) on how to configure Spring beans.

The `myBean` bean that we configured using the DSL would be configured with this syntax in the XML file:

```

<bean id="myBean" class="my.company.MyBeanImpl">
    <property name="someProperty" value="42" />
    <property name="otherProperty" value="blue" />
</bean>

```

Like the other bean it can be auto-wired into any class that supports dependency injection:

```

class ExampleController {
    def myBean
}

```

Referencing Existing Beans

Beans declared in `resources.groovy` OR `resources.xml` can reference other beans by convention. For example if you had a `BookService` class its Spring bean name would be `bookService`, so your bean would reference it like this in the DSL:

```

beans = {
    myBean(MyBeanImpl) {
        someProperty = 42
        otherProperty = "blue"
        bookService = ref("bookService")
    }
}

```

or like this in XML:

```

<bean id="myBean" class="my.company.MyBeanImpl">
    <property name="someProperty" value="42" />
    <property name="otherProperty" value="blue" />
    <property name="bookService" ref="bookService" />
</bean>

```

The bean needs a public setter for the bean reference (and also the two simple properties),

which in Groovy would be defined like this:

```
package my.company

class MyBeanImpl {
    Integer someProperty
    String otherProperty
    BookService bookService // or just "def bookService"
}
```

or in Java like this:

```
package my.company;

class MyBeanImpl {

    private BookService bookService;
    private Integer someProperty;
    private String otherProperty;

    public void setBookService(BookService theBookService) {
        this.bookService = theBookService;
    }

    public void setSomeProperty(Integer someProperty) {
        this.someProperty = someProperty;
    }

    public void setOtherProperty(String otherProperty) {
        this.otherProperty = otherProperty;
    }
}
```

Using `ref` (in XML or the DSL) is very powerful since it configures a runtime reference, so the referenced bean doesn't have to exist yet. As long as it's in place when the final application context configuration occurs, everything will be resolved correctly.

For a full reference of the available beans see the plugin reference in the reference guide.

18.2 Runtime Spring with the Beans DSL

This Bean builder in Grails aims to provide a simplified way of wiring together dependencies that uses Spring at its core.

In addition, Spring's regular way of configuration (via XML and annotations) is static and difficult to modify and configure at runtime, other than programmatic XML creation which is both error prone and verbose. Grails' [BeanBuilder](#) changes all that by making it possible to programmatically wire together components at runtime, allowing you to adapt the logic based on system properties or environment variables.

This enables the code to adapt to its environment and avoids unnecessary duplication of code (having different Spring configs for test, development and production environments)

The BeanBuilder class

Grails provides a [grails.spring.BeanBuilder](#) class that uses dynamic Groovy to construct bean definitions. The basics are as follows:

```
import org.apache.commons.dbcp.BasicDataSource
import org.grails.orm.hibernate.ConfigurableLocalSessionFactoryBean
import org.springframework.context.ApplicationContext
import grails.spring.BeanBuilder

def bb = new BeanBuilder()

bb.beans {

    dataSource(BasicDataSource) {
        driverClassName = "org.h2.Driver"
        url = "jdbc:h2:mem:grailsDB"
        username = "sa"
    }
}
```

```

        password = ""
    }

    sessionFactory(ConfigurableLocalSessionFactoryBean) {
        dataSource = ref('dataSource')
        hibernateProperties = ["hibernate.hbm2ddl.auto": "create-drop",
                             "hibernate.show_sql":      "true"]
    }
}

ApplicationContext appContext = bb.createApplicationContext()

```

Within [plugins](#) and the `grails-app/conf/spring/resources.groovy` file you don't need to create a new instance of `BeanBuilder`. Instead the DSL is implicitly available inside the `doWithSpring` and `beans` blocks respectively.

This example shows how you would configure Hibernate with a data source with the `BeanBuilder` class.

Each method call (in this case `dataSource` and `sessionFactory` calls) maps to the name of the bean in Spring. The first argument to the method is the bean's class, whilst the last argument is a block. Within the body of the block you can set properties on the bean using standard Groovy syntax.

Bean references are resolved automatically using the name of the bean. This can be seen in the example above with the way the `sessionFactory` bean resolves the `dataSource` reference.

Certain special properties related to bean management can also be set by the builder, as seen in the following code:

```

sessionFactory(ConfigurableLocalSessionFactoryBean) { bean ->
    // Autowiring behaviour. The other option is 'byType'. <<autowire>>
    bean.autowire = 'byName'
    // Sets the initialisation method to 'init'. [init-method]
    bean.initMethod = 'init'
    // Sets the destruction method to 'destroy'. [destroy-method]
    bean.destroyMethod = 'destroy'
    // Sets the scope of the bean. <<scope>>
    bean.scope = 'request'
    dataSource = ref('dataSource')
    hibernateProperties = ["hibernate.hbm2ddl.auto": "create-drop",
                         "hibernate.show_sql":      "true"]
}

```

The strings in square brackets are the names of the equivalent bean attributes in Spring's XML definition.

Using BeanBuilder with Spring MVC

Include the `grails-spring-<version>.jar` file in your classpath to use `BeanBuilder` in a regular Spring MVC application. Then add the following `<context-param>` values to your

`/WEB-INF/web.xml` file:

```

<context-param>
  <param-name>contextConfigLocation</param-name>
  <param-value>/WEB-INF/applicationContext.groovy</param-value>
</context-param>

<context-param>
  <param-name>contextClass</param-name>
  <param-value>
    grails.web.servlet.context.GrailsWebApplicationContext
  </param-value>
</context-param>

```

Then create a `/WEB-INF/applicationContext.groovy` file that does the rest:

```

import org.apache.commons.dbcp.BasicDataSource

beans {
    dataSource(BasicDataSource) {
        driverClassName = "org.h2.Driver"
        url = "jdbc:h2:mem:grailsDB"
    }
}

```

```

        username = "sa"
        password = ""
    }
}

```

Loading Bean Definitions from the File System

You can use the `BeanBuilder` class to load external Groovy scripts that define beans using the same path matching syntax defined here. For example:

```

def bb = new BeanBuilder()
bb.loadBeans("classpath:*SpringBeans.groovy")

def applicationContext = bb.createApplicationContext()

```

Here the `BeanBuilder` loads all Groovy files on the classpath ending with `SpringBeans.groovy` and parses them into bean definitions. An example script can be seen below:

```

import org.apache.commons.dbcp.BasicDataSource
import org.grails.orm.hibernate.ConfigurableLocalSessionFactoryBean

beans {

    dataSource(BasicDataSource) {
        driverClassName = "org.h2.Driver"
        url = "jdbc:h2:mem:grailsDB"
        username = "sa"
        password = ""
    }

    sessionFactory(ConfigurableLocalSessionFactoryBean) {
        dataSource = dataSource
        hibernateProperties = ["hibernate.hbm2ddl.auto": "create-drop",
                             "hibernate.show_sql": "true"]
    }
}

```

Adding Variables to the Binding (Context)

If you're loading beans from a script you can set the binding to use by creating a Groovy `Binding`:

```

def binding = new Binding()
binding.maxSize = 10000
binding.productGroup = 'finance'

def bb = new BeanBuilder()
bb.binding = binding
bb.loadBeans("classpath:*SpringBeans.groovy")

def ctx = bb.createApplicationContext()

```

Then you can access the `maxSize` and `productGroup` properties in your DSL files.

18.3 The BeanBuilder DSL Explained

Using Constructor Arguments

Constructor arguments can be defined using parameters to each bean-defining method. Put them after the first argument (the Class):

```

bb.beans {
    exampleBean(MyExampleBean, "firstArgument", 2) {
        someProperty = [1, 2, 3]
    }
}

```

This configuration corresponds to a `MyExampleBean` with a constructor that looks like this:

```

MyExampleBean(String foo, int bar) {
    ...
}

```

Configuring the BeanDefinition (Using factory methods)

The first argument to the closure is a reference to the bean configuration instance, which you can use to configure factory methods and invoke any method on the [AbstractBeanDefinition](#) class:

```
bb.beans {
    exampleBean(MyExampleBean) { bean ->
        bean.factoryMethod = "getInstance"
        bean.singleton = false
        someProperty = [1, 2, 3]
    }
}
```

As an alternative you can also use the return value of the bean defining method to configure the bean:

```
bb.beans {
    def example = exampleBean(MyExampleBean) {
        someProperty = [1, 2, 3]
    }
    example.factoryMethod = "getInstance"
}
```

Using Factory beans

Spring defines the concept of factory beans and often a bean is created not directly from a new instance of a Class, but from one of these factories. In this case the bean has no Class argument and instead you must pass the name of the factory bean to the bean defining method:

```
bb.beans {
    myFactory(ExampleFactoryBean) {
        someProperty = [1, 2, 3]
    }
    myBean(myFactory) {
        name = "blah"
    }
}
```

Another common approach is provide the name of the factory method to call on the factory bean. This can be done using Groovy's named parameter syntax:

```
bb.beans {
    myFactory(ExampleFactoryBean) {
        someProperty = [1, 2, 3]
    }
    myBean(myFactory: "getInstance") {
        name = "blah"
    }
}
```

Here the `getInstance` method on the `ExampleFactoryBean` bean will be called to create the `myBean` bean.

Creating Bean References at Runtime

Sometimes you don't know the name of the bean to be created until runtime. In this case you can use a string interpolation to invoke a bean defining method dynamically:

```
def beanName = "example"
bb.beans {
    "${beanName}Bean"(MyExampleBean) {
        someProperty = [1, 2, 3]
    }
}
```


In this case the `beanName` variable defined earlier is used when invoking a bean defining method. The example has a hard-coded value but would work just as well with a name that is generated programmatically based on configuration, system properties, etc.

Furthermore, because sometimes bean names are not known until runtime you may need to reference them by name when wiring together other beans, in this case using the `ref` method:

```
def beanName = "example"
bb.beans {
    "${beanName}Bean"(MyExampleBean) {
        someProperty = [1, 2, 3]
    }
    anotherBean(AnotherBean) {
        example = ref("${beanName}Bean")
    }
}
```

Here the example property of `AnotherBean` is set using a runtime reference to the `exampleBean`. The `ref` method can also be used to refer to beans from a parent `ApplicationContext` that is provided in the constructor of the `BeanBuilder`:

```
ApplicationContext parent = ...//
def bb = new BeanBuilder(parent)
bb.beans {
    anotherBean(AnotherBean) {
        example = ref("${beanName}Bean", true)
    }
}
```

Here the second parameter `true` specifies that the reference will look for the bean in the parent context.

Using Anonymous (Inner) Beans

You can use anonymous inner beans by setting a property of the bean to a block that takes an argument that is the bean type:

```
bb.beans {
    marge(Person) {
        name = "Marge"
        husband = { Person p ->
            name = "Homer"
            age = 45
            props = [overweight: true, height: "1.8m"]
        }
        children = [ref('bart'), ref('lisa')]
    }
    bart(Person) {
        name = "Bart"
        age = 11
    }
    lisa(Person) {
        name = "Lisa"
        age = 9
    }
}
```

In the above example we set the `marge` bean's `husband` property to a block that creates an inner bean reference. Alternatively if you have a factory bean you can omit the type and just use the specified bean definition instead to setup the factory:

```
bb.beans {
    personFactory(PersonFactory)
    marge(Person) {
        name = "Marge"
        husband = { bean ->
            bean.factoryBean = "personFactory"
            bean.factoryMethod = "newInstance"
            name = "Homer"
            age = 45
        }
    }
}
```

```

        props = [overweight: true, height: "1.8m"]
      }
      children = [ref('bart'), ref('lisa')]
    }
  }
}

```

Abstract Beans and Parent Bean Definitions

To create an abstract bean definition define a bean without a `class` parameter:

```

class HolyGrailQuest {
  def start() { println "lets begin" }
}

```

```

class KnightOfTheRoundTable {

  String name
  String leader
  HolyGrailQuest quest

  KnightOfTheRoundTable(String name) {
    this.name = name
  }

  def embarkOnQuest() {
    quest.start()
  }
}

```

```

import grails.spring.BeanBuilder

```

```

def bb = new BeanBuilder()
bb.beans {
  abstractBean {
    leader = "Lancelot"
  }
  ...
}

```

Here we define an abstract bean that has a `leader` property with the value of "Lancelot". To use the abstract bean set it as the parent of the child bean:

```

bb.beans {
  ...
  quest(HolyGrailQuest)

  knights(KnightOfTheRoundTable, "Camelot") { bean ->
    bean.parent = abstractBean
    quest = ref('quest')
  }
}

```

When using a parent bean you must set the parent property of the bean before setting any other properties on the bean!

If you want an abstract bean that has a `class` specified you can do it this way:

```

import grails.spring.BeanBuilder

```

```

def bb = new BeanBuilder()
bb.beans {

  abstractBean(KnightOfTheRoundTable) { bean ->
    bean.'abstract' = true
    leader = "Lancelot"
  }

  quest(HolyGrailQuest)

  knights("Camelot") { bean ->
    bean.parent = abstractBean
    quest = quest
  }
}

```

In this example we create an abstract bean of type `KnightOfTheRoundTable` and use the `bean` argument to set it to abstract. Later we define a `knights` bean that has no `class` defined, but inherits the `class` from the parent bean.

Using Spring Namespaces

Since Spring 2.0, users of Spring have had easier access to key features via XML namespaces. You can use a Spring namespace in BeanBuilder by declaring it with this syntax:

```
xmlns context:"http://www.springframework.org/schema/context"
```

and then invoking a method that matches the names of the Spring namespace tag and its associated attributes:

```
context.'component-scan'('base-package': "my.company.domain")
```

You can do some useful things with Spring namespaces, such as looking up a JNDI resource:

```
xmlns jee:"http://www.springframework.org/schema/jee"
jee.'jndi-lookup'(id: "dataSource", 'jndi-name': "java:comp/env/myDataSource")
```

This example will create a Spring bean with the identifier `dataSource` by performing a JNDI lookup on the given JNDI name. With Spring namespaces you also get full access to all of the powerful AOP support in Spring from BeanBuilder. For example given these two classes:

```
class Person {
    int age
    String name

    void birthday() {
        ++age;
    }
}

class BirthdayCardSender {
    List peopleSentCards = []

    void onBirthday(Person person) {
        peopleSentCards << person
    }
}
```

You can define an aspect that uses a pointcut to detect whenever the `birthday()` method is called:

```
xmlns aop:"http://www.springframework.org/schema/aop"

fred(Person) {
    name = "Fred"
    age = 45
}

birthdayCardSenderAspect(BirthdayCardSender)

aop {
    config("proxy-target-class": true) {
        aspect(id: "sendBirthdayCard", ref: "birthdayCardSenderAspect") {
            after method: "onBirthday",
            pointcut: "execution(void ..Person.birthday()) and this(person)"
        }
    }
}
```

18.4 Property Placeholder Configuration

Grails supports the notion of property placeholder configuration through an extended version of Spring's [PropertyPlaceholderConfigurer](#).

Settings defined in either [ConfigSlurper](#) scripts or Java properties files can be used as placeholder values for Spring configuration in `grails-app/conf/spring/resources.xml` and `grails-app/conf/spring/resources.groovy`. For example given the following entries in

grails-app/conf/application.groovy (or an externalized config):

```
database.driver="com.mysql.jdbc.Driver"
database.dbname="mysql:mydb"
```

You can then specify placeholders in `resources.xml` as follows using the familiar `${..}` syntax:

```
<bean id="dataSource"
      class="org.springframework.jdbc.datasource.DriverManagerDataSource">
  <property name="driverClassName">
    <value>${database.driver}</value>
  </property>
  <property name="url">
    <value>jdbc:${database.dbname}</value>
  </property>
</bean>
```

To specify placeholders in `resources.groovy` you need to use single quotes:

```
dataSource(org.springframework.jdbc.datasource.DriverManagerDataSource) {
  driverClassName = '${database.driver}'
  url = 'jdbc:${database.dbname}'
}
```

This sets the property value to a literal string which is later resolved against the config by Spring's `PropertyPlaceholderConfigurer`.

A better option for `resources.groovy` is to access properties through the `grailsApplication` variable:

```
dataSource(org.springframework.jdbc.datasource.DriverManagerDataSource) {
  driverClassName = grailsApplication.config.getProperty('database.driver', String)
  url = "jdbc:${grailsApplication.config.getProperty('database.dbname', String)}"
}
```

Using this approach will keep the types as defined in your config.

18.5 Property Override Configuration

Grails supports setting of bean properties via [configuration](#).

You define a `beans` block with the names of beans and their values:

```
beans {
  bookService {
    webServiceURL = "http://www.amazon.com"
  }
}
```

The general format is:

```
<<bean name>>.<<property name>> = <<value>>
```

The same configuration in a Java properties file would be:

```
beans.bookService.webServiceURL=http://www.amazon.com
```

18.6 Spring Boot Actuators

[Spring Boot Actuator endpoints](#) allow you to monitor and interact with your application. Spring Boot includes a number of built-in endpoints. For example the `health` endpoint provides basic application health information.

These endpoints are disabled by default since Grails 3.1.8.

You can enable actuator endpoints in your `application.yml` as follows:

`grails-app/conf/application.yml`

```
management:
  endpoints:
    enabled-by-default: true
```

19 Scaffolding

Scaffolding lets you generate some basic CRUD interfaces for a domain class, including:

- The necessary [views](#)
- Controller actions for create/read/update/delete (CRUD) operations

The way for an application to express a dependency on the scaffolding plugin is by including the following in `build.gradle`.

```
dependencies {
    // ...
    implementation "org.grails.plugins:scaffolding"
    // ...
}
```

Dynamic Scaffolding

The simplest way to get started with scaffolding is to enable it by setting the `scaffold` property in the controller to a specific domain class:

```
class BookController {
    static scaffold = Book // Or any other domain class such as "Author", "Publisher"
}
```

With this configured, when you start your application the actions and views will be autogenerated at runtime. The following actions are dynamically implemented by default by the runtime scaffolding mechanism:

- `index`
- `show`
- `edit`
- `delete`
- `create`
- `save`
- `update`

A CRUD interface will also be generated. To access this open <http://localhost:8080/book> in a browser.

Note: The old alternative of defining `scaffold` property:

```
class BookController {
    static scaffold = true
}
```

is no longer supported above Grails 3.0.

If you prefer to keep your domain model in Java and [mapped with Hibernate](#) you can still use scaffolding, simply import the domain class and set its name as the `scaffold` argument.

You can add new actions to a scaffolded controller, for example:

```
class BookController {  
    static scaffold = Book  
  
    def changeAuthor() {  
        def b = Book.get(params.id)  
        b.author = Author.get(params["author.id"])  
        b.save()  
  
        // redirect to a scaffolded action  
        redirect(action:show)  
    }  
}
```

You can also override the scaffolded actions:

```
class BookController {  
    static scaffold = Book  
  
    // overrides scaffolded action to return both authors and books  
    def index() {  
        [bookInstanceList: Book.list(),  
         bookInstanceTotal: Book.count(),  
         authorInstanceList: Author.list()]  
    }  
  
    def show() {  
        def book = Book.get(params.id)  
        log.error("{} ", book)  
        [bookInstance : book]  
    }  
}
```

All of this is what is known as "dynamic scaffolding" where the CRUD interface is generated dynamically at runtime.

By default, the size of text areas in scaffolded views is defined in the CSS, so adding 'rows' and 'cols' attributes will have no effect.

Also, the standard scaffold views expect model variables of the form

<propertyName>InstanceList for collections and <propertyName>Instance for single instances. It's tempting to use properties like 'books' and 'book', but those won't work.

Static Scaffolding

Grails lets you generate a controller and the views used to create the above interface from the command line. To generate a controller type:

```
grails generate-controller Book
```

or to generate the views:

```
grails generate-views Book
```

or to generate everything:

```
grails generate-all Book
```

If you have a domain class in a package or are generating from a [Hibernate mapped class](#) remember to include the fully qualified package name:

```
grails generate-all com.bookstore.Book
```

Customizing the Generated Views

The views adapt to [Validation constraints](#). For example you can change the order that fields appear in the views simply by re-ordering the constraints in the builder:

```
def constraints = {
    title()
    releaseDate()
}
```

You can also get the generator to generate lists instead of text inputs if you use the `inList` constraint:

```
def constraints = {
    title()
    category(inList: ["Fiction", "Non-fiction", "Biography"])
    releaseDate()
}
```

Or if you use the `range` constraint on a number:

```
def constraints = {
    age(range:18..65)
}
```

Restricting the size with a constraint also effects how many characters can be entered in the view:

```
def constraints = {
    name(size:0..30)
}
```

The Fields Plugin

The Grails scaffolding templates make use of the [The Fields Plugin](#). Once you've generated the scaffold views, you can customize the forms and tables using the `TagLib` provided by the plugin (see the [Fields plugin docs](#) for details).

```
<%- Generate an HTML table from bookInstanceList, showing only 'title' and 'category' columns -%>
<f:table collection="bookInstanceList" properties="['title', 'category']"/>
```

Customizing the Scaffolding templates

The templates used by Grails to generate the controller and views can be customized by installing the templates with the [install-templates](#) command.

20 Deployment

Grails applications can be deployed in a number of ways, each of which has its pros and cons.

20.1 Standalone

"./gradlew bootRun"

You should be very familiar with this approach by now, since it is the most common method of running an application during the development phase. An embedded Tomcat server is launched that loads the web application from the development sources, thus allowing it to pick up any changes to application files.

You can run the application in the production environment using:

```
./gradlew bootRun -Dgrails.env=prod
```

You can run the app using the `bootRun` Gradle task. The next command uses the [Gradle Wrapper](#).

```
./gradlew bootRun
```

You can specify an environment supplying `grails.env` system property.

```
./gradlew -Dgrails.env=prod bootRun
```

Runnable WAR or JAR file

Another way to deploy in Grails 3.0 or above is to use the new support for runnable JAR or WAR files. To create runnable archives, run `grails package`:

```
grails package
```

Alternatively, you could use the `assemble` Gradle task.

```
./gradlew assemble
```

You can then run either the WAR file or the JAR using your Java installation:

```
java -Dgrails.env=prod -jar build/libs/mywar-0.1.war (or .jar)
```

A TAR/ZIP distribution

Note: TAR/ZIP distribution assembly has been removed from Grails 3.1.

20.2 Container Deployment (e.g. Tomcat)

Grails apps can be deployed to a Servlet Container or Application Server.

WAR file

A common approach to Grails application deployment in production is to deploy to an existing Servlet container via a WAR file. Containers allow multiple applications to be deployed on the same port with different paths.

Creating a WAR file is as simple as executing the [war](#) command:

```
grails war
```

This will produce a WAR file that can be deployed to a container, in the `build/libs` directory.

Note that by default Grails will include an embeddable version of Tomcat inside the WAR file so that it is runnable (see the previous section), this can cause problems if you deploy to a different version of Tomcat. If you don't intend to use the embedded container then you should either remove the Tomcat dependencies or change the scope to `testImplementation` prior to deploying to your production container in `build.gradle`:

```
testImplementation "org.springframework.boot:spring-boot-starter-tomcat"
```

Application servers

The Grails framework requires that runtime containers support Servlet 3.0 and above. By default, Grails framework applications are bundled with an embeddable Tomcat and testing is primarily done with Tomcat. Any servlet container meeting the minimum requirements should be able to run Grails framework applications, but some workarounds may be required for container-specific bugs or configurations.

20.3 Deployment Configuration Tasks

Setting up HTTPS and SSL certificates for standalone deployment

To configure an SSL certificate and to listen on an HTTPS port instead of HTTP, add properties like these to `application.yml`:

```
server:
  port: 8443                                # The port to listen on
  ssl:
    enabled: true                           # Activate HTTPS mode on the server port
    key-store: <the-location-of-your-keystore> # e.g. /etc/tomcat7/keystore/tomcat.keystore
    key-store-password: <your-key-store-password> # e.g. changeit
    key-alias: <your-key-alias>                # e.g. tomcat
    key-password: <usually-the-same-as-your-key-store-password>
```

These settings control the embedded Tomcat container for a production deployment.

Alternatively, the properties can be specified on the command-line. Example:

```
-Dserver.ssl.enabled=true -Dserver.ssl.key-store=/path/to/keystore.
```

Configuration of both an HTTP and HTTPS connector via application properties is not supported. If you want to have both, then you'll need to configure one of them programmatically. (More information on how to do this can be found in the how-to guide below.)

There are other relevant settings. Further reference:

- [Spring Boot: Embed Webservers - Configure SSL](#)
- [Spring Book: Enable Multiple Connectors with Tomcat](#)
- [Spring Boot: Common Application Properties](#)

21 Contributing to Grails

Grails is an open source project with an active community and we rely heavily on that community to help make Grails better. As such, there are various ways in which people can contribute to Grails. One of these is by [writing useful plugins](#) and making them publicly available. In this chapter, we'll look at some of the other options.

21.1 Report Issues in Github's issue tracker

Grails uses Github to track issues in the [core framework](#). Similarly for its documentation there is a [separate tracker](#). If you've found a bug or wish to see a particular feature added, these are the places to start. You'll need to create a (free) github account in order to either submit an issue or comment on an existing one in either of these.

When submitting issues, please provide as much information as possible and in the case of bugs, make sure you explain which versions of Groovy, Grails and various plugins you are using. Other environment details - OS version, JDK, Gradle etc. should also be included.

Also, an issue is much more likely to be dealt with if you upload a reproducible sample application on a github repository and provide a link in the issue.

Reviewing issues

There are quite a few old issues in github, some of which may no longer be valid. The core team can't track down these alone, so a very simple contribution that you can make is to verify one or two issues occasionally.

Which issues need verification? Going to the [issue tracker](#) will display all issues that haven't been resolved.

Once you've verified an issue, simply add a short comment explaining what you found. Be sure to mention your environment details and grails version.

21.2 Build From Source and Run Tests

If you're interested in contributing fixes and features to any part of grails, you will have to learn how to get hold of the project's source, build it and test it with your own applications. Before you start, make sure you have:

- A JDK (11 or above)
- A git client

Once you have all the pre-requisite packages installed, the next step is to download the Grails source code, which is hosted at [GitHub](#) in several repositories owned by the "[grails](#)" [GitHub user](#). This is a simple case of cloning the repository you're interested in. For example, to get the core framework run:

```
git clone http://github.com/grails/grails-core.git
```

This will create a "grails-core" directory in your current working directory containing all the project source files. The next step is to get a Grails installation from the source.

Creating a Grails installation

If you look at the project structure, you'll see that it doesn't look much like a standard `GRAILS_HOME` installation. But, it's very simple to turn it into one. Just run this from the root directory of the project:

```
./gradlew install
```

This will fetch all the standard dependencies required by Grails and then build a `GRAILS_HOME` installation. Note that this target skips the extensive collection of Grails test classes, which can take some time to complete.

Once the above command has finished, simply set the `GRAILS_HOME` environment variable to the checkout directory and add the "bin" directory to your path. When you next type `grails` command to run, you'll be using the version you just built.

If you are using [SDKMAN](#) then that can also be used to work with this local installation via the following:

```
sdk install grails dev /path/to/grails-core
```

You will also need to publish your local installation to your local maven.

```
./gradlew pTML
```

Now you will have a dev version in your local which you can use to test your features.

Running the test suite

All you have to do to run the full suite of tests is:

```
./gradlew test
```

These will take a while (15-30 mins), so consider running individual tests using the command line. For example, to run the test spec `BinaryPluginSpec` simply execute the following command:

```
./gradlew :grails-core:test --tests *.BinaryPluginSpec
```

Note that you need to specify the sub-project that the test case resides in, because the top-level "test" target won't work....

Developing in IntelliJ IDEA

You need to run the following gradle task:

```
./gradlew idea
```

Then open the project file which is generated in IDEA. Simple!

Developing in STS / Eclipse

You need to run the following gradle task:

```
./gradlew cleanEclipse eclipse
```

Before importing projects to STS do the following action:

- Edit `grails-scripts/.classpath` and remove the line `"<classpathentry kind="src" path="../scripts"/>"`.

Use "ImportGeneralExisting Projects into Workspace" to import all projects to STS. There will be a few build errors. To fix them do the following:

- Add the `springloaded-core` JAR file in `$GRAILS_HOME/lib/org.springframework.springloaded/springloaded-core/jars` to `grails-core`'s classpath.
- Remove `"src/test/groovy"` from `grails-plugin-testing`'s source path [GRECLIPSE-1067](#)
- Add the `jsp-api` JAR file in `$GRAILS_HOME/lib/javax.servlet.jsp/jsp-api/jars` to the classpath of `grails-web`
- Fix the source path of `grails-scripts`. Add linked source folder linking to `"../scripts"`. If you get build errors in `grails-scripts`, do `"./gradlew cleanEclipse eclipse"` in that directory and edit the `.classpath` file again (remove the line `"<classpathentry kind="src" path="../scripts"/>"`). Remove possible empty `"scripts"` directory under `grails-scripts` if you are not able to add the linked folder.
- Do a clean build for the whole workspace.

- To use Eclipse GIT scm team provider: Select all projects (except "Servers") in the navigation and right click Team Share project (not "Share projects"). Choose "Git". Then check "Use or create repository in parent folder of project" and click "Finish".
- Get the recommended code style settings from the [mailing list thread](#) (final style not decided yet, currently [profile.xml](#)). Import the code style xml file to STS in WindowPreferencesJavaCode StyleFormatterImport . Grails code uses spaces instead of tabs for indenting.

Debugging Grails or a Grails application

To enable debugging, run:

```
./gradlew bootRun --debug-jvm
```

By default Grails forks a JVM to run the application in. The `-debug-jvm` argument causes the debugger to be associated with the forked JVM.

21.3 Submit Patches to Grails Core

If you want to submit patches to the project, you simply need to fork the repository on GitHub rather than clone it directly. Then you will commit your changes to your fork and send a pull request for a core team member to review.

Forking and Pull Requests

One of the benefits of [GitHub](#) is the way that you can easily contribute to a project by [forking the repository](#) and [sending pull requests](#) with your changes.

What follows are some guidelines to help ensure that your pull requests are speedily dealt with and provide the information we need. They will also make your life easier!

Make sure your fork is up to date

Making changes to outdated sources is not a good idea. Someone else may have already made the change.

```
git pull upstream master
```

Create a local branch for your changes

Your life will be greatly simplified if you create a local branch to make your changes on. For example, as soon as you fork a repository and clone the fork locally, execute

```
git checkout -b issue_123
```

This will create a new local branch called "issue_123" based off the "master" branch. Of course, you can name the branch whatever you like, but a good idea would be to reference the GitHub issue number that the change is relevant to. Each Pull Request should have its own branch.

Create Github issues for non-trivial changes

For any non-trivial changes, raise an issue on github if one doesn't already exist. That helps us keep track of what changes go into each new version of Grails.

Include github issue ID in commit messages

This may not seem particularly important, but having a github issue ID in a commit message means that we can find out at a later date why a change was made. Include the ID in any and all commits that relate to that issue. If a commit isn't related to an issue, then there's no need to include an issue ID.

Make sure your fork is up to date again and rebase

Since the core developers must merge your commits into the main repository, it makes life much easier if your fork on GitHub is up to date before you send a pull request.

Let's say you have the main repository set up as a remote called "upstream" and you want to submit a pull request. Also, all your changes are currently on the local "issue_123" branch but not on "master". The first step involves pulling any changes from the main repository that have been added since you last fetched and merged:

```
git checkout master
git pull upstream master
```

This should complete without any problems or conflicts. Next, rebase your local branch against the now up-to-date master:

```
git checkout issue_123
git rebase master
```

What this does is rearrange the commits such that all of your changes come after the most recent one in master. Think adding some cards to the top of a deck rather than shuffling them into the pack.

Push your branch to GitHub and send Pull Request

Finally, you must push your changes to your fork on GitHub, otherwise the core developers won't be able to pick them up:

```
git push origin issue_123
```

You should not merge your branch to your forks master. If the Pull Request is not accepted, your master will then be out of sync with upstream forever.

You're now ready to send the pull request from the GitHub user interface.

Say what your pull request is for

A pull request can contain any number of commits and it may be related to any number of issues. In the pull request message, please specify the IDs of all issues that the request relates to. Also give a brief description of the work you have done, such as: "I refactored the data binder and added support for custom number editors. Fixes #xxxx".

21.4 Submit Patches to Grails Documentation

Contributing Simple Changes

The user guide is written using [Asciidoctor](#). The simplest way to contribute fixes is to simply click on the "Improve this doc" link that is to the right of each section of the documentation.

This will link to the Github edit screen where you can make changes, preview them and create a pull request.

Building the Guide

If you want to make significant changes, such as changing the structure of the table of contents etc. then we recommend you build the user guide. To do that simply checkout the sources from github:

```
$ git clone https://github.com/grails/grails-doc/  
$ cd grails-doc
```

The source files can be found in the `src/en/guide` directory. Whilst the Table of Contents (TOC) is defined in the `src/en/guide/toc.yml` file.

Each YAML key points to a Asciidoc template. For example consider the following YAML:

```
introduction:  
  title: Introduction  
  whatsNew:  
    title: What's new in Grails 3.2?  
  ...
```

The `introduction` key points to `src/en/guide/introduction.adoc`. The `title` key defines the title that is displayed in the TOC. Because `whatsNew` key is nested underneath the `introduction` key it points to `src/en/guide/introduction/whatsNew.adoc`, which is nested in a directory called `introduction`.

Essentially, using the `toc.yml` file and the directory structure you can manipulate the structure of the user guide.

To generate the documentation run the `publishGuide` task:

```
$ ./gradlew publishGuide -x apiDocs
```

In the above example we skip the `apiDocs` task to speed up building of the guide, otherwise all Groovydoc documentation will be built too!

Once the guide is built simply open the `build/docs/index.html` file in a browser to review your changes.

Copies of this document may be made for your own use and for distribution to others, provided that you do not charge any fee for such copies and further provided that each copy contains this Copyright Notice, whether distributed in print or electronically.